

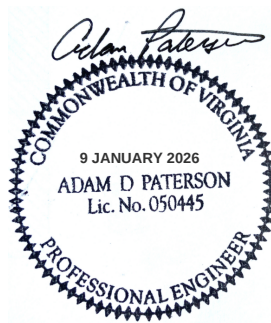
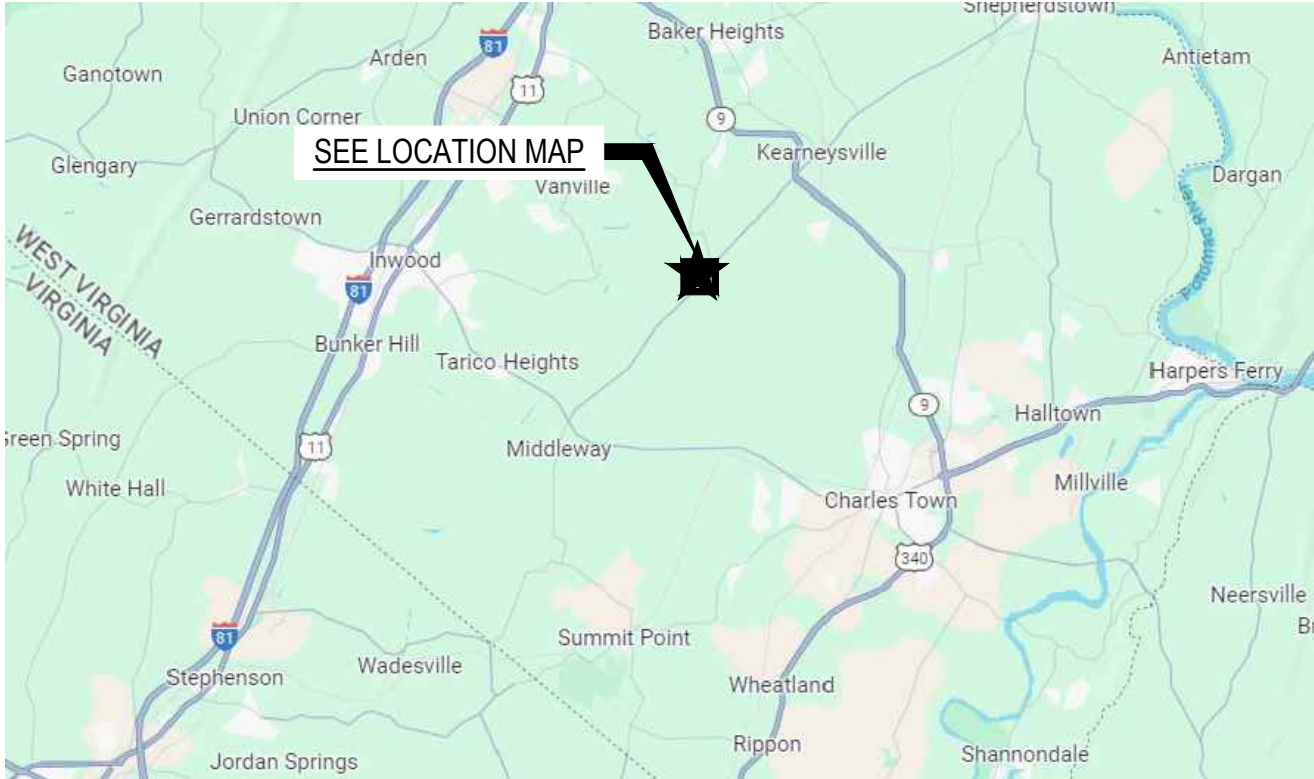
NATIONAL CENTER FOR COOL AND COLD WATER  
AQUACULTURE (NCCCCWA)  
11861 LEETOWN RD, KEARNEYSVILLE, WV 25430

NCCCCWA FIRE SYSTEM UPGRADES

LOCATION MAP



VICINITY MAP



|                                                                                                                                                                                       |  |                 |  |                  |  |                                                     |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|--|------------------|--|-----------------------------------------------------|--|
| <div><div>SUMMER</div><div>CONSULTANTS</div><div>INCORPORATED</div></div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div> |  |                 |  |                  |  |                                                     |  |
|                                                                                                                                                                                       |  |                 |  |                  |  |                                                     |  |
|                                                                                                                                                                                       |  | NO.             |  | DATE             |  | BY                                                  |  |
| A-E FIRM                                                                                                                                                                              |  | USDA            |  |                  |  | REVISIONS                                           |  |
| PROJECT MANAGER                                                                                                                                                                       |  | EPM             |  | U<br>S<br>D<br>A |  | Agricultural Research Service                       |  |
| PATERSON, A                                                                                                                                                                           |  | C. MORRIS       |  |                  |  | FIRE SYSTEM UPGRADES                                |  |
|                                                                                                                                                                                       |  |                 |  |                  |  | KEARNEYSVILLE, WEST VIRGINIA                        |  |
| DESIGNER                                                                                                                                                                              |  | PPM             |  |                  |  | NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE |  |
| ROBERTSON, A                                                                                                                                                                          |  | T. VIGIL        |  |                  |  |                                                     |  |
| CHECKED BY                                                                                                                                                                            |  | SAFETY & HEALTH |  | SOLICITATION NO. |  | DWG. NO.                                            |  |
| NICHOLS, J                                                                                                                                                                            |  | TBD             |  |                  |  | G-001                                               |  |
| DRAWN BY                                                                                                                                                                              |  | REAL PROPERTY   |  | DATE             |  | COVER SHEET                                         |  |
| ROBERTSON, A                                                                                                                                                                          |  | TBD             |  | 2025-12-12       |  |                                                     |  |
|                                                                                                                                                                                       |  |                 |  |                  |  | SHEET 1 OF 29                                       |  |



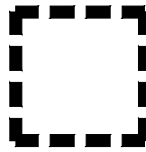
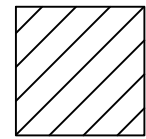




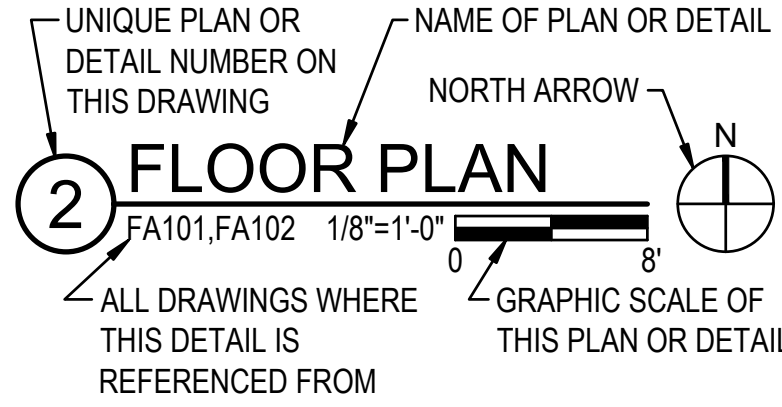
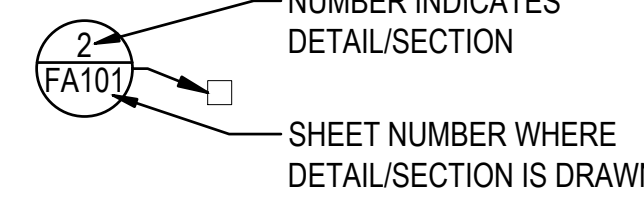
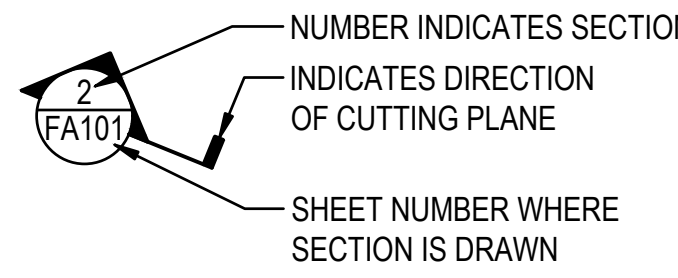
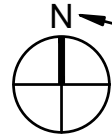
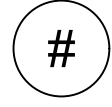
ABBREVIATIONS

|        |                                              |
|--------|----------------------------------------------|
| A      | AMPERES.                                     |
| AC     | ABOVE CEILING                                |
| ADS    | ACOUSTICALLY DISTINGUISHABLE SPACE           |
| AFF    | ABOVE FINISHED FLOOR                         |
| AHJ    | AUTHORITY HAVING JURISDICTION                |
| AMP    | AMPLIFIER                                    |
| AS     | ASPIRATING SMOKE                             |
| AWG    | AMERICAN WIRE GAUGE                          |
| C      | CEILING MOUNT                                |
| CA     | CLEAN AGENT                                  |
| CL     | CEILING LEVEL                                |
| CONT.  | CONTINUED                                    |
| D      | DRAIN                                        |
| DN     | DOWN                                         |
| ELEV.  | ELEVATOR                                     |
| EMT    | ELECTRICAL METALLIC TUBING                   |
| (E)    | EXISTING                                     |
| F      | FIRE LINE                                    |
| FA     | FIRE ALARM                                   |
| FAAP   | FIRE ALARM ANNUNCIATOR PANEL                 |
| FACP   | FIRE ALARM CONTROL PANEL                     |
| FARP   | FIRE ALARM RELEASING PANEL                   |
| FP     | FIRE PUMP                                    |
| FPL    | FIRE PUMP PHASE LOSS                         |
| FPP    | FIRE PUMP POWER                              |
| FPR    | FIRE PUMP PHASE REVERSAL                     |
| FPS    | FIRE PUMP SECONDARY POWER                    |
| FMCP   | FIRE ALARM / MASS NOTIFICATION CONTROL PANEL |
| FATC   | FIRE ALARM TERMINAL CABINET                  |
| LOC    | LOCAL OPERATING CONSOLE                      |
| MNS    | MASS NOTIFICATION SYSTEM                     |
| NAC    | NOTIFICATION APPLIANCE CIRCUIT               |
| N.I.C. | NOT IN CONTRACT                              |
| P      | PHOTOELECTRIC                                |
| (R)    | RELOCATE                                     |
| SLC    | SIGNALING LINE CIRCUIT                       |
| SP     | SPRINKLER PIPING                             |
| SPD    | SURGE PROTECTIVE DEVICE                      |
| TYP.   | TYPICAL                                      |
| WP     | WEATHERPROOF                                 |
| ZCA    | SPRINKLER ZONE CONTROL ASSEMBLY              |

SYMBOLS

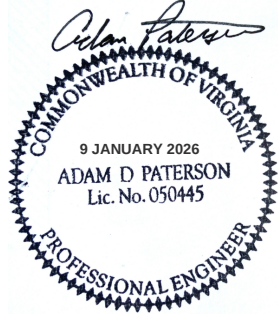
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|    | AREA OF WORK / FIRE ALARM ZONE                                                                                                                                                                                                                                                |
|    | MODIFICATION OF THE FIRE ALARM SYSTEM IN THIS AREA IS NOT IN CONTRACT (N.I.C.)                                                                                                                                                                                                |
|    | MANUAL PULL STATION                                                                                                                                                                                                                                                           |
|    | WATERFLOW SWITCH                                                                                                                                                                                                                                                              |
|    | TAMPER SWITCH                                                                                                                                                                                                                                                                 |
|    | FIRE ALARM CONTROL PANEL                                                                                                                                                                                                                                                      |
|    | FIRE ALARM ANNUNCIATOR PANEL                                                                                                                                                                                                                                                  |
|    | FIRE ALARM TERMINAL CABINET                                                                                                                                                                                                                                                   |
|    | BATTERY CABINET                                                                                                                                                                                                                                                               |
|    | DIGITAL ALARM COMMUNICATOR TRANSMITTER                                                                                                                                                                                                                                        |
|    | SURGE PROTECTIVE DEVICE                                                                                                                                                                                                                                                       |
|    | TYPE                                                                                                                                                                                                                                                                          |
|    | SMOKE DETECTOR - CEILING MOUNTED                                                                                                                                                                                                                                              |
|    | DUCT SMOKE DETECTOR                                                                                                                                                                                                                                                           |
|    | HORN - WALL MOUNTED                                                                                                                                                                                                                                                           |
|    | HORN / STROBE - WALL MOUNTED<br>CD = CANDELA RATING                                                                                                                                                                                                                           |
|    | HORN - CEILING MOUNTED                                                                                                                                                                                                                                                        |
|   | HORN / STROBE - CEILING MOUNTED<br>CD = CANDELA RATING                                                                                                                                                                                                                        |
|  | STROBE - WALL MOUNTED<br>CD = CANDELA RATING                                                                                                                                                                                                                                  |
|  | STROBE - CEILING MOUNTED<br>CD = CANDELA RATING                                                                                                                                                                                                                               |
|  | ADDRESSABLE INPUT (MONITOR) MODULE <ul style="list-style-type: none"><li>TS = TAMPER SWITCH</li><li>PS = PRESSURE SWITCH</li><li>WF = WATER FLOW SWITCH</li><li>PW = POWER MONITOR</li></ul>                                                                                  |
|  | ADDRESSABLE OUTPUT (CONTROL) MODULE <ul style="list-style-type: none"><li>FPR = FIRE PUMP RUNNING</li><li>FPP = FIRE PUMP PHASE REVERSAL</li><li>FPS = FIRE PUMP SECONDARY POWER</li><li>FPL = FIRE PUMP POWER LOSS</li><li>FP = GENERAL GROUPING OF ABOVE NOTATION</li></ul> |
|  | NAC                                                                                                                                                                                                                                                                           |
|  | SLC                                                                                                                                                                                                                                                                           |

GENERAL SYMBOLS

|                                                                                       |                                                  |
|---------------------------------------------------------------------------------------|--------------------------------------------------|
| FLOOR PLAN AND DETAIL TITLES:                                                         |                                                  |
|    |                                                  |
| 2-PART REFERENCE DETAIL IDENTIFICATION:                                               |                                                  |
|    |                                                  |
| SECTION IDENTIFICATION                                                                |                                                  |
|    |                                                  |
| NORTH ARROW                                                                           |                                                  |
|    | ARROW INDICATES TRUE NORTH AND/OR PROJECT NORTH. |
|    | COLUMN LINE NUMBER                               |
|    | REVISION CLOUD WITH REVISION NUMBER DESIGNATION  |
|  | BREAKLINE                                        |
|  | MATCHLINE DESIGNATION                            |
|  | EXISTING                                         |
|  | POINT OF CONNECTION TO EXISTING                  |
|  | POINT OF TERMINATION OF REMOVAL                  |
|  | NOTE NUMBER                                      |
|  | ROOM NUMBER                                      |

FIRE ALARM REQUIREMENTS

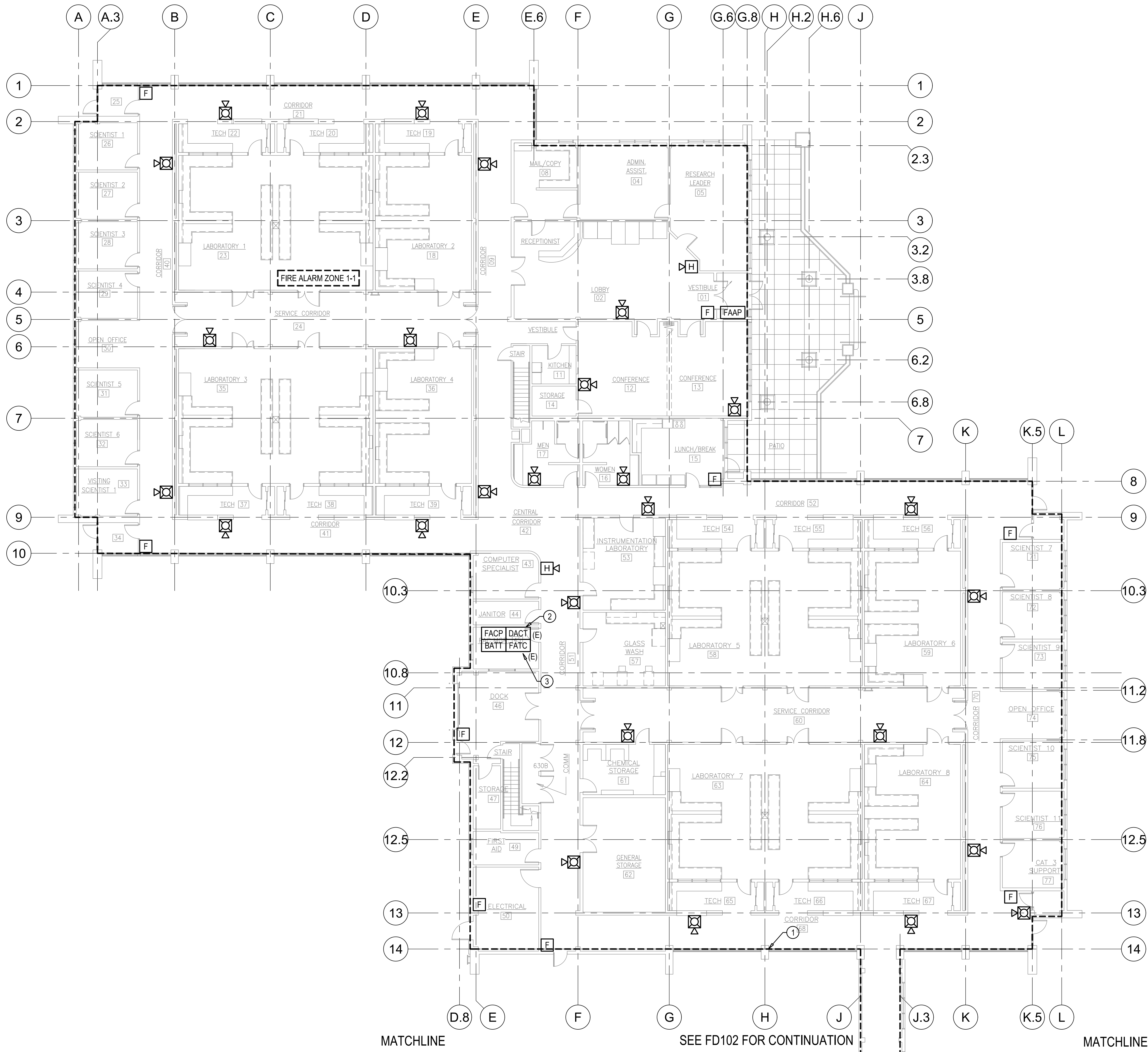
- DISCONNECT AND REMOVE THE EXISTING FIRE ALARM AND DETECTION SYSTEM FOLLOWING THE INSTALLATION OF THE NEW FIRE ALARM SYSTEM.
- PROVIDE FIRE ALARM AND DETECTION SYSTEM AS INDICATED IN THE DRAWINGS, SPECIFICATIONS IN COMPLIANCE WITH ARS 242.1 2024 EDITION AND NFPA 72 2025 EDITION COMPLIANT SYSTEM.
- PROVIDE CLASS B, SOLID WIRING FOR RISER, SLC, AND NAC CIRCUIT. FOLLOW MANUFACTURER GUIDELINES FOR CIRCUIT SIZING.
- PROVIDE SURVIVABILITY LEVEL 0 WIRING AT MINIMUM.
- PROVIDE FIRE ALARM CONDUCTORS IN EMT CONDUIT IN MAIN/TANK BUILDING. PROVIDE GRAY CPVC CONDUIT IN BROODSTOCK BUILDING.
- PLACE EQUIPMENT AT CENTER OF ACOUSTICAL CEILING TILES.
- DRAWINGS ARE DIAGRAMMATIC IN NATURE AND ARE NOT INTENDED TO BE USED FOR COMPLETE SYSTEM DESIGN. COORDINATE SYSTEM COMPONENTS WITH ALL OTHER DISCIPLINES AND TRADES.
- PROVIDE A FIRE ALARM SYSTEM DESIGNER WITH A MINIMUM NICET LEVEL II CERTIFICATION IN FIRE ALARM SYSTEM TECHNOLOGY OR A LICENSED PROFESSIONAL FIRE PROTECTION ENGINEER RESPONSIBLE FOR THE FIRE ALARM SYSTEM DESIGN.
- CHANGES TO THE DESIGN AND EQUIPMENT INSTALLATION IS THE RESPONSIBILITY OF CONTRACTOR.
- EXPOSED WIRING WILL NOT BE PERMITTED IN ANY PORTION OF THE BUILDING. FLEXIBLE METALLIC CONDUIT NOT EXCEEDING 6 FEET IS PERMITTED FOR TERMINATIONS AT DEVICES. ROUTE EMT CONDUIT TIGHT TO STRUCTURE.
- T-TAPS ARE PERMITTED. CONDUIT TERMINATIONS MUST BE ON SCREW TERMINALS OF SYSTEM COMPONENT. WIRE NUTS ARE PROHIBITED.
- PROVIDE A LABEL FOR EACH WIRE/CIRCUIT (WITHIN ALL PANELS, CABINETS AND JUNCTION BOXES) INDICATING ITS PURPOSE AND LOCATION. LABELS MUST BE A MINIMUM FONT SIZE OF 1/4-INCH AND ARIAL NARROW. EXAMPLE OF A FIRST FLOOR SIGNALING LINE CIRCUIT ONE OF TWO IS SLC-C1R1-FL1.
- PROVIDE 25% ADDITIONAL CIRCUIT CAPACITY FOR STROBES AND HORNS.
- SEAL FIRE BARRIER/PARTITIONS AND FIRE RATED FLOOR PENETRATIONS WITH AN UL LISTED OR FM APPROVED FIRE STOPPING ASSEMBLY. FIRE STOPPING SYSTEM MUST AT MINIMUM MATCH THE EXISTING RATING OF THE PENETRATED WALL/FLOOR.
- ALL REQUIRED FIRE ALARM SYSTEM OUTAGES SHALL BE COORDINATED THROUGH THE CONTRACTING OFFICER REPRESENTATIVE NOT LESS THAN 15 WORKING DAYS IN ADVANCE.
- A FIRE WATCH MUST BE CONDUCTED CONTINUOUSLY THROUGHOUT THE DURATION OF THE OUTAGE IN ALL AREAS AFFECTED BY THE OUTAGE. SEE USDA DIV 1 SPEC FOR INFORMATION AND FORMS. FINDINGS/COMMENTS WILL BE DOCUMENTED HOURLY ON FIRE WATCH FORMS. DOCUMENTATION OF FINDINGS WILL BE MADE AVAILABLE TO THE FIRE DEPARTMENT OFFICIALS UPON THEIR REQUEST. IN THE EVENT OF ANY FIRE/SMOKE EMERGENCIES DISCOVERED DURING THE FIRE WATCH, EMERGENCY NOTIFICATION SHALL BE PROVIDED TO THE PREDETERMINED POINTS OF CONTACT.
- OVERNIGHT OUTAGES WILL NOT BE PERMITTED UNLESS WORK IS BEING DONE DURING THAT TIME PERIOD AND THE FIRE ALARM SYSTEM IS PUT BACK INTO SERVICE WITHIN AN HOUR FROM COMPLETION OF WORK. OUTAGES SHALL BE LIMITED TO 8 HOURS.
- THE FIRE ALARM SYSTEM IS NOT PERMITTED TO HAVE AN OUTAGE FOR MORE THAN AN HOUR BEFORE STARTING WORK. THE FIRE ALARM SYSTEM SHALL BE RESTORED WITHIN AN HOUR OF COMPLETION OF WORK. THE CONTRACTOR CANNOT LEAVE THE CONSTRUCTION SITE UNPROTECTED.
- THE FIRE ALARM SYSTEM SHALL PRODUCE A SOUND LEVEL AT LEAST 15 dB ABOVE THE AVERAGE AMBIENT SOUND LEVEL, OR 5 dB ABOVE THE MAXIMUM SOUND LEVEL HAVING A DURATION OF AT LEAST 60 SECONDS, WHICHEVER IS GREATER, MEASURED 5 FT ABOVE THE FLOOR IN THE ADS REQUIRING AUDIBILITY.
- BIOSECURITY MUST BE FOLLOWED WHEN OPERATING IN THE FOLLOWING LOCATIONS TO PROTECT FISH HEALTH: TANK ROOMS A-G, FISH HEALTH LABS 07, 08, 17, 18 AND ALONG THE BROODSTOCK BUILDING. FURTHER INFORMATION CAN BE FOUND ON ATTACHMENT BIOSECURITY STATEMENT.



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FILE NAME: 212.012 FD101.DWG  
OUR REF: 212.012  
PLOT DATE: 12/12/2025 8:46:05 AM



1 FIRST FLOOR MAIN BUILDING - DEMOLITION

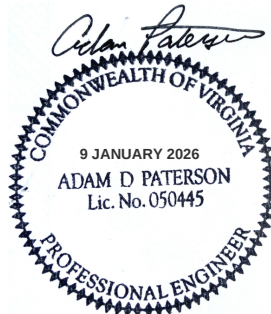
3/32"=1'-0"  
0 8 16 24'

GENERAL NOTES

G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

NOTES

1. DISCONNECT AND REMOVE THE EXISTING FIRE ALARM SYSTEM, ASSOCIATED 24V POWER, CONDUCTORS, CONDUIT, ELECTRICAL BACK-BOXES, MOUNTS, BRACKETS, SCREWS, ETC IN ITS ENTIRETY. PATCH AND REPAIR ALL WALLS.
2. THE DACT PROVIDES COMMUNICATION PRIMARILY FOR SECURITY. THERE ARE CONNECTIONS FOR FIRE ALARM, SUPERVISORY, AND TROUBLE REPORTING. DISCONNECT AND REMOVE ALL FIRE ALARM CONDUCTORS, STRAPS, AND REPROGRAM THE PANEL.
3. DISCONNECT FIRE ALARM CONDUCTORS FROM TERMINAL CABINET.



|                                                                                                                                                        |  |                        |  |                    |                                                                                                                               |                                           |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|--|--------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--|
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|                                                                                                                                                        |  |                        |  |                    |                                                                                                                               |                                           |  |
|                                                                                                                                                        |  | NO.                    |  | DATE               |                                                                                                                               | BY                                        |  |
| A-E FIRM                                                                                                                                               |  | USDA                   |  |                    |                                                                                                                               | REVISIONS                                 |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                         |  | EPM<br>C. MORRIS       |  | U<br>S<br>D<br>A   | <div>Agricultural Research Service</div> |                                           |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                               |  | PPM<br>T. VIGIL        |  |                    | FIRE SYSTEM UPGRADES<br><small>KEARNEYVILLE, WEST VIRGINIA</small>                                                            |                                           |  |
|                                                                                                                                                        |  |                        |  |                    | NATIONAL CENTER FOR COOL AND COLD WATER<br>AQUACULTURE                                                                        |                                           |  |
| CHECKED BY<br>PATERSON, A                                                                                                                              |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.   |                                                                                                                               | DWG. NO.                                  |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                               |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12 |                                                                                                                               | FIRST FLOOR MAIN BUILDING -<br>DEMOLITION |  |
|                                                                                                                                                        |  |                        |  |                    |                                                                                                                               | FD101                                     |  |
|                                                                                                                                                        |  |                        |  |                    |                                                                                                                               | SHEET 4 of 29                             |  |

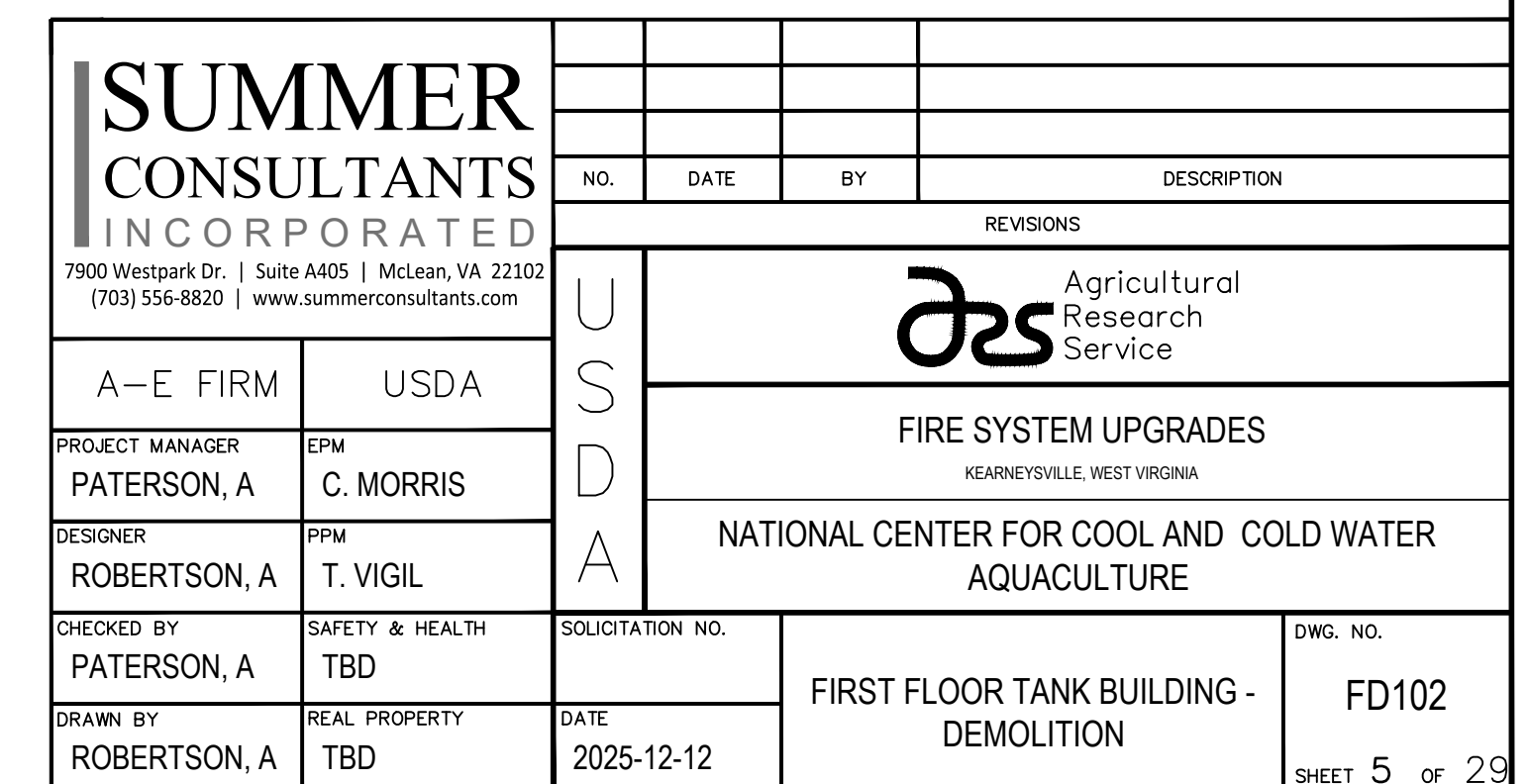
ISSUED FOR CONSTRUCTION - 9 JANUARY 2026



## MATCHLINE



1. DISCONNECT AND REMOVE THE EXISTING FIRE ALARM CONTROL PANEL AND ASSOCIATED POWER, CONDUIT, MOUNTS, ETC.



ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

FILE NAME: 212.012 FD102.DWG  
OUR REF: 212.012  
PLOT DATE: 12/12/2025 8:46:18 AM

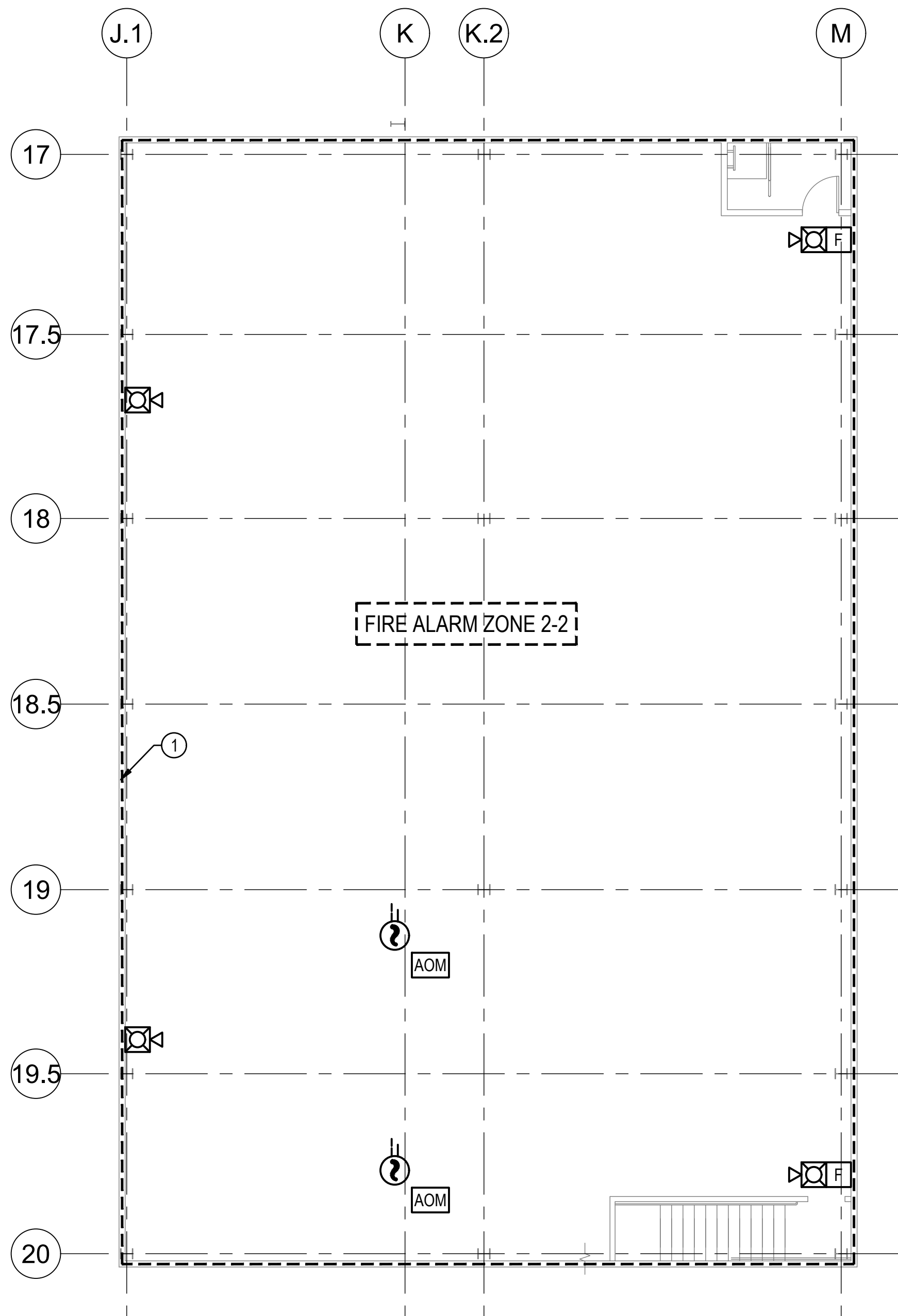


GENERAL NOTES

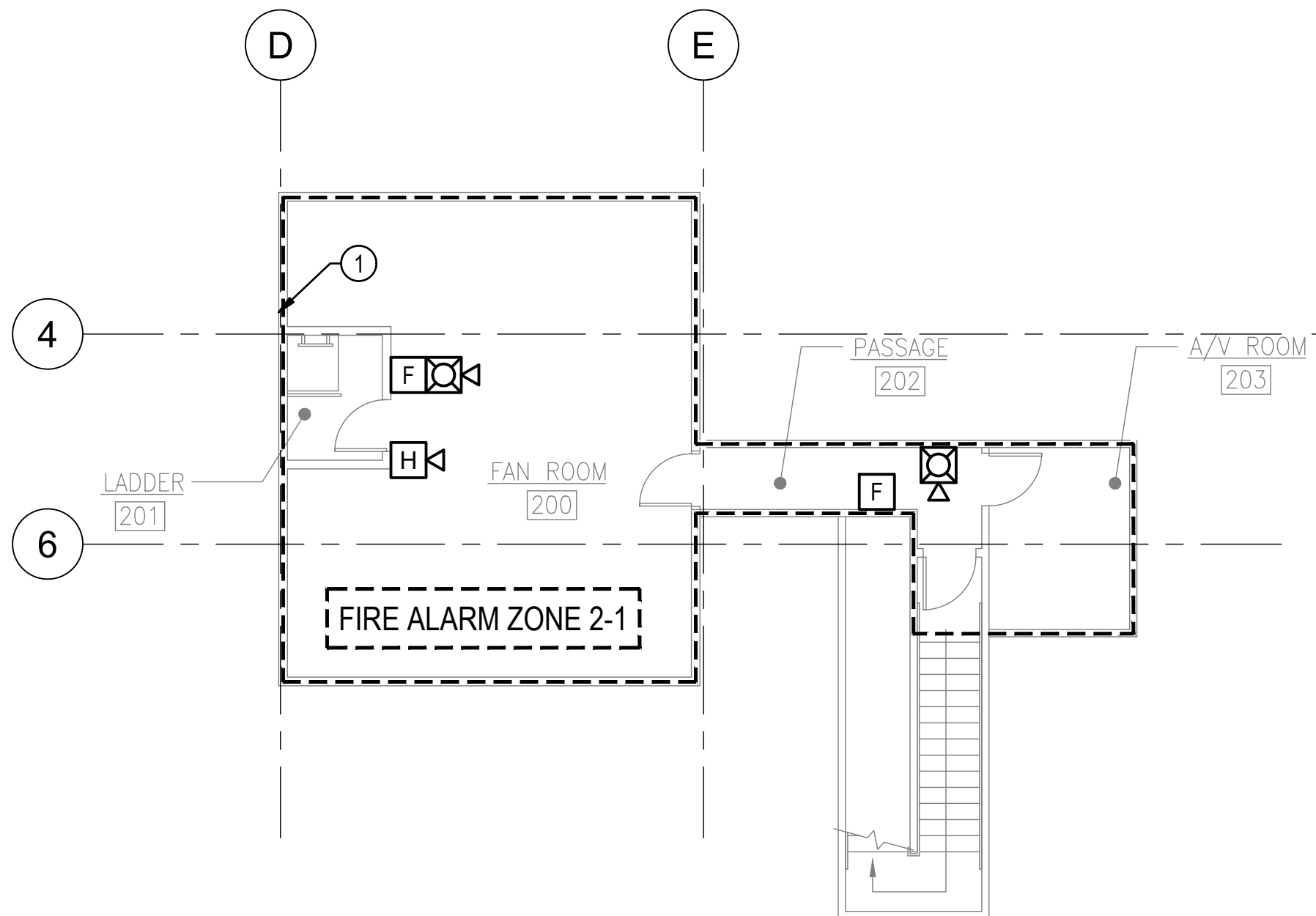
G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

NOTES

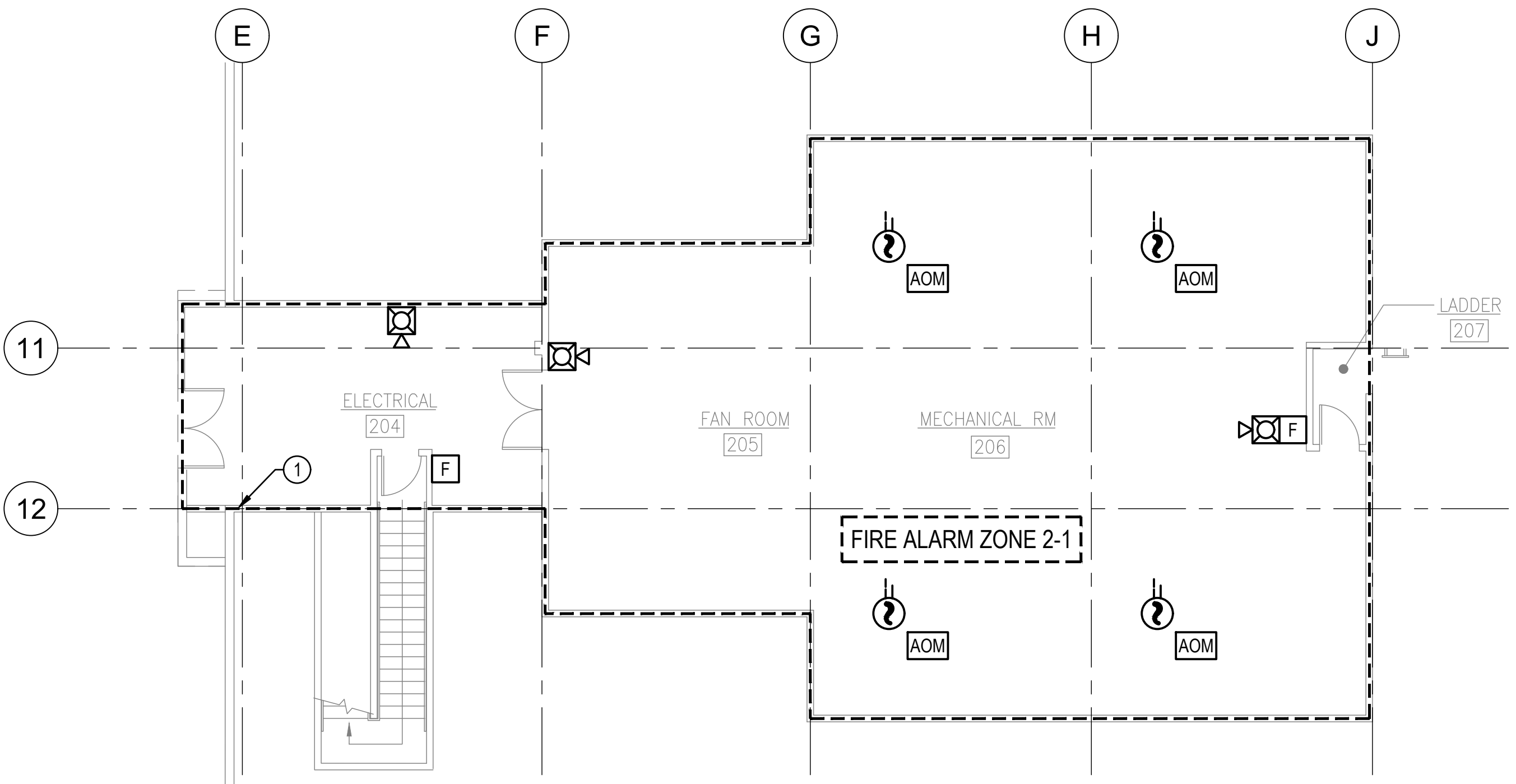
1. DISCONNECT AND REMOVE THE EXISTING FIRE ALARM CONTROL PANEL AND ASSOCIATED POWER, CONDUIT, MOUNTS, ETC.



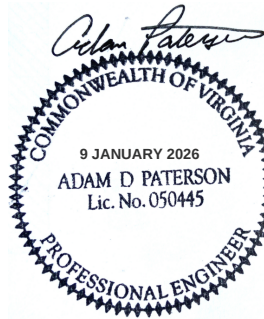
1 SECOND FLOOR TANK PENTHOUSE - DEMOLITION



2 SECOND FLOOR NORTH PENTHOUSE - DEMOLITION



3 SECOND FLOOR SOUTH PENTHOUSE - DEMOLITION



|                                                                                                                                                                |  |                        |  |                    |  |                                                                                                                                       |  |
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|                                                                                                                                                                |  |                        |  |                    |  |                                                                                                                                       |  |
|                                                                                                                                                                |  |                        |  |                    |  |                                                                                                                                       |  |
| NO.                                                                                                                                                            |  | DATE                   |  | BY                 |  | DESCRIPTION                                                                                                                           |  |
|                                                                                                                                                                |  |                        |  |                    |  | REVISIONS                                                                                                                             |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |  | USDA               |  | <div>Agricultural<br/>Research<br/>Service</div> |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |  |                    |  | FIRE SYSTEM UPGRADES                                                                                                                  |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |  |                    |  | KEARNEYVILLE, WEST VIRGINIA                                                                                                           |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.   |  | DWS. NO.                                                                                                                              |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12 |  | SECOND FLOOR - DEMOLITION                                                                                                             |  |
|                                                                                                                                                                |  |                        |  |                    |  | FD103                                                                                                                                 |  |
|                                                                                                                                                                |  |                        |  |                    |  | SHEET 6 OF 29                                                                                                                         |  |

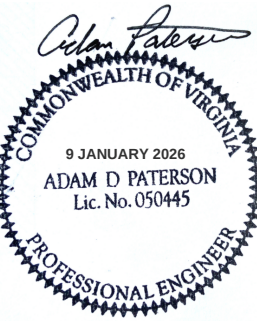


GENERAL NOTES

G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

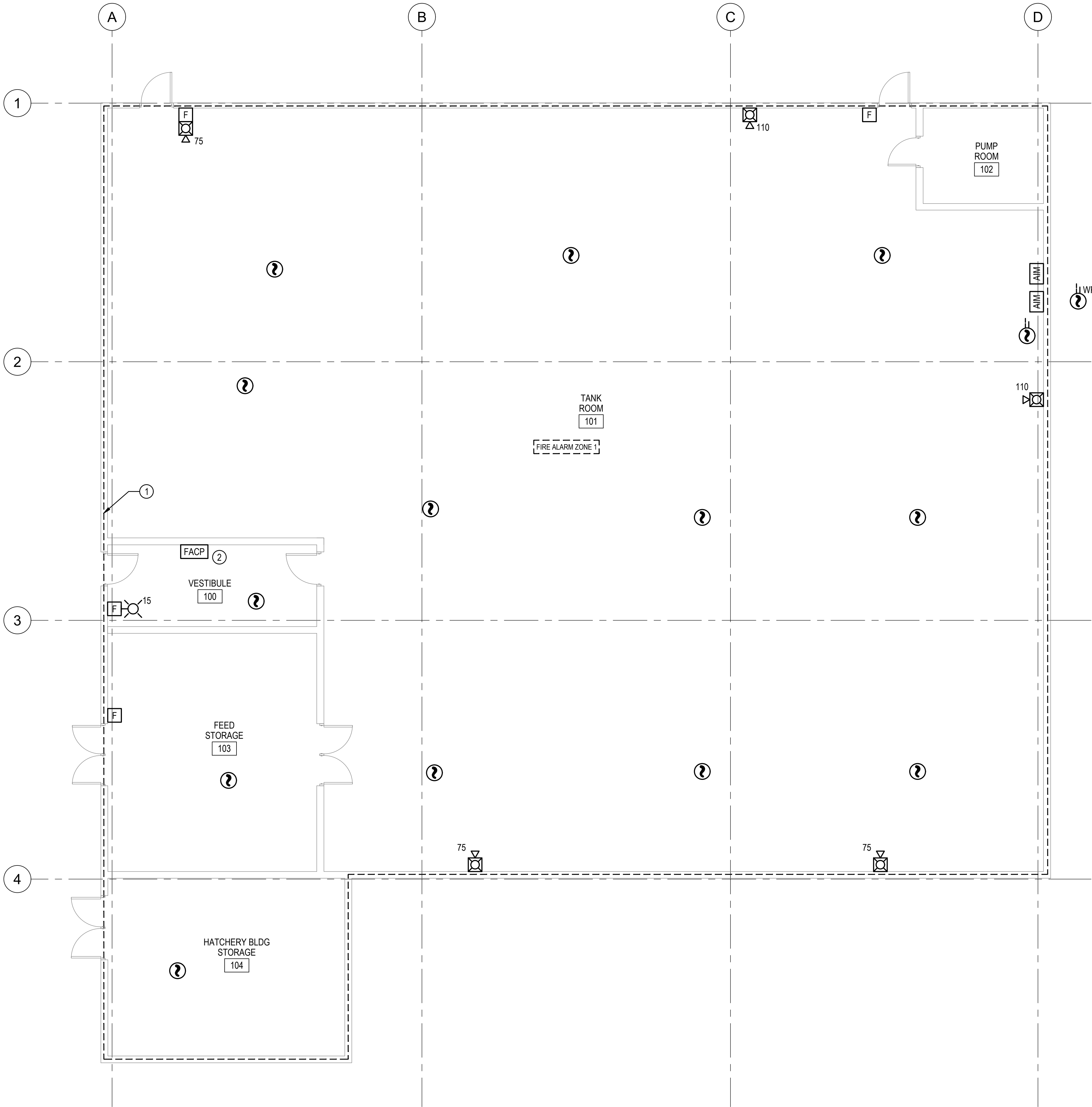
NOTES

1. DISCONNECT AND REMOVE THE EXISTING FIRE ALARM CONTROL PANEL AND ASSOCIATED POWER, CONDUIT, MOUNTS, ETC.
2. RETAIN POWER AND TELEPHONE FOR FIRE ALARM CONTROL PANEL INSTALLATION.



|                                                                                                                                                                |                        |                    |                                                                                                                                                                                                                                                                              |                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <div><div>SUMMERCONSULTANTSINCORPORATED</div><div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div></div> | U                      |                    | <div><div>Agricultural Research Service</div><div>FIRE SYSTEM UPGRADES</div><div>KEARNEYSVILLE, WEST VIRGINIA</div><div>NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE</div></div> |                                  |
|                                                                                                                                                                | S                      |                    |                                                                                                                                                                                                                                                                              |                                  |
|                                                                                                                                                                | D                      |                    |                                                                                                                                                                                                                                                                              |                                  |
|                                                                                                                                                                | A                      |                    |                                                                                                                                                                                                                                                                              |                                  |
| A-E FIRM                                                                                                                                                       | USDA                   |                    |                                                                                                                                                                                                                                                                              |                                  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 | EPM<br>C. MORRIS       | SOLICITATION NO.   |                                                                                                                                                                                                                                                                              | BROODSTOCK BUILDING - DEMOLITION |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       | PPM<br>T. VIGIL        | DATE<br>2025-12-12 |                                                                                                                                                                                                                                                                              |                                  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      | SAFETY & HEALTH<br>TBD | DWG. NO.<br>FD104  |                                                                                                                                                                                                                                                                              |                                  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       | REAL PROPERTY<br>TBD   | SHEET 7 OF 29      |                                                                                                                                                                                                                                                                              |                                  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026



1 BROODSTOCK BUILDING - DEMOLITION  
3/16"=1'-0"  
0 4' 8' 12'

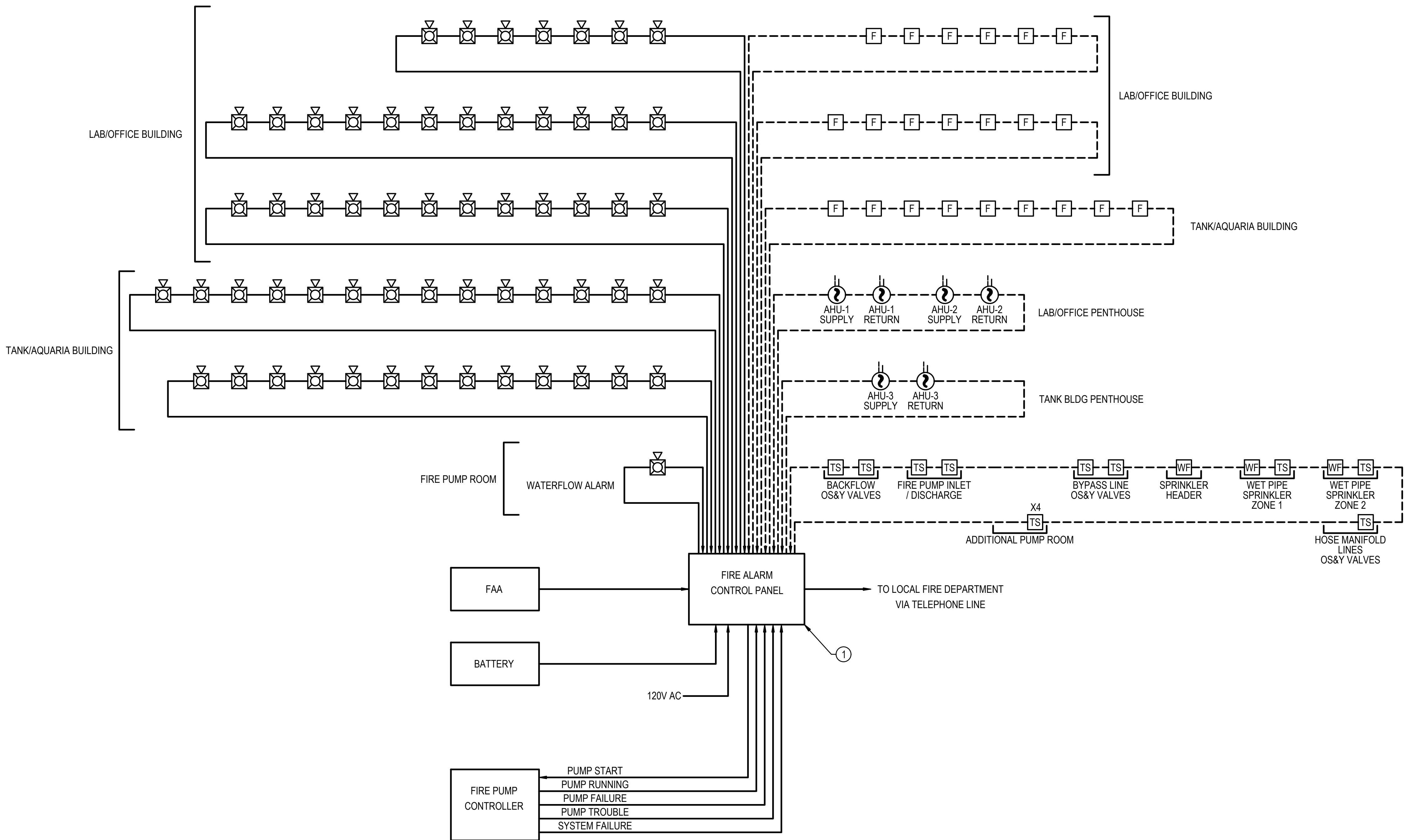


GENERAL NOTES

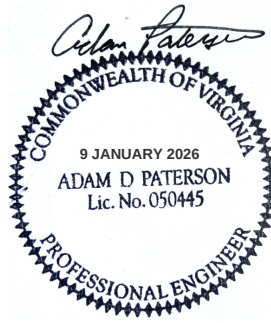
- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. EQUIPMENT SHOWN IS SCHEMATIC.

NOTES

1. DEMOLISH EXISTING FIRE ALARM SYSTEM.



1 FIRE ALARM RISER DIAGRAM - DEMOLITION  
NOT TO SCALE



|                                                                                                                                                                                       |                                |            |                        |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------|------------------------|---------------------------|
| <div><div>SUMMER</div><div>CONSULTANTS</div><div>INCORPORATED</div></div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div> |                                |            |                        |                           |
|                                                                                                                                                                                       |                                |            |                        |                           |
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| NO.                                                                                                                                                                                   |                                | DATE       | BY                     | DESCRIPTION               |
| REVISIONS                                                                                                                                                                             |                                |            |                        |                           |
| U<br>S<br>D<br>A                                                                                                                                                                      | A-E FIRM                       |            | USDA                   |                           |
|                                                                                                                                                                                       | PROJECT MANAGER<br>PATERSON, A |            | EPM<br>C. MORRIS       |                           |
|                                                                                                                                                                                       | DESIGNER<br>ROBERTSON, A       |            | PPM<br>T. VIGIL        |                           |
|                                                                                                                                                                                       | CHECKED BY<br>PATERSON, A      |            | SAFETY & HEALTH<br>TBD |                           |
|                                                                                                                                                                                       | DRAWN BY<br>ROBERTSON, A       |            | REAL PROPERTY<br>TBD   |                           |
| SOLICITATION NO.                                                                                                                                                                      |                                | DATE       |                        | RISER DETAIL - DEMOLITION |
|                                                                                                                                                                                       |                                | 2025-12-12 |                        |                           |
|                                                                                                                                                                                       |                                |            |                        |                           |
|                                                                                                                                                                                       |                                |            |                        | DWG. NO.<br>FD601         |
|                                                                                                                                                                                       |                                |            |                        | SHEET 8 OF 29             |

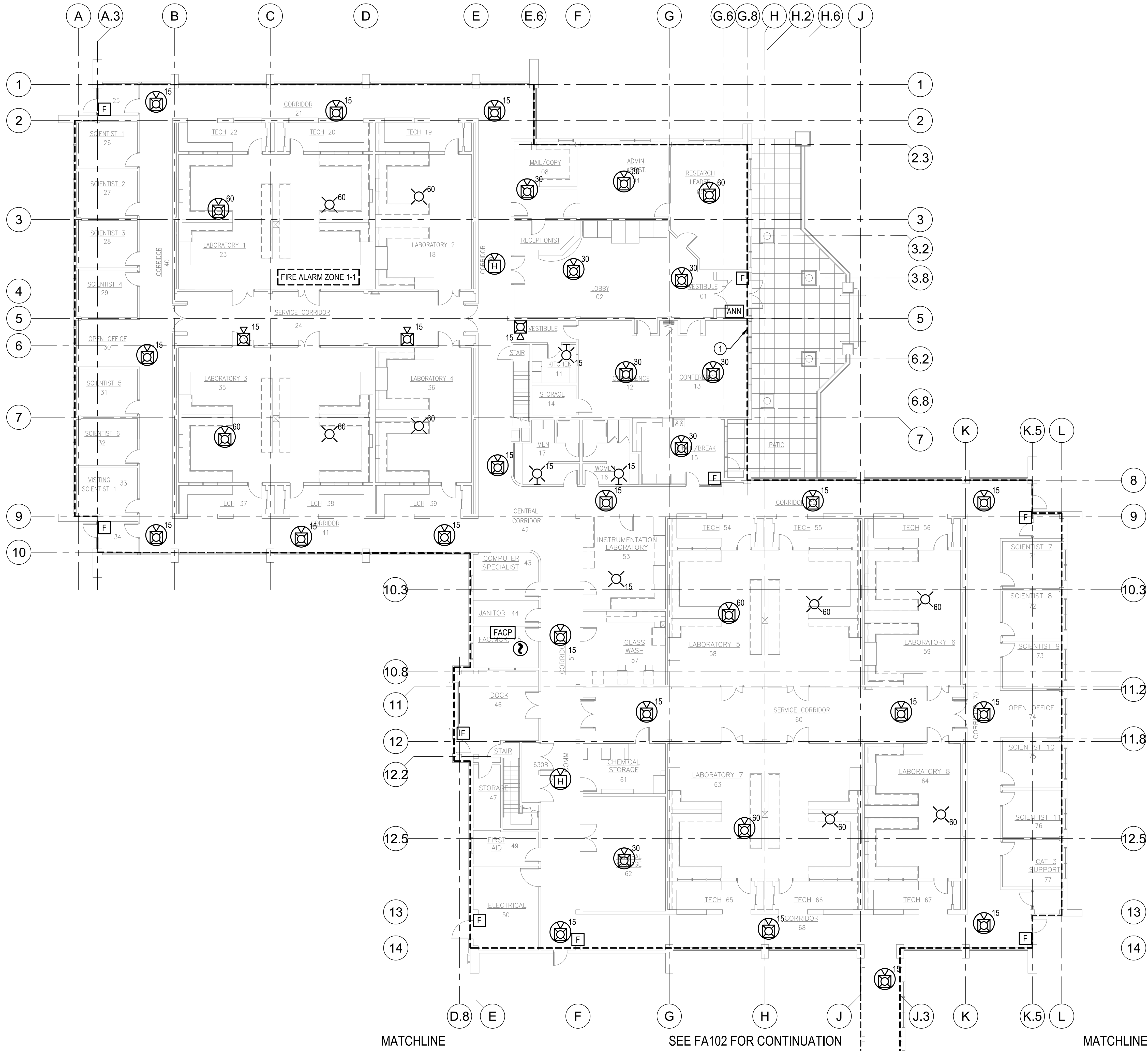


GENERAL NOTES

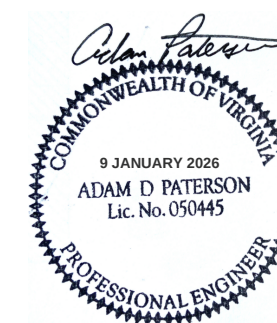
- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "E".

NOTES

1. PROVIDE NFPA 72 AND ARS 242.1 MANUAL COMPLIANT FIRE ALARM SYSTEM THROUGHOUT THE MAIN BUILDING.



1 FIRST FLOOR MAIN BUILDING  
3/32"=1'-0"  
0 8 16 24'



|                                                                                                                                                                                     |  |                        |  |                                                                                                                                                       |  |                                    |  |
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|                                                                                                                                                                                     |  |                        |  |                                                                                                                                                       |  |                                    |  |
|                                                                                                                                                                                     |  | NO.                    |  | DATE                                                                                                                                                  |  | BY                                 |  |
|                                                                                                                                                                                     |  | REVISIONS              |  |                                                                                                                                                       |  |                                    |  |
| A-E FIRM                                                                                                                                                                            |  | USDA                   |  | <div>USDA</div> <div><div></div><div>Agricultural Research Service</div></div> <div>FIRE SYSTEM UPGRADES</div> <div>KEARNEYVILLE, WEST VIRGINIA</div> |  |                                    |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                                      |  | EPM<br>C. MORRIS       |  |                                                                                                                                                       |  |                                    |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                                            |  | PPM<br>T. VIGIL        |  | NATIONAL CENTER FOR COOL AND COLD WATER<br>AQUACULTURE                                                                                                |  |                                    |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                                           |  | SAFETY & HEALTH<br>TBD |  |                                                                                                                                                       |  |                                    |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                                            |  | REAL PROPERTY<br>TBD   |  | SOLICITATION NO.                                                                                                                                      |  | DWG. NO.                           |  |
|                                                                                                                                                                                     |  |                        |  | DATE<br>2025-12-12                                                                                                                                    |  | FIRST FLOOR MAIN BUILDING<br>FA101 |  |
|                                                                                                                                                                                     |  |                        |  |                                                                                                                                                       |  | SHEET 9 of 29                      |  |

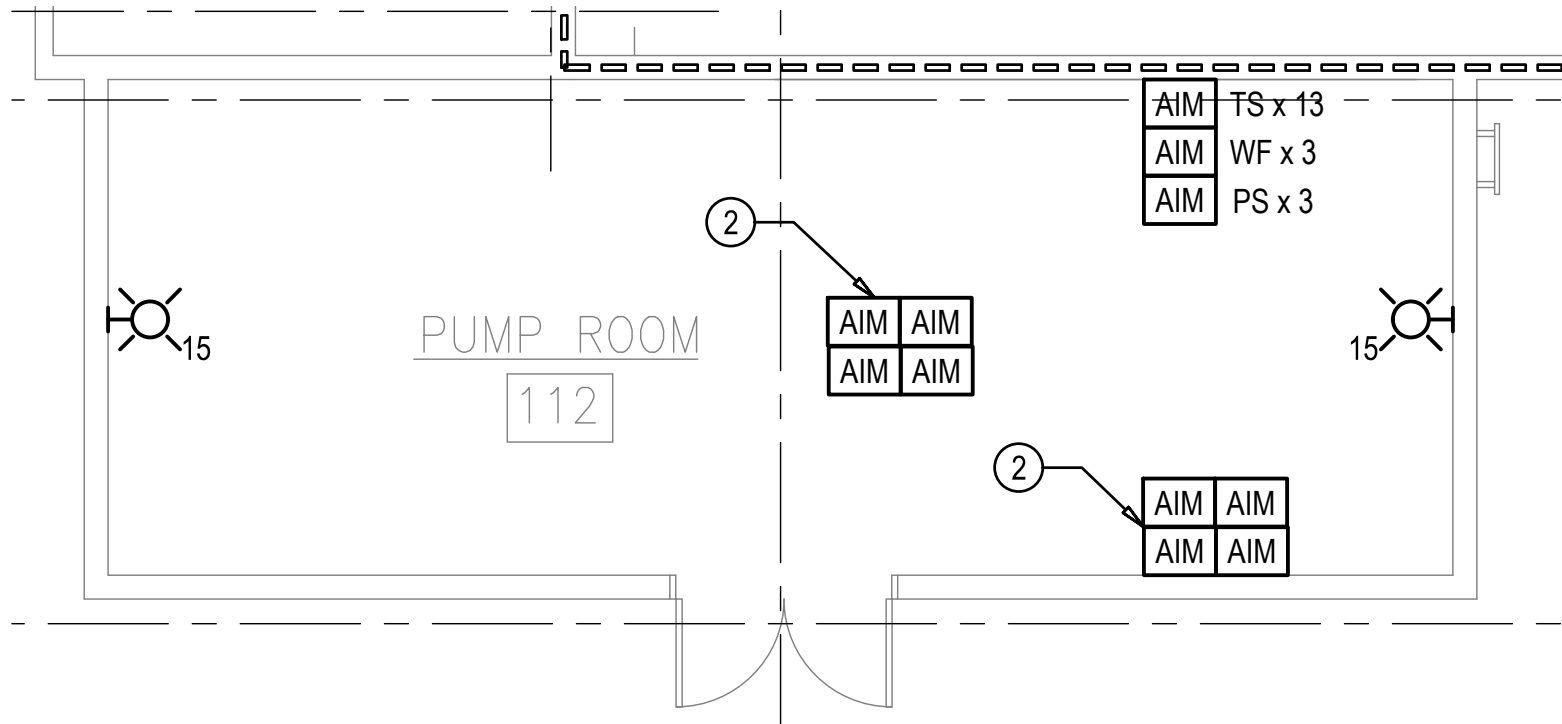


GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "E".

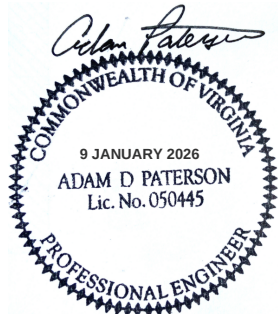
NOTES

1. PROVIDE FIRE ALARM SYSTEM IN ACCORDANCE WITH NFPA 72.
2. PROVIDE FOUR ADDRESSABLE INPUT MODULES FOR FIRE PUMP RUNNING, CONTROLLER OR SYSTEM TROUBLE, LOSS OF PHASE, AND PHASE REVERSAL.



2 ENLARGED PUMP ROOM

FD102 3/16"=1'-0" 0 4' 8' 12'



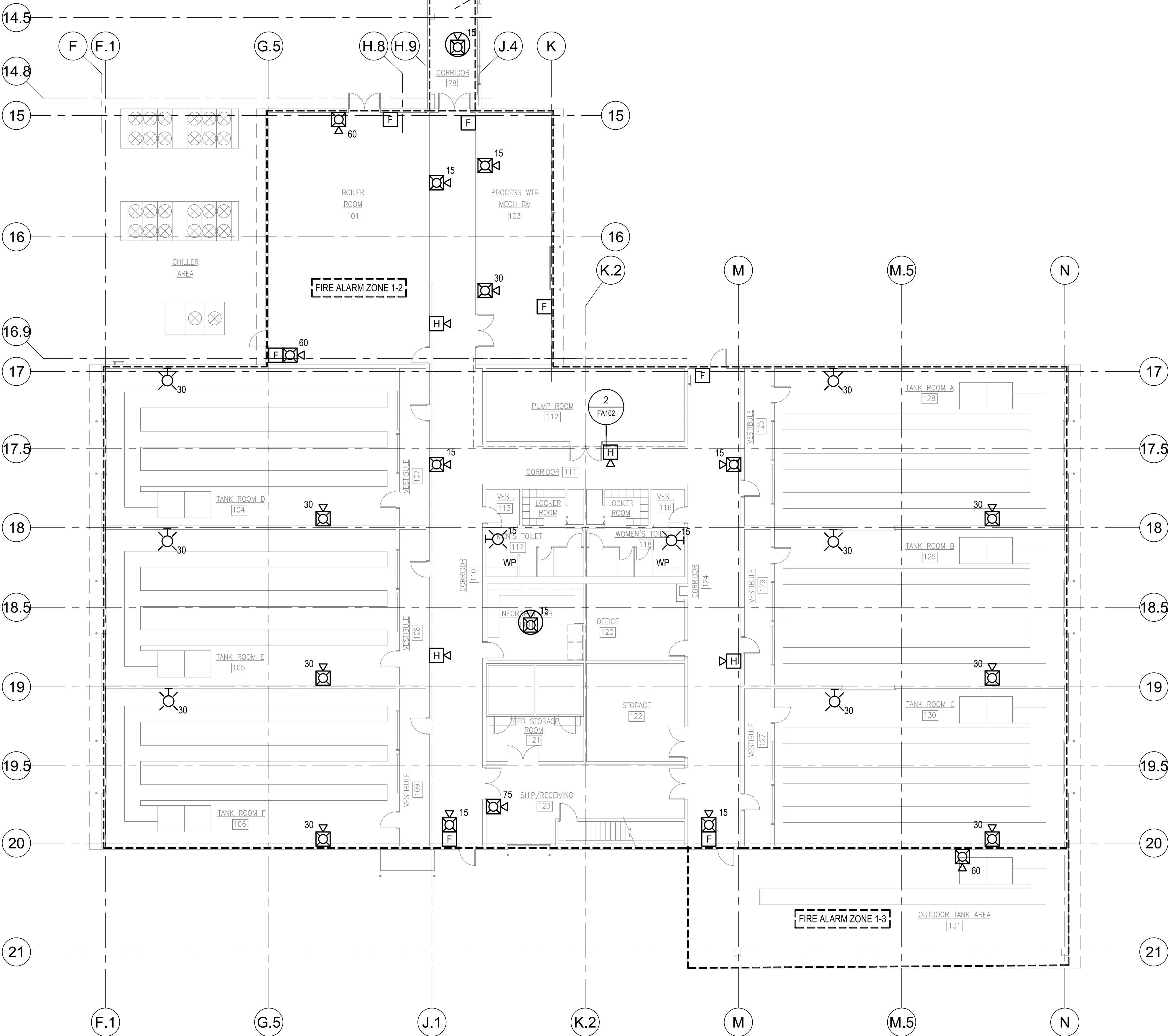
|                                                                                                                                                                |  |                        |  |                                                                                                                                                                                                                                                                                                 |  |                                    |  |
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| <div>SUMMER<br/>CONSULTANTS<br/>INCORPORATED</div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div> |  |                        |  |                                                                                                                                                                                                                                                                                                 |  |                                    |  |
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|                                                                                                                                                                |  | NO.                    |  | DATE                                                                                                                                                                                                                                                                                            |  | BY                                 |  |
|                                                                                                                                                                |  | REVISIONS              |  |                                                                                                                                                                                                                                                                                                 |  |                                    |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |  | <div>USDA</div> <div>Agricultural<br/>Research<br/>Service</div> <div>FIRE SYSTEM UPGRADES</div> <div>KEARNEYVILLE, WEST VIRGINIA</div> <div>NATIONAL CENTER FOR COOL AND COLD WATER<br/>AQUACULTURE</div> |  |                                    |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |  |                                                                                                                                                                                                                                                                                                 |  |                                    |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |  |                                                                                                                                                                                                                                                                                                 |  |                                    |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.                                                                                                                                                                                                                                                                                |  | DWG. NO.                           |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12                                                                                                                                                                                                                                                                              |  | FIRST FLOOR TANK BUILDING<br>FA102 |  |
|                                                                                                                                                                |  |                        |  |                                                                                                                                                                                                                                                                                                 |  | SHEET 10 OF 29                     |  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

MATCHLINE

SEE FA101 FOR CONTINUATION

MATCHLINE



1 FIRST FLOOR TANK BUILDING

3/32"=1'-0" 0 8' 16' 24'

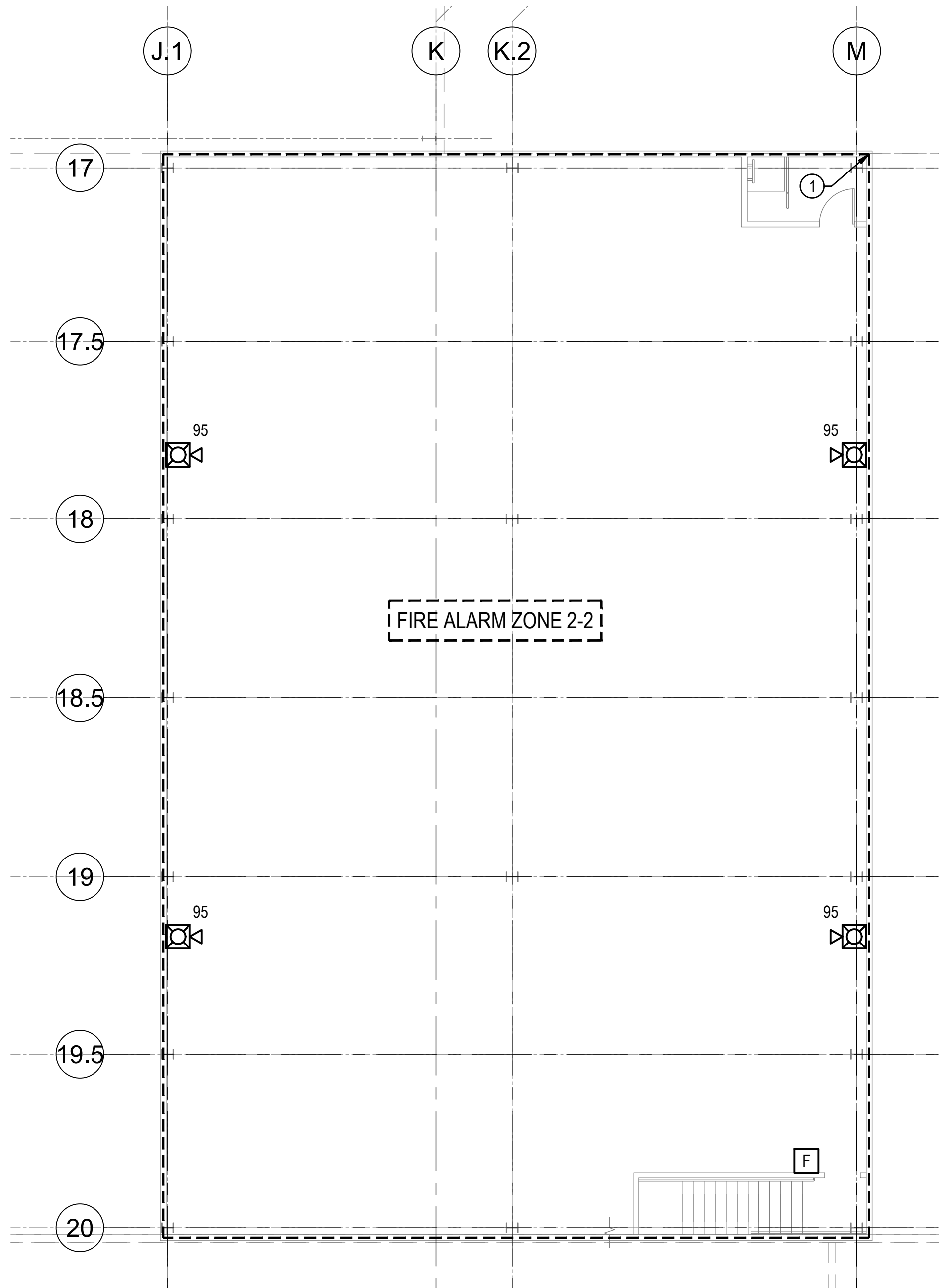


GENERAL NOTES

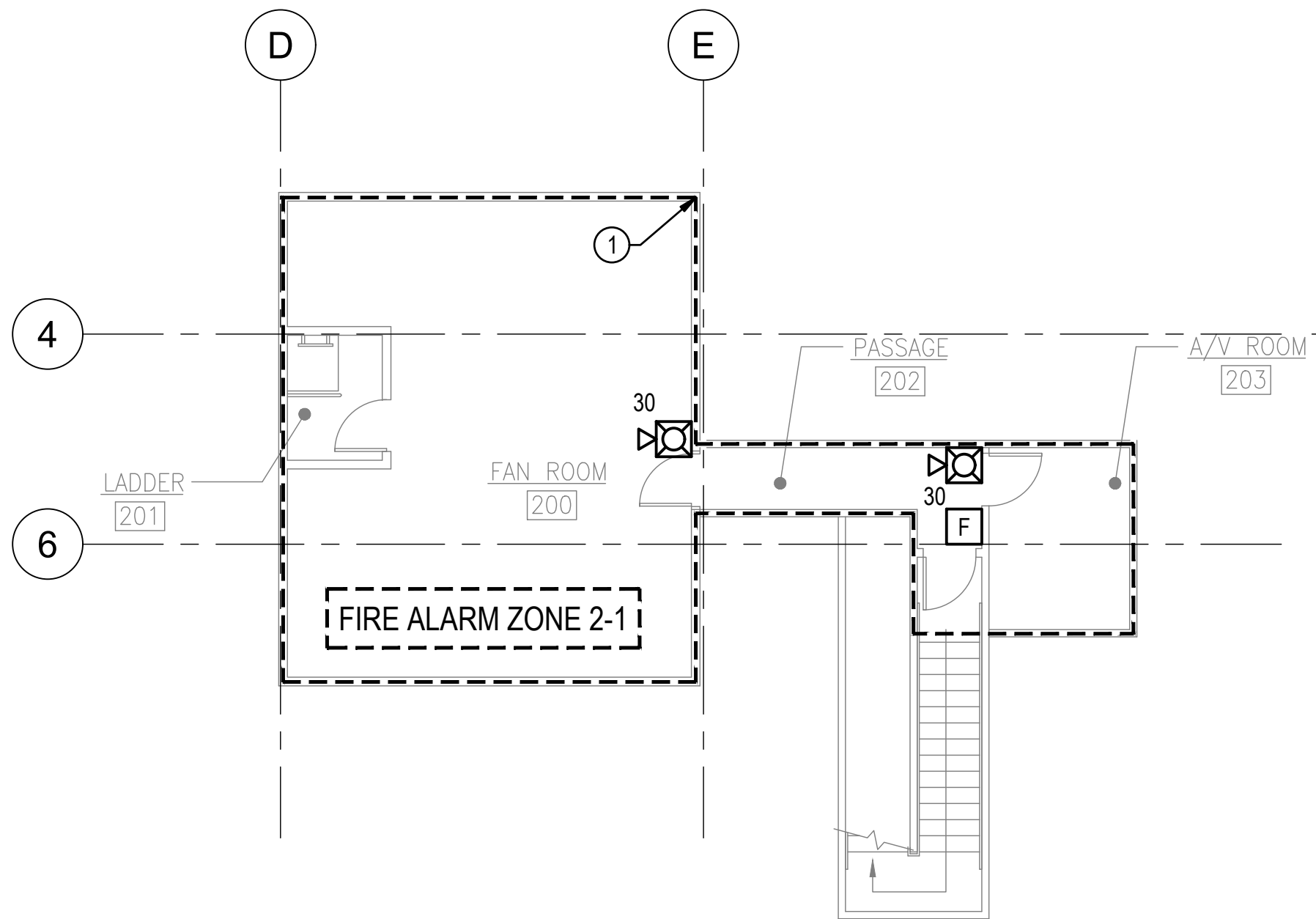
- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".

NOTES

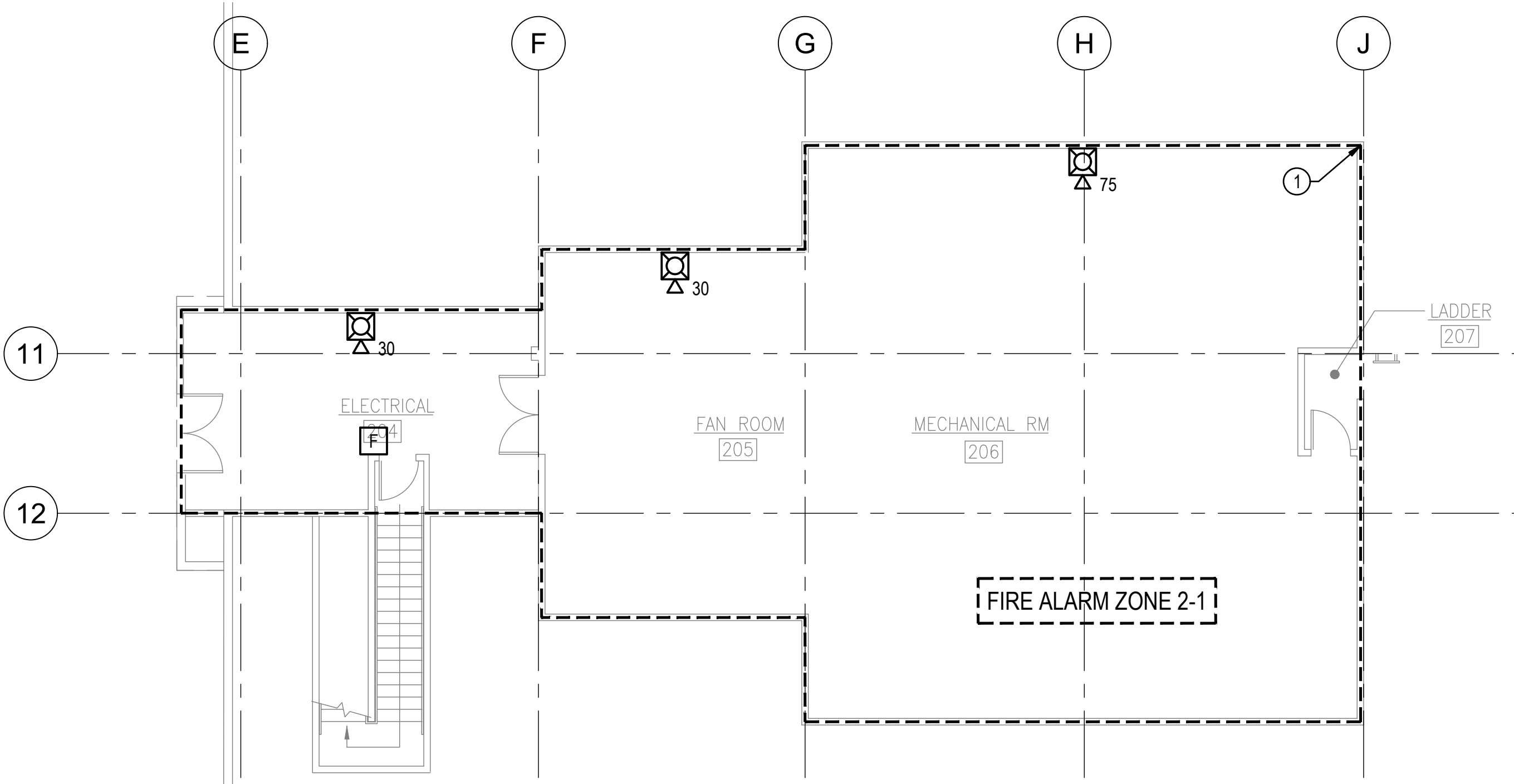
1. PROVIDE FIRE ALARM SYSTEM IN ACCORDANCE WITH NFPA 72.



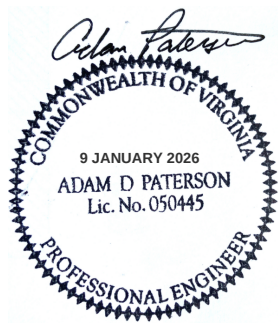
1 SECOND FLOOR TANK PENTHOUSE



2 SECOND FLOOR NORTH PENTHOUSE



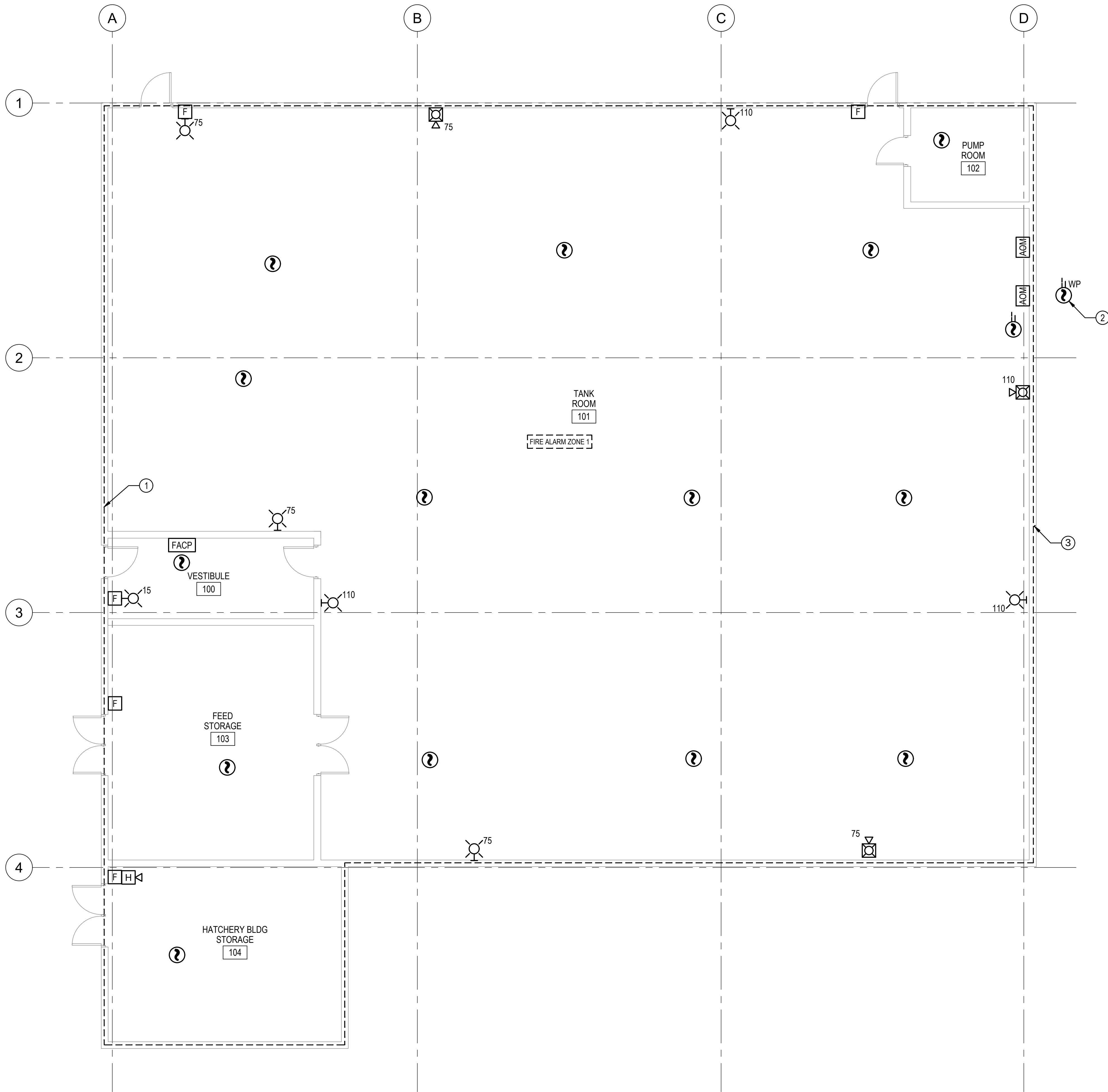
3 SECOND FLOOR SOUTH PENTHOUSE



|                                                                                                                                                                |  |                        |  |                    |  |                                                                                                                           |  |                                                                                                                       |  |
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|                                                                                                                                                                |  |                        |  |                    |  |                                                                                                                           |  |                                                                                                                       |  |
|                                                                                                                                                                |  | NO.                    |  | DATE               |  | BY                                                                                                                        |  | DESCRIPTION                                                                                                           |  |
|                                                                                                                                                                |  | REVISIONS              |  |                    |  |                                                                                                                           |  |                                                                                                                       |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |  | U<br>S<br>D<br>A   |  |  Agricultural<br>Research<br>Service |  | FIRE SYSTEM UPGRADES<br><br>KEARNEYVILLE, WEST VIRGINIA<br><br>NATIONAL CENTER FOR COOL AND COLD WATER<br>AQUACULTURE |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |  |                    |  |                                                                                                                           |  |                                                                                                                       |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |  |                    |  |                                                                                                                           |  |                                                                                                                       |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.   |  | SECOND FLOOR                                                                                                              |  | DWG. NO.<br>FA103                                                                                                     |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12 |  |                                                                                                                           |  | SHEET 11 OF 29                                                                                                        |  |



FILE NAME: 212.012 FA104.DWG  
OUR REF: 212.012  
PLOT DATE: 12/12/2025 8:47:05 AM



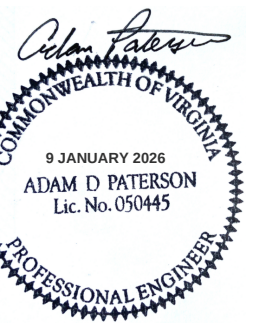
1 BROODSTOCK BUILDING  
3/16"=1'-0"  
0 4' 8' 12'

## GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".

## NOTES

1. PROVIDE FIRE ALARM AND DETECTION SYSTEM.
2. PROVIDE DUCT DETECTION FOR EXTERIOR AIR HANDLING UNIT.
3. PROVIDE REMOTE TEST SWITCHES FOR CEILING MOUNTED DETECTORS.



|                                                                                                                                                                |  |                        |  |                                                                                                                                       |  |                              |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------|--|
| <div>SUMMER<br/>CONSULTANTS<br/>INCORPORATED</div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div> |  |                        |  |                                                                                                                                       |  |                              |  |
|                                                                                                                                                                |  |                        |  |                                                                                                                                       |  |                              |  |
|                                                                                                                                                                |  |                        |  |                                                                                                                                       |  |                              |  |
|                                                                                                                                                                |  | NO.                    |  | DATE                                                                                                                                  |  | BY                           |  |
|                                                                                                                                                                |  | REVISIONS              |  |                                                                                                                                       |  |                              |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |  | <div>Agricultural<br/>Research<br/>Service</div> |  |                              |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |  | FIRE SYSTEM UPGRADES                                                                                                                  |  |                              |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |  | KEARNEYVILLE, WEST VIRGINIA                                                                                                           |  |                              |  |
|                                                                                                                                                                |  |                        |  | NATIONAL CENTER FOR COOL AND COLD WATER<br>AQUACULTURE                                                                                |  |                              |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.                                                                                                                      |  | DWG. NO.                     |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12                                                                                                                    |  | BROODSTOCK BUILDING<br>FA104 |  |
|                                                                                                                                                                |  |                        |  |                                                                                                                                       |  | SHEET 12 OF 29               |  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

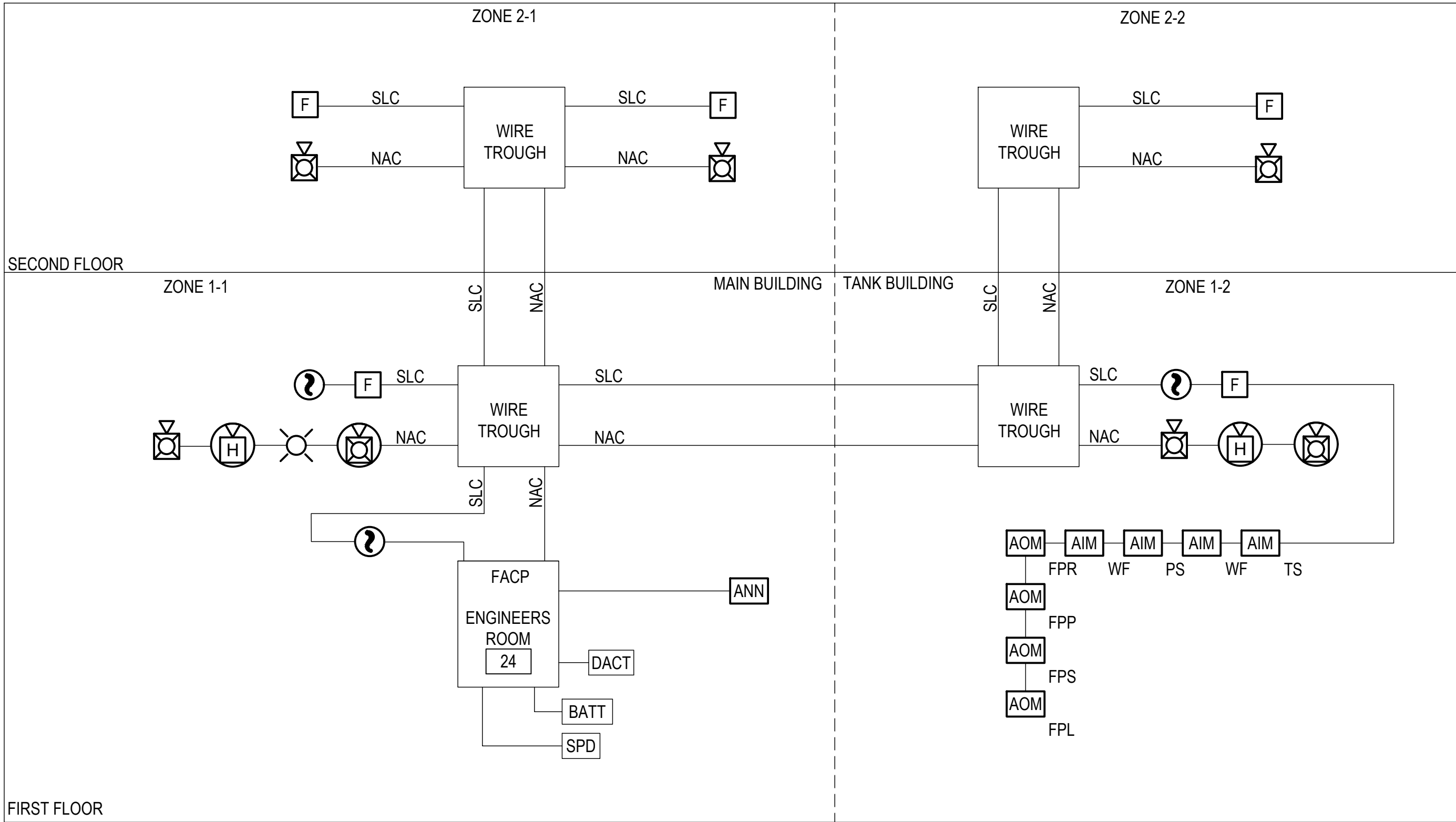


GENERAL NOTES

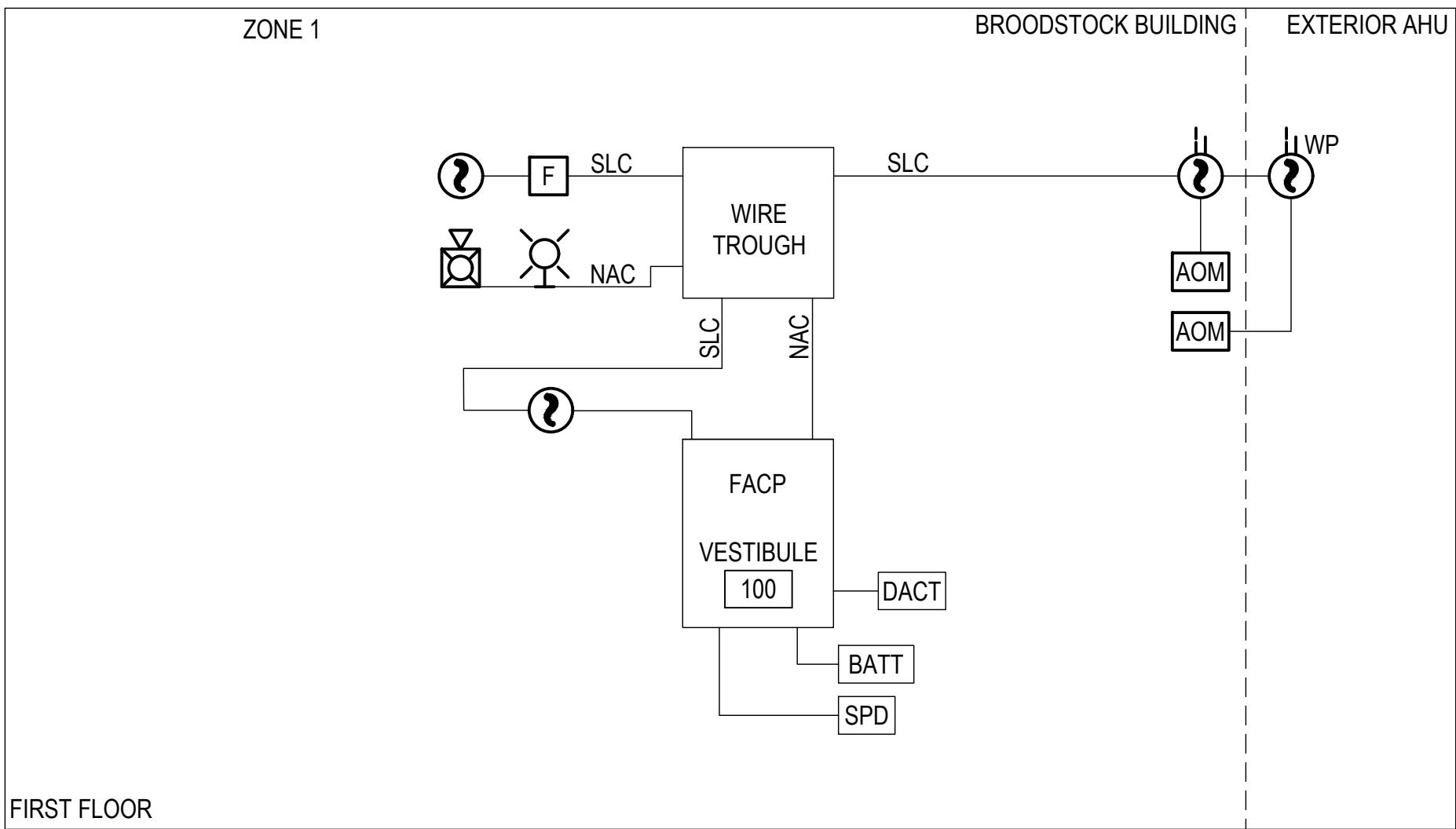
- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".

NOTES

1. EQUIPMENT SHOWN IS SCHEMATIC.



1 FIRE ALARM RISER DIAGRAM  
NOT TO SCALE



3 FIRE ALARM RISER DIAGRAM - BROODSTOCK  
NOT TO SCALE

|                                             | ALARM ANNUNCIATION AT FACP/FAAP AND SEND STATUS TO FIRE STATION | SUPERVISORY ANNUNCIATION AT FACP AND SEND STATUS TO FIRE STATION | TROUBLE ANNUNCIATION AT FACP AND SEND STATUS TO FIRE STATION | ACTIVATE VISUAL AND AUDIBLE NOTIFICATION - GENERAL EVACUATION | SHUT DOWN ASSOCIATED HVAC UNIT |
|---------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|--------------------------------|
| MANUAL PULL STATION                         | ●                                                               |                                                                  |                                                              | ●                                                             |                                |
| SMOKE DETECTOR                              | ●                                                               |                                                                  |                                                              | ●                                                             |                                |
| SPRINKLER SYSTEM - WATER FLOW SWITCH        | ●                                                               |                                                                  |                                                              | ●                                                             |                                |
| SPRINKLER SYSTEM - ALARM PRESSURE SWITCH    | ●                                                               |                                                                  |                                                              | ●                                                             |                                |
| SPRINKLER SYSTEM - HIGH/LOW PRESSURE SWITCH |                                                                 | ●                                                                |                                                              |                                                               |                                |
| SPRINKLER SYSTEM - TAMPER SWITCH            |                                                                 | ●                                                                |                                                              |                                                               |                                |
| DUCT SMOKE DETECTOR                         |                                                                 | ●                                                                |                                                              |                                                               | ●                              |
| FIRE PUMP RUNNING                           |                                                                 | ●                                                                |                                                              |                                                               |                                |
| FIRE PUMP PHASE LOSS                        |                                                                 | ●                                                                |                                                              |                                                               |                                |
| FIRE PUMP POWER LOSS                        |                                                                 | ●                                                                |                                                              |                                                               |                                |
| FIRE PUMP SECONDARY POWER                   |                                                                 | ●                                                                |                                                              |                                                               |                                |
| FACP POWER FAILURE                          |                                                                 |                                                                  | ●                                                            |                                                               |                                |
| LOW BATTERY                                 |                                                                 |                                                                  | ●                                                            |                                                               |                                |
| OPEN CIRCUIT                                |                                                                 |                                                                  | ●                                                            |                                                               |                                |
| GROUND FAULT                                |                                                                 |                                                                  | ●                                                            |                                                               |                                |

2 FIRE ALARM MATRIX  
NO SCALE



|                                                                                                                                                                                       |  |                        |  |                                                                                                                                                                                                                                                                                                                   |  |                          |  |             |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------|--|-------------|--|
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|                                                                                                                                                                                       |  |                        |  |                                                                                                                                                                                                                                                                                                                   |  |                          |  |             |  |
|                                                                                                                                                                                       |  | NO.                    |  | DATE                                                                                                                                                                                                                                                                                                              |  | BY                       |  | DESCRIPTION |  |
|                                                                                                                                                                                       |  | REVISIONS              |  |                                                                                                                                                                                                                                                                                                                   |  |                          |  |             |  |
| A-E FIRM                                                                                                                                                                              |  | USDA                   |  | <div><div>USDA</div><div><div>Agricultural<br/>Research<br/>Service</div><div>FIRE SYSTEM UPGRADES</div><div>KEARNEYSVILLE WEST VIRGINIA</div><div>NATIONAL CENTER FOR COOL AND COLD WATER<br/>AQUACULTURE</div></div></div> |  |                          |  |             |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                                        |  | EPM<br>C. MORRIS       |  |                                                                                                                                                                                                                                                                                                                   |  |                          |  |             |  |
| DESIGNER<br>PIRTLE, R                                                                                                                                                                 |  | PPM<br>T. VIGIL        |  |                                                                                                                                                                                                                                                                                                                   |  |                          |  |             |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                                             |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.                                                                                                                                                                                                                                                                                                  |  |                          |  |             |  |
| DRAWN BY<br>PIRTLE, R                                                                                                                                                                 |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12                                                                                                                                                                                                                                                                                                |  | RISER DIAGRAM AND MATRIX |  |             |  |
|                                                                                                                                                                                       |  |                        |  |                                                                                                                                                                                                                                                                                                                   |  | DWS. NO.<br>FA601        |  |             |  |
|                                                                                                                                                                                       |  |                        |  |                                                                                                                                                                                                                                                                                                                   |  | SHEET 13 OF 29           |  |             |  |



ABBREVIATIONS

|        |                                 |
|--------|---------------------------------|
| A      | AMPERES.                        |
| AC     | ABOVE CEILING                   |
| AFF    | ABOVE FINISHED FLOOR            |
| AHJ    | AUTHORITY HAVING JURISDICTION   |
| ATC    | ACOUSTIC TILE CEILING           |
| AWG    | AMERICAN WIRE GAUGE             |
| CA     | CLEAN AGENT                     |
| CL     | CEILING LEVEL                   |
| CONT.  | CONTINUED                       |
| D      | DRAIN                           |
| DN     | DOWN                            |
| ELEV.  | ELEVATOR                        |
| EMT    | ELECTRICAL METALLIC TUBING      |
| (E)    | EXISTING TO REMAIN              |
| F      | FIRE LINE                       |
| FDC    | FIRE DEPARTMENT CONNECTION      |
| FH     | FIRE HYDRANT                    |
| FHV    | 2-1/2" FIRE HOSE VALVE          |
| N.I.C. | NOT IN CONTRACT                 |
| PA     | PRE-ACTION                      |
| (R)    | RELOCATE EXISTING EQUIPMENT     |
| SP     | SPRINKLER PIPING                |
| TYP.   | TYPICAL                         |
| ZCA    | SPRINKLER ZONE CONTROL ASSEMBLY |

FIRE PUMP TEST RESULTS

TEST DATE: 25 NOVEMBER 2024

RUN TIME: 5 MINUTES AT 100%

EQUIPMENT USED: TWO (2) 50 FT 2-1/2-INCH HOSES WITH DIFFUSER

FIRE PUMP GAUGES ARE BROKEN AND UNABLE TO REGISTER PRESSURE ABOVE 50 PSI. TEST CONFIRMED FIRE PUMP IS ABLE TO RUN.

THE PREVIOUS FIRE PUMP TEST WAS RUN BY ARK SYSTEMS, INC. ON 12 FEBRUARY 2024, YIELDING THE FOLLOWING DISCHARGE PRESSURES AND FLOWS.

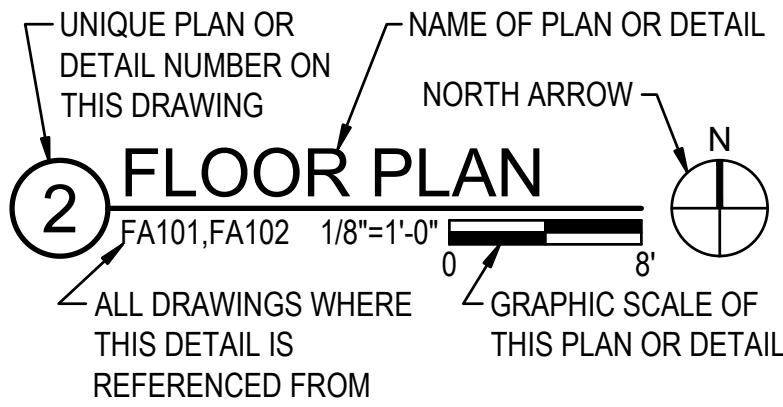
|       |         |         |
|-------|---------|---------|
| CHURN | 100 PSI | 0 GPM   |
| 100%  | 100 PSI | 500 GPM |
| 150%  | 70 PSI  | 750 GPM |

SYMBOLS

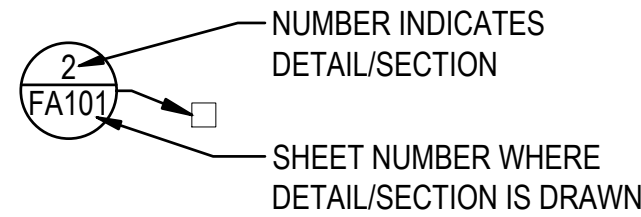
|  |                                                                               |                        |
|--|-------------------------------------------------------------------------------|------------------------|
|  | LIGHT HAZARD<br>DISCHARGE DENSITY<br>REMOTE AREA                              | 0.1 GPM/SF<br>1500 SF  |
|  | ORDINARY HAZARD GROUP 1<br>DISCHARGE DENSITY<br>REMOTE AREA                   | 0.15 GPM/SF<br>1500 SF |
|  | MODIFICATION OF THE SPRINKLER SYSTEM IN THIS AREA IS NOT IN CONTRACT (N.I.C.) |                        |
|  | AREA OF WORK                                                                  |                        |
|  | SPRINKLER ZONE                                                                |                        |
|  | WATERFLOW SWITCH                                                              |                        |
|  | TAMPER SWITCH                                                                 |                        |
|  | FIRE PUMP                                                                     |                        |
|  | ZONE CONTROL ASSEMBLY                                                         |                        |
|  | DOUBLE CHECK BACKFLOW PREVENTER                                               |                        |
|  | FIRE SUPPRESSION PIPE<br>GATE VALVE                                           |                        |
|  | GLOBE VALVE                                                                   |                        |
|  | SWING CHECK VALVE                                                             |                        |
|  | OS&Y VALVE                                                                    |                        |
|  | BALL VALVE                                                                    |                        |
|  | BUTTERFLY VALVE                                                               |                        |
|  | BACKFLOW PREVENTER                                                            |                        |
|  | PIPE TURN UP                                                                  |                        |
|  | PIPE TURN DOWN                                                                |                        |
|  | CONCENTRIC PIPE REDUCER                                                       |                        |
|  | ECCENTRIC PIPE REDUCER                                                        |                        |
|  | PIPE CAP                                                                      |                        |
|  | WET PIPE SPRINKLER RISER                                                      |                        |
|  | DRY PIPE SPRINKLER RISER                                                      |                        |
|  | EXISTING SPRINKLER                                                            |                        |
|  | UPRIGHT SPRINKLER                                                             |                        |
|  | PENDENT SPRINKLER                                                             |                        |
|  | SIDEWALL SPRINKLER                                                            |                        |
|  | ELECTRIC WATER FLOW ALARM BELL                                                |                        |

GENERAL SYMBOLS

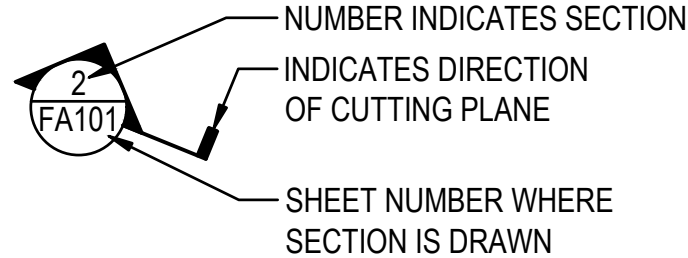
FLOOR PLAN AND DETAIL TITLES:



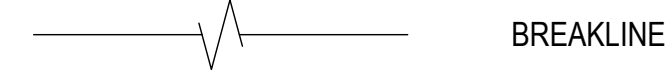
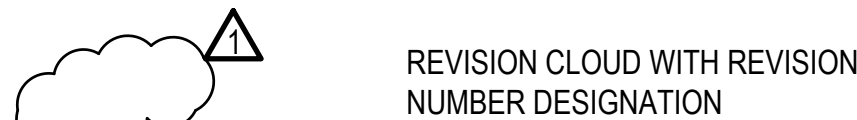
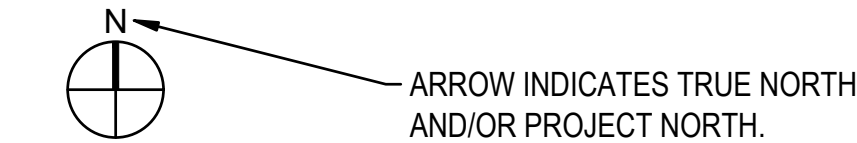
2-PART REFERENCE DETAIL IDENTIFICATION:



SECTION IDENTIFICATION



NORTH ARROW



MATCHLINE DESIGNATION

EXISTING

POINT OF CONNECTION TO EXISTING

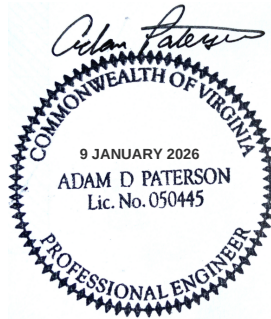
POINT OF TERMINATION OF REMOVAL

NOTE NUMBER

ROOM NUMBER

FIRE SUPPRESSION REQUIREMENTS

- MODIFY THE EXISTING WET-PIPE SPRINKLER SYSTEM DESIGNED AND INSTALLED PER ARS 242.1 2024 EDITION AND NFPA 13 2025 EDITION CRITERIA.
- UNLESS OTHERWISE NOTED BY HATCHING, PROVIDE SPRINKLER SYSTEM DESIGN IN ACCORDANCE WITH LIGHT HAZARD CRITERIA.
- PROVIDE QUICK RESPONSE CHROME SPRINKLERS IN AREAS WITH FINISHED CEILINGS. PROVIDE QUICK RESPONSE BRASS UPRIGHT SPRINKLERS IN AREAS WITHOUT FINISHED CEILINGS. PROVIDE A MINIMUM SPRINKLER TEMPERATURE OF 155 FAHRENHEIT OR HIGHER FOR ORDINARY TEMPERATURE SPRINKLERS.
- PROVIDE SPRINKLERS IN CENTER OF ACOUSTIC CEILING TILES.
- PROVIDE BLACK STEEL SPRINKLER PIPE. PIPE 2-INCH AND BELOW MUST BE SCHEDULE 40 THREADED PIPE. PIPE 2.5-INCHES AND GREATER MUST BE MINIMUM SCHEDULE 10 GROOVED JOINT PIPE.
- RETAIN OR PROVIDE IDENTIFICATION SIGNAGE FOR ALL VALVES LOCATED ABOVE SUSPENDED CEILINGS.
- ALL SPRINKLER SYSTEM OUTAGES SHALL BE COORDINATED THROUGH THE CONTRACTING OFFICER REPRESENTATIVE NOT LESS THAN 15 WORKING DAYS IN ADVANCE.
- A FIRE WATCH MUST BE CONDUCTED CONTINUOUSLY THROUGHOUT THE DURATION OF THE OUTAGE IN ALL AREAS AFFECTED BY THE OUTAGE. FINDINGS/COMMENTS WILL BE DOCUMENTED HOURLY ON FIRE WATCH FORMS. IN THE EVENT OF ANY FIRE/SMOKE EMERGENCIES DISCOVERED DURING THE FIRE WATCH, EMERGENCY NOTIFICATION WILL BE PROVIDED TO THE PREDETERMINED POINT OF CONTACT.
- OVERNIGHT OUTAGES WILL NOT BE PERMITTED UNLESS WORK IS BEING DONE DURING THAT TIME PERIOD AND THE SPRINKLER SYSTEM IS PUT BACK INTO SERVICE WITHIN AN HOUR FROM COMPLETION OF WORK. OUTAGES SHALL BE LIMITED TO 8 HOURS INCLUDING TIME FOR DRAINING AND REFILLING.
- THE SPRINKLER SYSTEM IS NOT PERMITTED TO HAVE AN OUTAGE FOR MORE THAN AN HOUR BEFORE STARTING WORK. THE SPRINKLER SYSTEM SHALL BE RESTORED WITHIN AN HOUR OF COMPLETION OF WORK. THE CONTRACTOR CANNOT LEAVE THE CONSTRUCTION SITE UNPROTECTED. THE SPRINKLER SYSTEM SHALL BE FILLED WITHIN AN HOUR OF COMPLETION OF WORK.
- PAINT PIPE RED WHERE PIPE AND FITTINGS IS PROVIDED.
- BIOSECURITY MUST BE FOLLOWED WHEN OPERATING IN THE FOLLOWING LOCATIONS TO PROTECT FISH HEALTH: TANK ROOMS A-G, FISH HEALTH LABS 07, 08, 17, 18 AND ALONG THE BROODSTOCK BUILDING. FURTHER INFORMATION CAN BE FOUND ON ATTACHMENT BIOSECURITY STATEMENT.



|                                                                                                                                                                 |                        |                                                                                                                           |      |                    |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------|------|--------------------|-------------|
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|                                                                                                                                                                 |                        | REVISIONS                                                                                                                 |      |                    |             |
|                                                                                                                                                                 |                        |  Agricultural<br>Research<br>Service |      |                    |             |
|                                                                                                                                                                 |                        |                                                                                                                           |      |                    |             |
| A-E FIRM                                                                                                                                                        | USDA                   | FIRE SYSTEM UPGRADES<br><small>KEARNEYSVILLE, WEST VIRGINIA</small>                                                       |      |                    |             |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                  | EPW<br>C. MORRIS       | NATIONAL CENTER FOR COOL AND COLD WATER<br>AQUACULTURE                                                                    |      |                    |             |
| DESIGNER<br>ROBERTSON, A                                                                                                                                        | PPM<br>T. VIGIL        | SYMBOLS & ABBREVIATIONS                                                                                                   |      | DWG. NO.<br>FX001  |             |
| CHECKED BY<br>PATERSON, A                                                                                                                                       | SAFETY & HEALTH<br>TBD | SOLICITATION NO.                                                                                                          |      | DATE<br>2025-12-12 |             |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                        | REAL PROPERTY<br>TBD   | SHEET 14 OF 29                                                                                                            |      |                    |             |

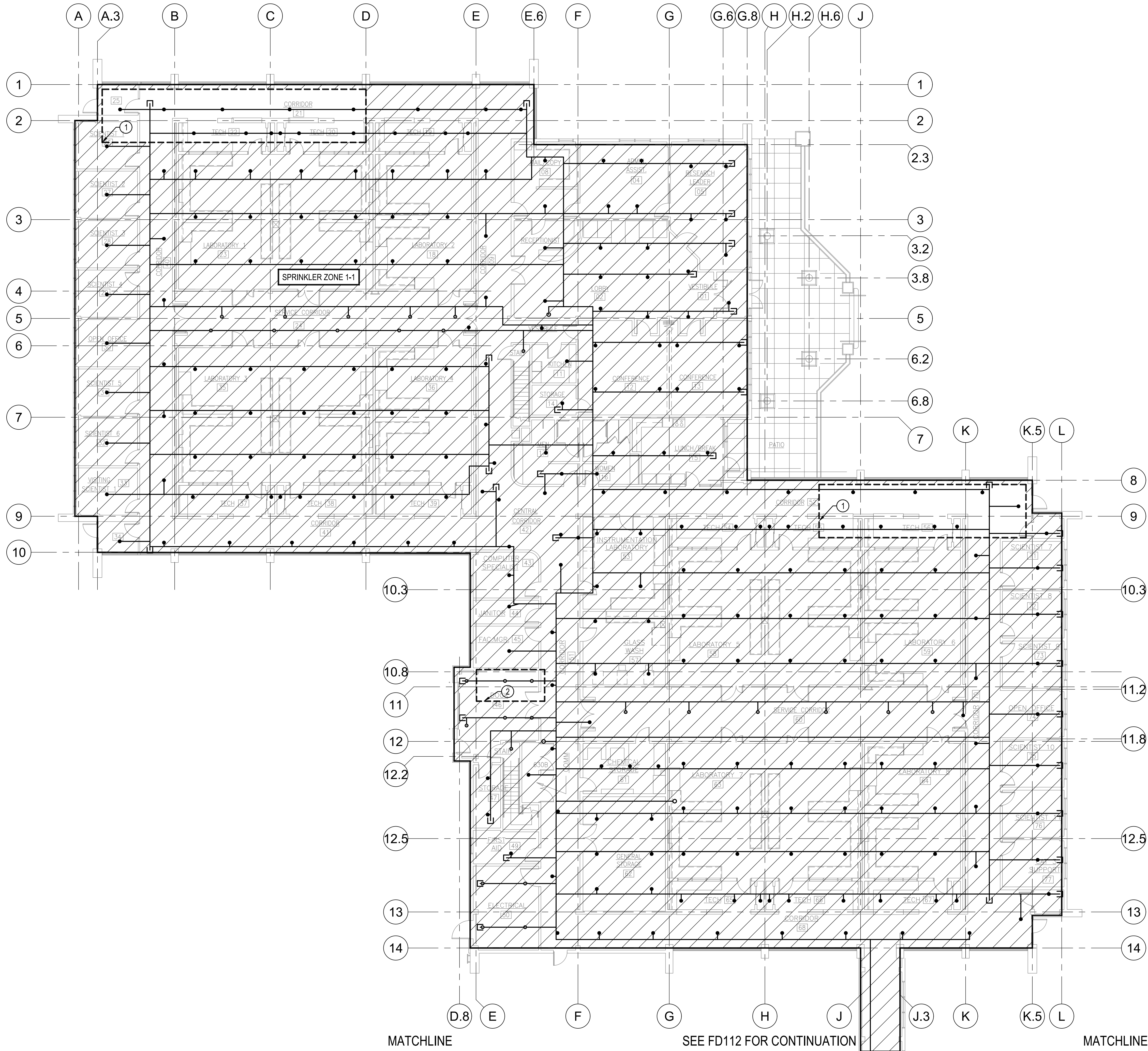


GENERAL NOTES

G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

NOTES

1. DISCONNECT AND REMOVE CORRODED ARM-OVERS AND HANGERS.
2. DISCONNECT AND REMOVE CORRODED SPRIG UPS AND ASSOCIATED SUPPORTS.

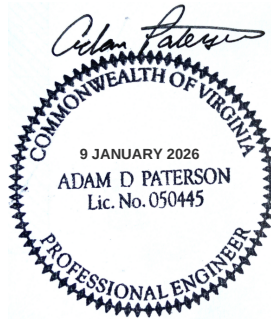


1 FIRST FLOOR MAIN BUILDING - DEMOLITION

3/32"=1'-0"

|                                                                                                                                                                |  |                 |    |                                           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|----|-------------------------------------------|--|
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|                                                                                                                                                                |  |                 |    |                                           |  |
| NO.                                                                                                                                                            |  | DATE            | BY | DESCRIPTION                               |  |
| REVISIONS                                                                                                                                                      |  |                 |    |                                           |  |
| A-E FIRM                                                                                                                                                       |  | USDA            |    |                                           |  |
| PROJECT MANAGER                                                                                                                                                |  | EPM             |    |                                           |  |
| PATERSON, A                                                                                                                                                    |  | C. MORRIS       |    |                                           |  |
| DESIGNER                                                                                                                                                       |  | PPM             |    |                                           |  |
| ROBERTSON, A                                                                                                                                                   |  | T. VIGIL        |    |                                           |  |
| CHECKED BY                                                                                                                                                     |  | SAFETY & HEALTH |    | SOLICITATION NO.                          |  |
| PATERSON, A                                                                                                                                                    |  | TBD             |    | FIRST FLOOR MAIN BUILDING -<br>DEMOLITION |  |
| DRAWN BY                                                                                                                                                       |  | REAL PROPERTY   |    |                                           |  |
| ROBERTSON, A                                                                                                                                                   |  | TBD             |    | FD111                                     |  |
|                                                                                                                                                                |  | DATE            |    | SHEET 15 OF 29                            |  |
|                                                                                                                                                                |  | 2025-12-12      |    |                                           |  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026



**ds** Agricultural Research Service

FIRE SYSTEM UPGRADES

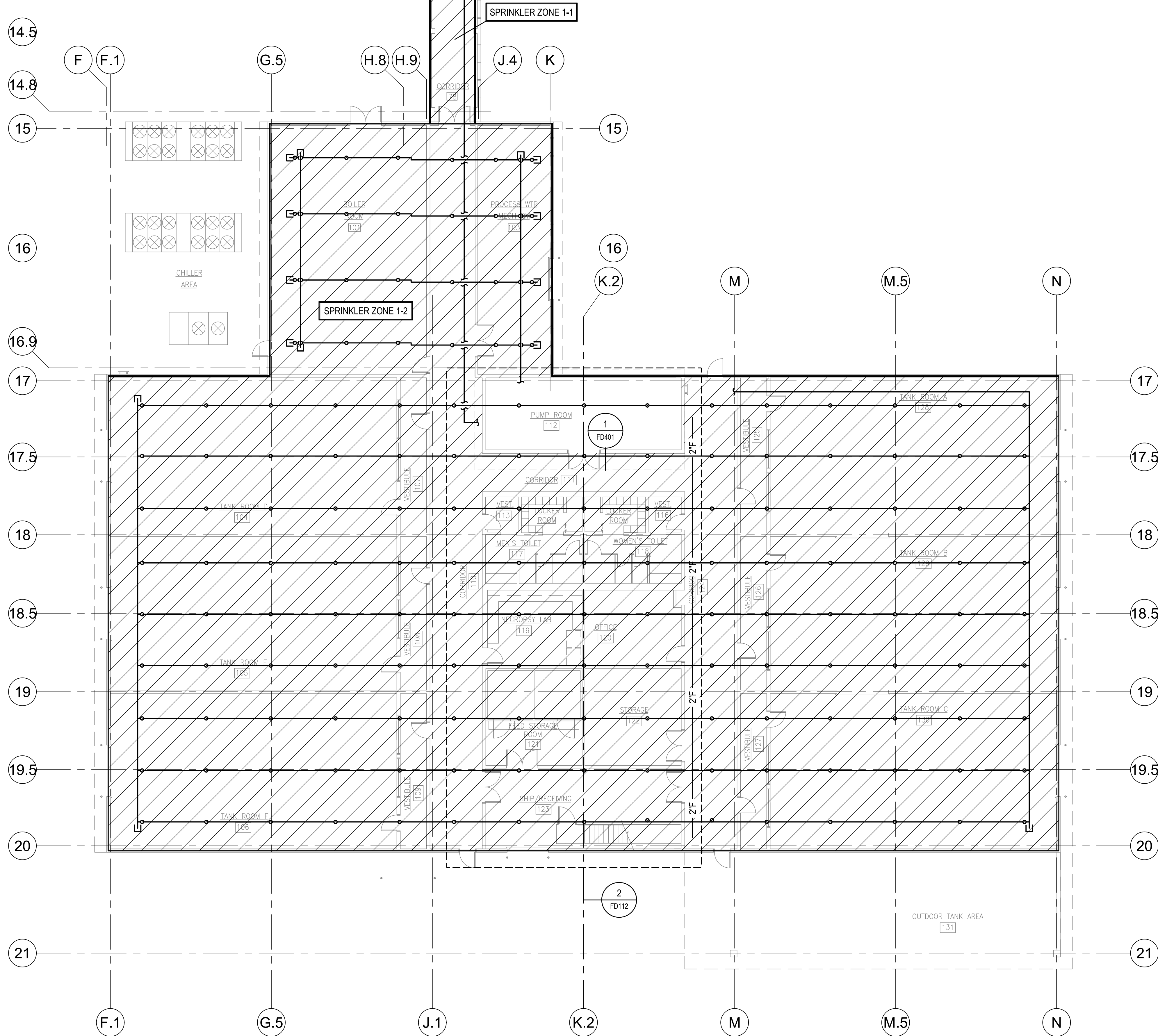
NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE

DWS. NO.

FD111



MATCHLINE SEE FD111 FOR CONTINUATION MATCHLINE



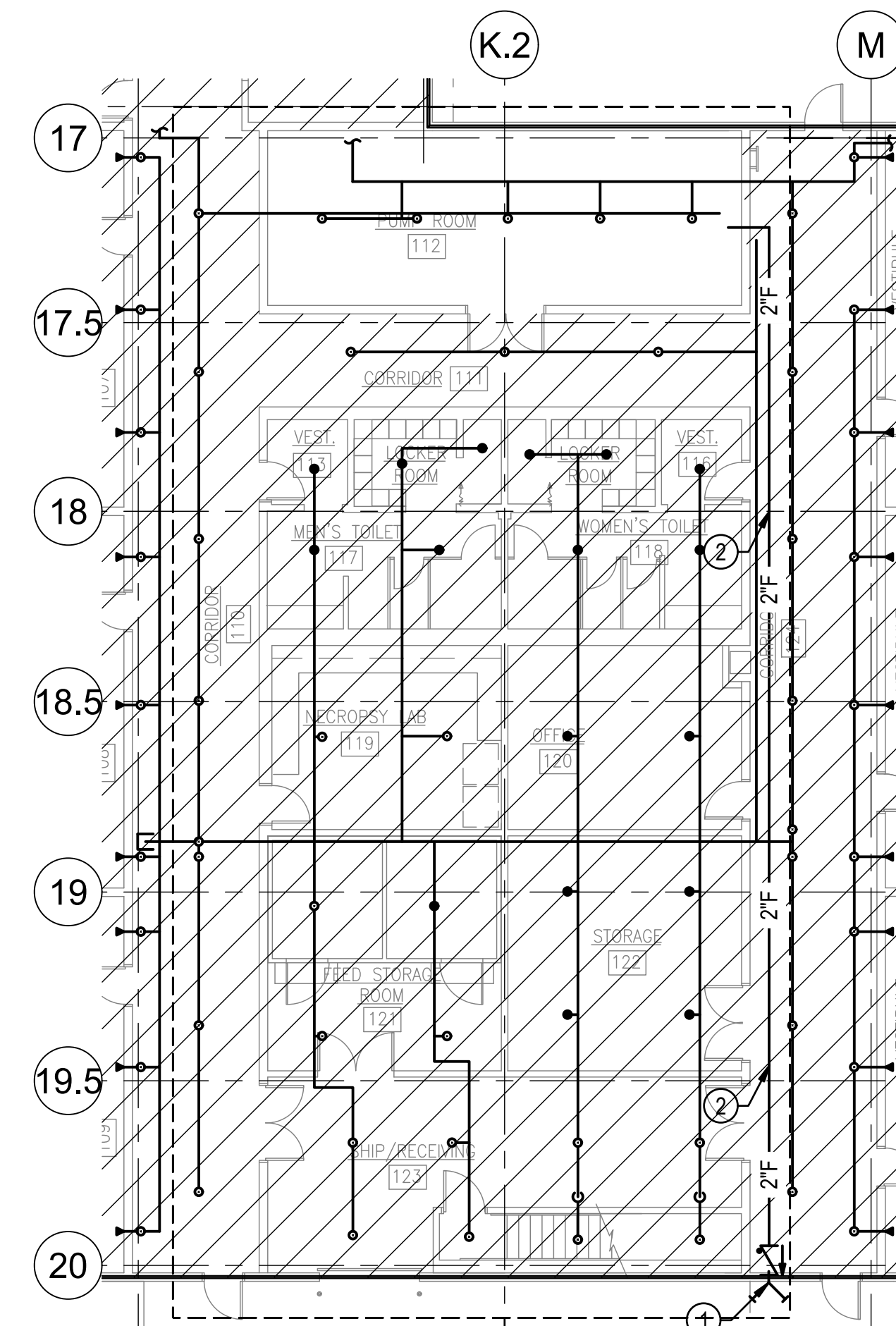
1 ROOF SPRINKLER SYSTEM  
FIRST FLOOR TANK BUILDING - DEMOLITION

## NOTES

1. DISCONNECT AND REMOTE THE EXISTING FIRE DEPARTMENT CONNECTION.
2. DISCONNECT AND REMOVE EXISTING 2-INCH PIPE AND ASSOCIATED EQUIPMENT/SUPPORTS.

## GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.



CEILING SPRINKLER SYSTEM  
FIRST FLOOR TANK BUILDING - DEMOLITION

|                                                                                                                                                        |  |                 |  |                                                                                                                                                                                                |    |                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------------------------------|
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|                                                                                                                                                        |  | REVISIONS       |  |                                                                                                                                                                                                |    |                                        |
| A-E FIRM                                                                                                                                               |  | USDA            |  | <div>USDA</div> <div>Agricultural Research Service</div> <div>FIRE SYSTEM UPGRADES</div> <div>KEARNEYVILLE, WEST VIRGINIA</div> <div>NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE</div> |    |                                        |
| PROJECT MANAGER                                                                                                                                        |  | EPM             |  |                                                                                                                                                                                                |    |                                        |
| PATERSON, A                                                                                                                                            |  | C. MORRIS       |  |                                                                                                                                                                                                |    |                                        |
| DESIGNER                                                                                                                                               |  | PPM             |  |                                                                                                                                                                                                |    |                                        |
| ROBERTSON, A                                                                                                                                           |  | T. VIGIL        |  |                                                                                                                                                                                                |    |                                        |
| CHECKED BY                                                                                                                                             |  | SAFETY & HEALTH |  | SOLICITATION NO.                                                                                                                                                                               |    | FIRST FLOOR TANK BUILDING - DEMOLITION |
| PATERSON, A                                                                                                                                            |  | TBD             |  | DATE                                                                                                                                                                                           |    |                                        |
| DRAWN BY                                                                                                                                               |  | REAL PROPERTY   |  | 2025-12-12                                                                                                                                                                                     |    | DWS. NO.<br>FD112                      |
| ROBERTSON, A                                                                                                                                           |  | TBD             |  |                                                                                                                                                                                                |    |                                        |
|                                                                                                                                                        |  |                 |  |                                                                                                                                                                                                |    | SHEET 16 OF 29                         |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

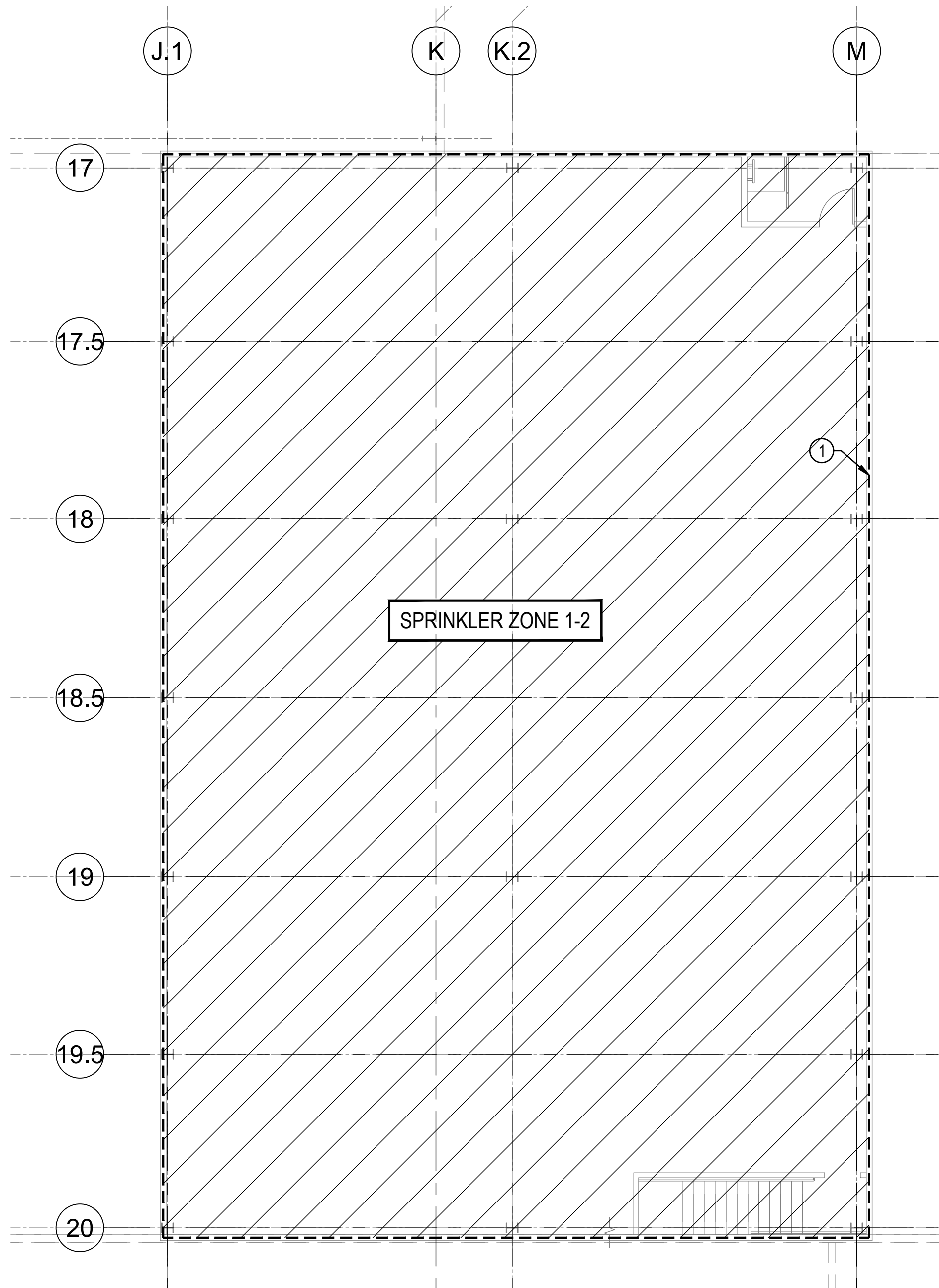


GENERAL NOTES

G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

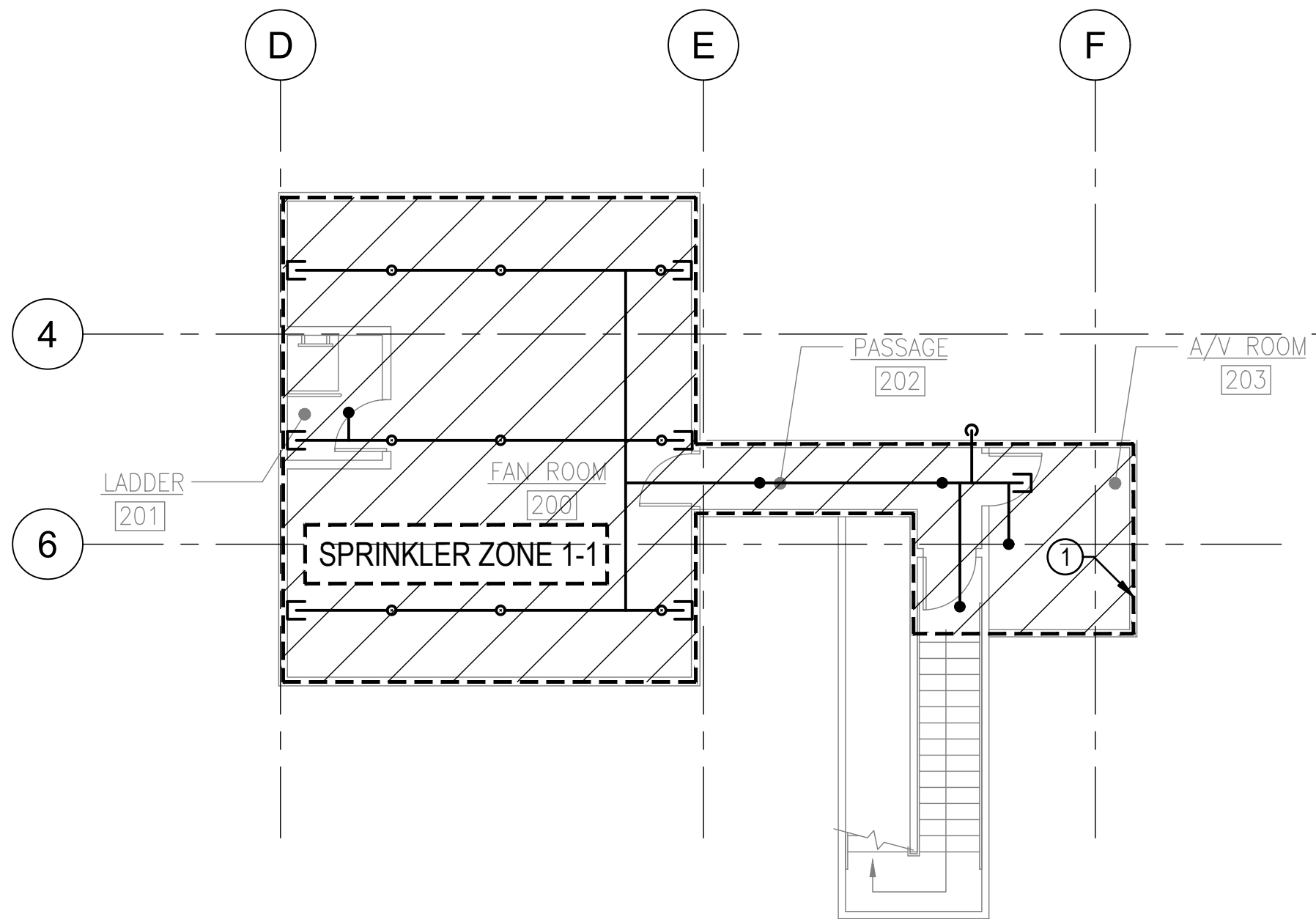
NOTES

1. SHEET PROVIDED FOR INFORMATION ONLY. NO DEMOLITION REQUIRED.



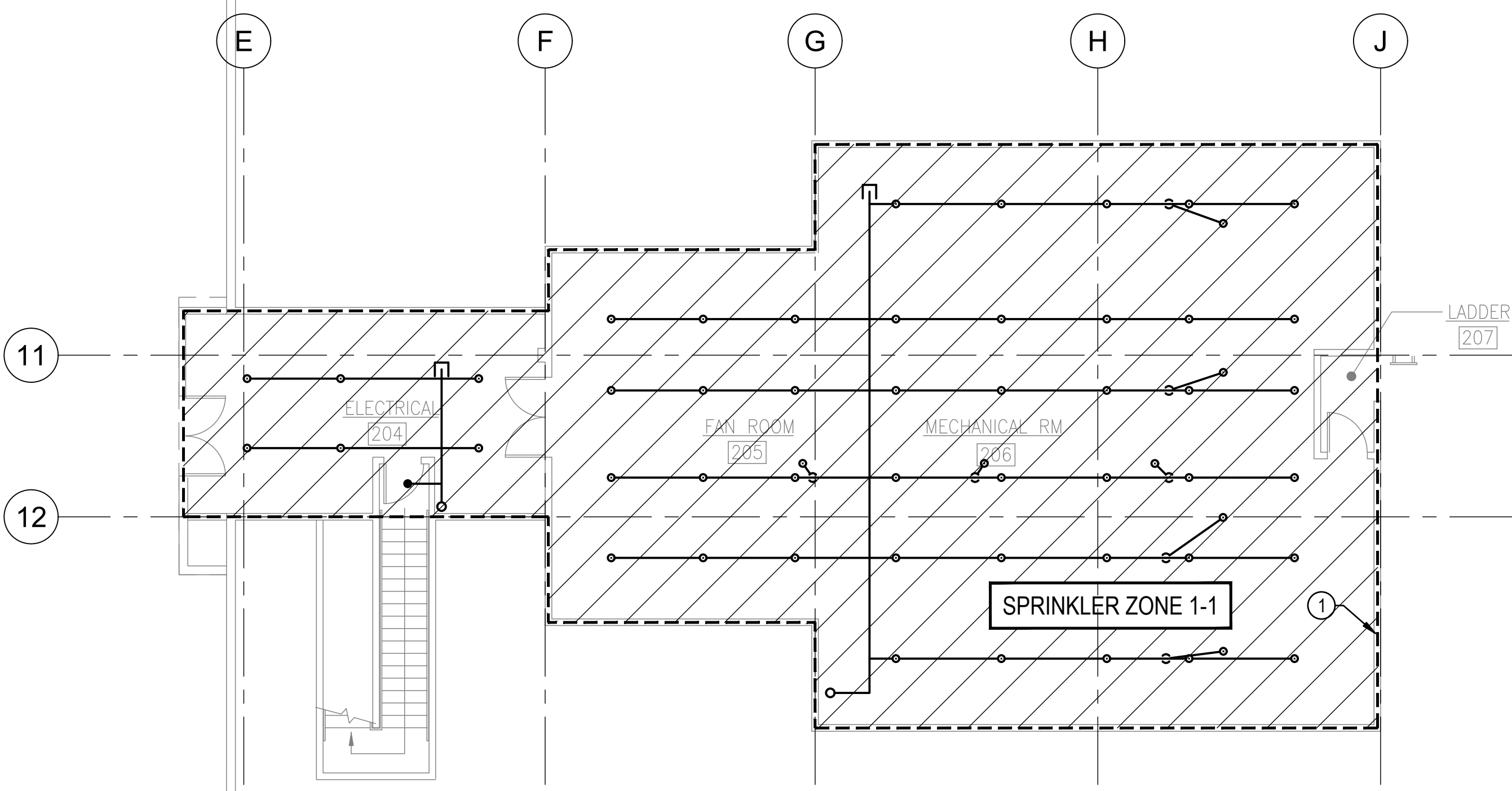
1 SECOND FLOOR TANK PENTHOUSE - DEMOLITION

1/8"=1'-0" 0 8' 16'



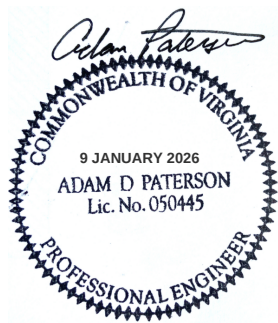
2 SECOND FLOOR NORTH PENTHOUSE - DEMOLITION

1/8"=1'-0" 0 8' 16'



3 SECOND FLOOR SOUTH PENTHOUSE - DEMOLITION

1/8"=1'-0" 0 8' 16'



|                                                                                                                                                                |  |                        |  |                    |  |                                     |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|--|--------------------|--|-------------------------------------|--|
| <div>SUMMER<br/>CONSULTANTS<br/>INCORPORATED</div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div> |  |                        |  |                    |  |                                     |  |
|                                                                                                                                                                |  |                        |  |                    |  |                                     |  |
|                                                                                                                                                                |  |                        |  |                    |  |                                     |  |
| NO.                                                                                                                                                            |  | DATE                   |  | BY                 |  | DESCRIPTION                         |  |
|                                                                                                                                                                |  |                        |  |                    |  | REVISIONS                           |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |  | U<br>S<br>D<br>A   |  | Agricultural<br>Research<br>Service |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |  |                    |  | FIRE SYSTEM UPGRADES                |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |  |                    |  | KEARNEYSVILLE, WEST VIRGINIA        |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.   |  | DWS. NO.                            |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12 |  | SECOND FLOOR - DEMOLITION<br>FD113  |  |
|                                                                                                                                                                |  |                        |  |                    |  | SHEET 17 of 29                      |  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

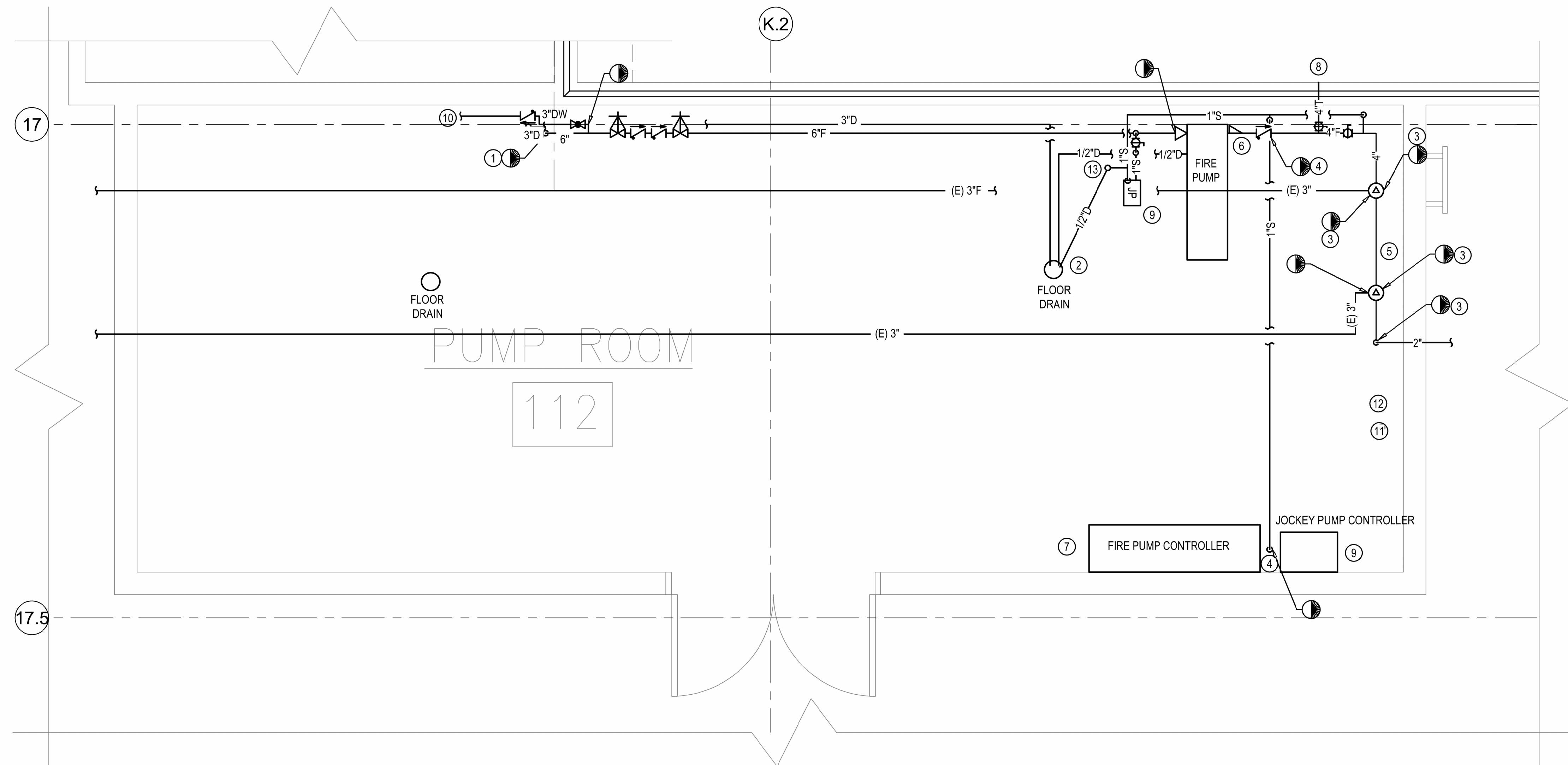


## GENERAL NOTES

G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

## NOTES

1. DISCONNECT AND REMOVE EXISTING 6" DOUBLE CHECK BACKFLOW PREVENTER. RETAIN FOR REINSTALLATION. DEMOLISH TEST PORTS AND GAUGES.
2. DISCONNECT AND REMOVE EXISTING DRAIN LINES TO FLOOR DRAIN FOR FUTURE FIRE PUMP ROOM BUILD OUT.
3. DISCONNECT AND REMOVE SPRINKLER PIPE, HEADER, AND VALVES.
4. DISCONNECT AND REMOVE SENSING LINES TO FIRE PUMP CONTROLLER.
5. RETAIN EXISTING MAIN BUILDING AND TANK BUILDING SPRINKLER RISERS. DISCONNECT AT HEADER AND AT CEILING MAIN LEVEL TO RELOCATE SPRINKLER RISER.
6. RETAIN FIRE PUMP AND CHECK VALVE. DISCONNECT AND REMOVE FIRE PUMP MOTOR POWER.
7. DISCONNECT AND REMOVE THE FIRE PUMP CONTROLLER PIPING, VALVES, FITTINGS, POWER, CONTROLLER, CONDUIT, CONDUCTORS, SCREWS, BOLTS, ETC. RETAIN THE FIRE PUMP TO RELOCATE. KEEP INCOMING CONDUIT FOR PRIMARY AND SECONDARY POWER. SEE ELECTRICAL DRAWINGS FOR POWER WORK.
8. DISCONNECT AND REMOVE FIRE PUMP TEST CONNECTION.
9. DISCONNECT AND REMOVE JOCKEY PUMP AND JOCKEY PUMP CONTROLLER AND ALL ASSOCIATED FOUNDATION, PIPING, VALVES, FITTINGS, POWER, CONTROLLER, CONDUIT, CONDUCTORS, SCREWS, BOLTS, ETC.
10. DISCONNECT EXISTING DOMESTIC LINE FROM SHARED INCOMING WATER LINE. RETAIN THE ISOLATION VALVE AND BACKFLOW VALVE. DEMOLISH THE 3 INCH DRAIN CONNECTION FOR BACKFLOW TO THE FLOOR DRAIN. COORDINATE OUTAGE WITH BUILDING AND RECONNECT SAME DAY.
11. DISCONNECT AND RELOCATE FIRE EXTINGUISHER, MOUNT, AND SIGN.
12. DISCONNECT AND RELOCATE THE SPRINKLER CABINET.
13. DISCONNECT AND REMOVE THE PRESSURE RELIEF VALVE.



1 ENLARGED PUMP ROOM



|                                                                                                 |  |                        |  |                                                                                                                     |  |                                            |  |
|-------------------------------------------------------------------------------------------------|--|------------------------|--|---------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------|--|
|            |  |                        |  |                                                                                                                     |  |                                            |  |
| 7500 Westpark Dr.   Suite A405   McLean, VA 22102<br>(703) 556-8820   www.summerconsultants.com |  | NO.                    |  | DATE                                                                                                                |  | BY                                         |  |
|                                                                                                 |  |                        |  |                                                                                                                     |  | DESCRIPTION                                |  |
|                                                                                                 |  |                        |  |                                                                                                                     |  | REVISIONS                                  |  |
| A-E FIRM      USDA                                                                              |  | U<br>S<br>D<br>A       |  |  Agricultural Research Service |  |                                            |  |
| PROJECT MANAGER<br>PATERSON, A                                                                  |  |                        |  |                                                                                                                     |  | EPM<br>C. MORRIS                           |  |
| DESIGNER<br>ROBERTSON, A                                                                        |  |                        |  |                                                                                                                     |  | PPM<br>T. VIGIL                            |  |
| CHECKED BY<br>PATERSON, A                                                                       |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.                                                                                                    |  | DWG. NO.                                   |  |
| DRAWN BY<br>ROBERTSON, A                                                                        |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12                                                                                                  |  | FIRE PUMP DETAIL - DEMOLITION<br><br>FD401 |  |
|                                                                                                 |  |                        |  |                                                                                                                     |  | SHEET 18 OF 29                             |  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

FILE NAME: 212.012 FD401.DWG  
OUR REF: 212.012  
PLOT DATE: 12/12/2025 2:28:49 PM

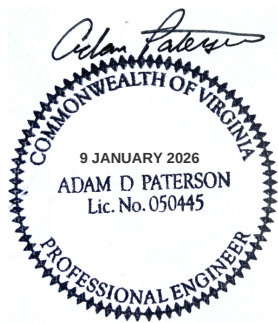
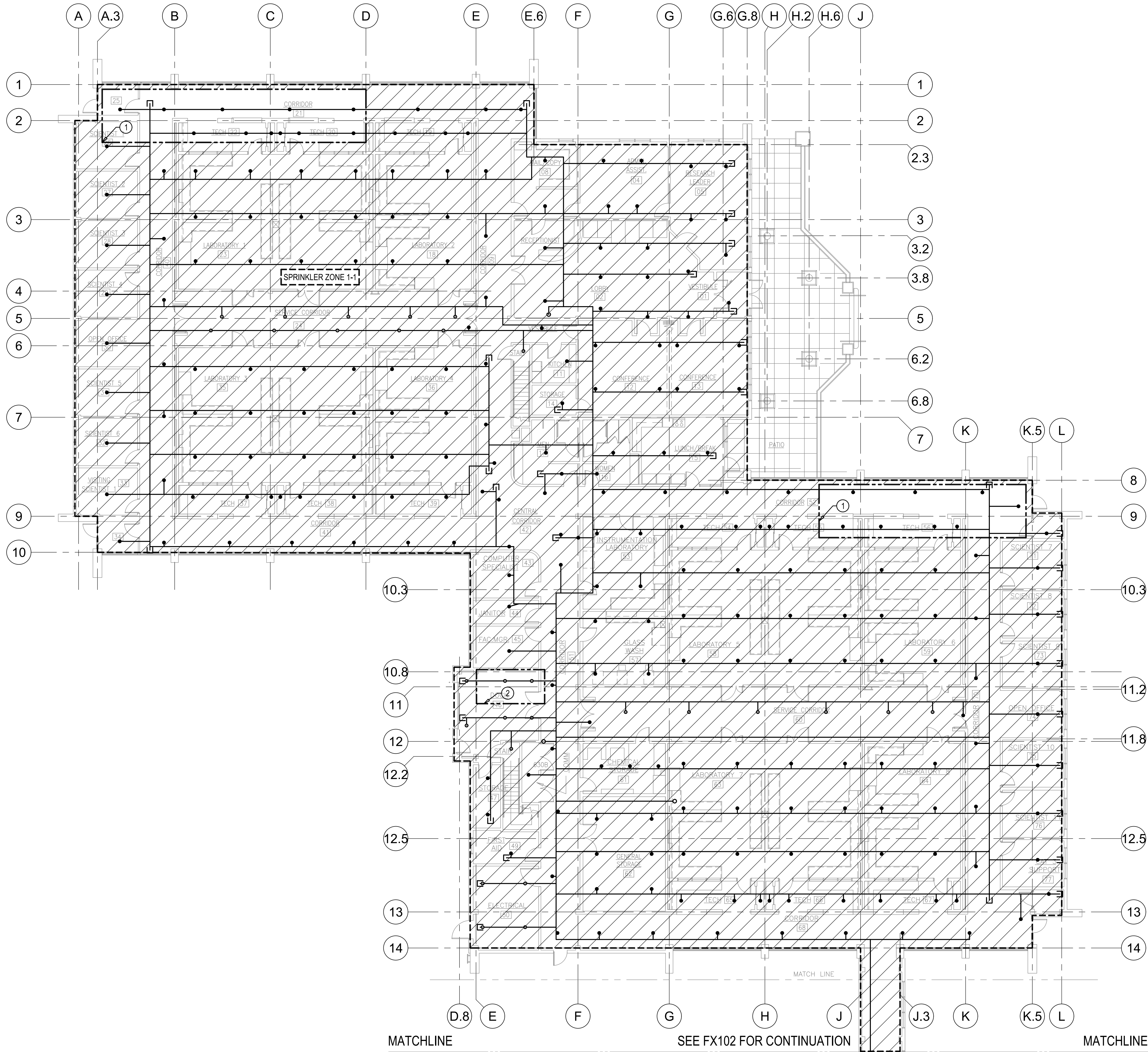


GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "E".

NOTES

1. PROVIDE SPRINKLER HEADS AND ARMOVERS WITHIN AREA OF WORK.
2. PROVIDE SPRINKLER HEADS AND SPRIG UPS WITHIN AREA OF WORK.



|                                                                                                                                                                |  |                        |      |                                                                                                                                                                                                            |  |                                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------|--|
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|                                                                                                                                                                |  |                        |      |                                                                                                                                                                                                            |  |                                    |  |
|                                                                                                                                                                |  |                        |      |                                                                                                                                                                                                            |  |                                    |  |
|                                                                                                                                                                |  | NO.                    | DATE | BY                                                                                                                                                                                                         |  | DESCRIPTION                        |  |
|                                                                                                                                                                |  | REVISIONS              |      |                                                                                                                                                                                                            |  |                                    |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |      | <div>USDA</div> <div>Agricultural<br/>Research<br/>Service</div> <div>FIRE SYSTEM UPGRADES</div> <div>KEARNEYVILLE, WEST VIRGINIA</div> <div>NATIONAL CENTER FOR COOL AND COLD WATER<br/>AQUACULTURE</div> |  |                                    |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |      |                                                                                                                                                                                                            |  |                                    |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |      |                                                                                                                                                                                                            |  |                                    |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |      |                                                                                                                                                                                                            |  |                                    |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |      | SOLICITATION NO.<br>DATE<br>2025-12-12                                                                                                                                                                     |  | FIRST FLOOR MAIN BUILDING<br>FX101 |  |
|                                                                                                                                                                |  |                        |      |                                                                                                                                                                                                            |  | SHEET 19 OF 29                     |  |





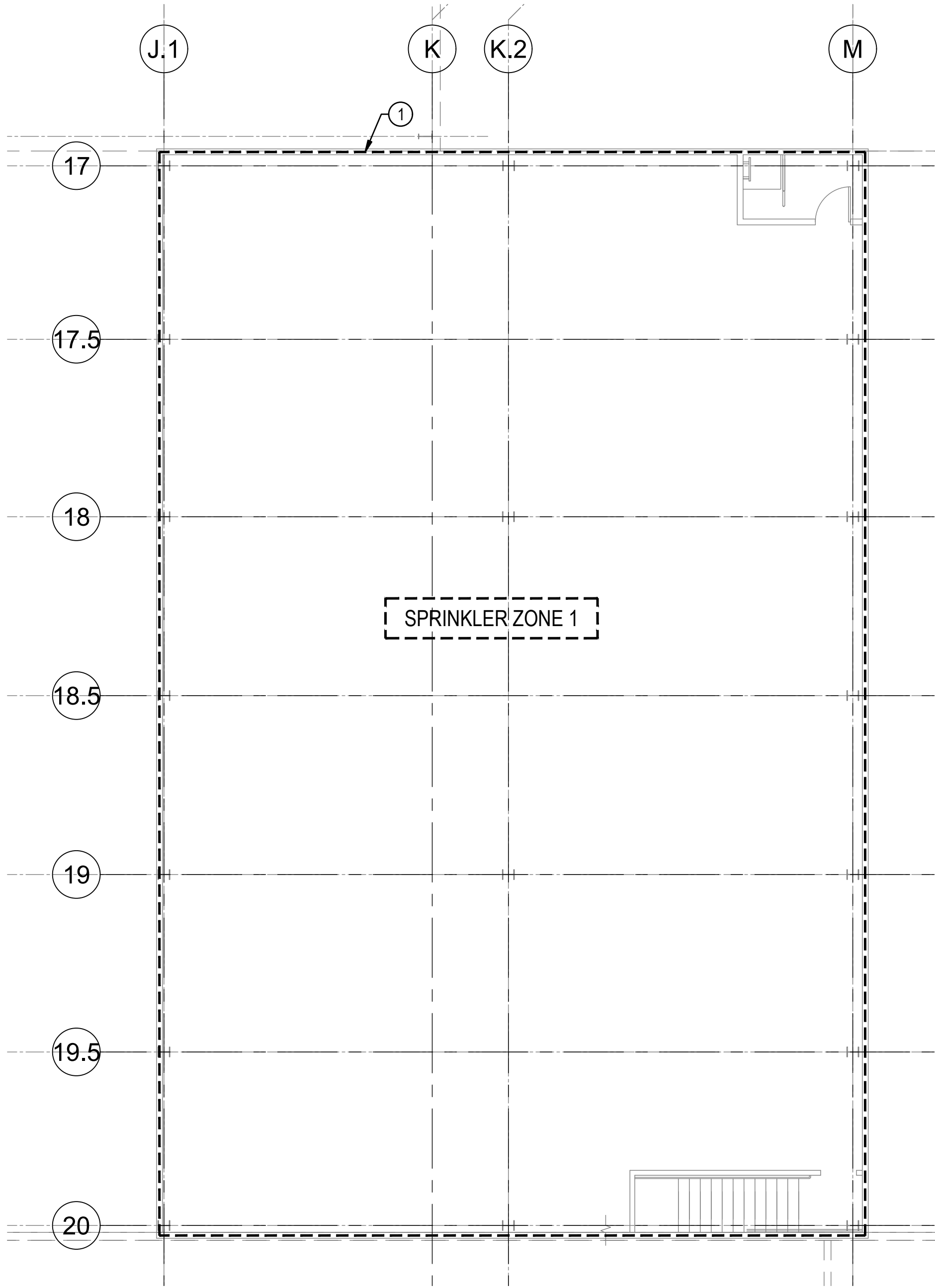


GENERAL NOTES

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- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".

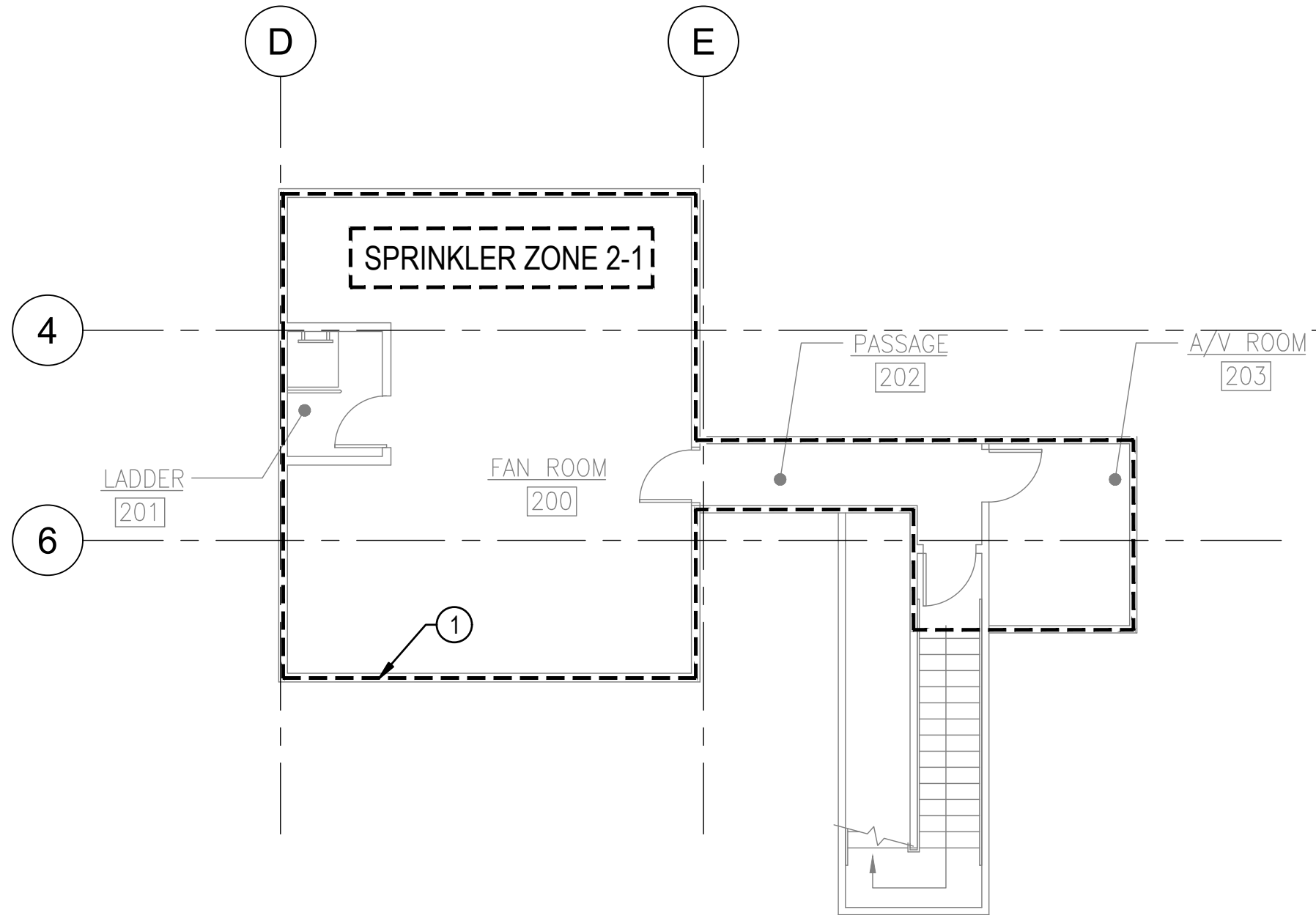
NOTES

1. EXISTING TO REMAIN WET-PIPE SPRINKLER SYSTEM. PROVIDED FOR INFORMATION ONLY. NO NEW WORK.



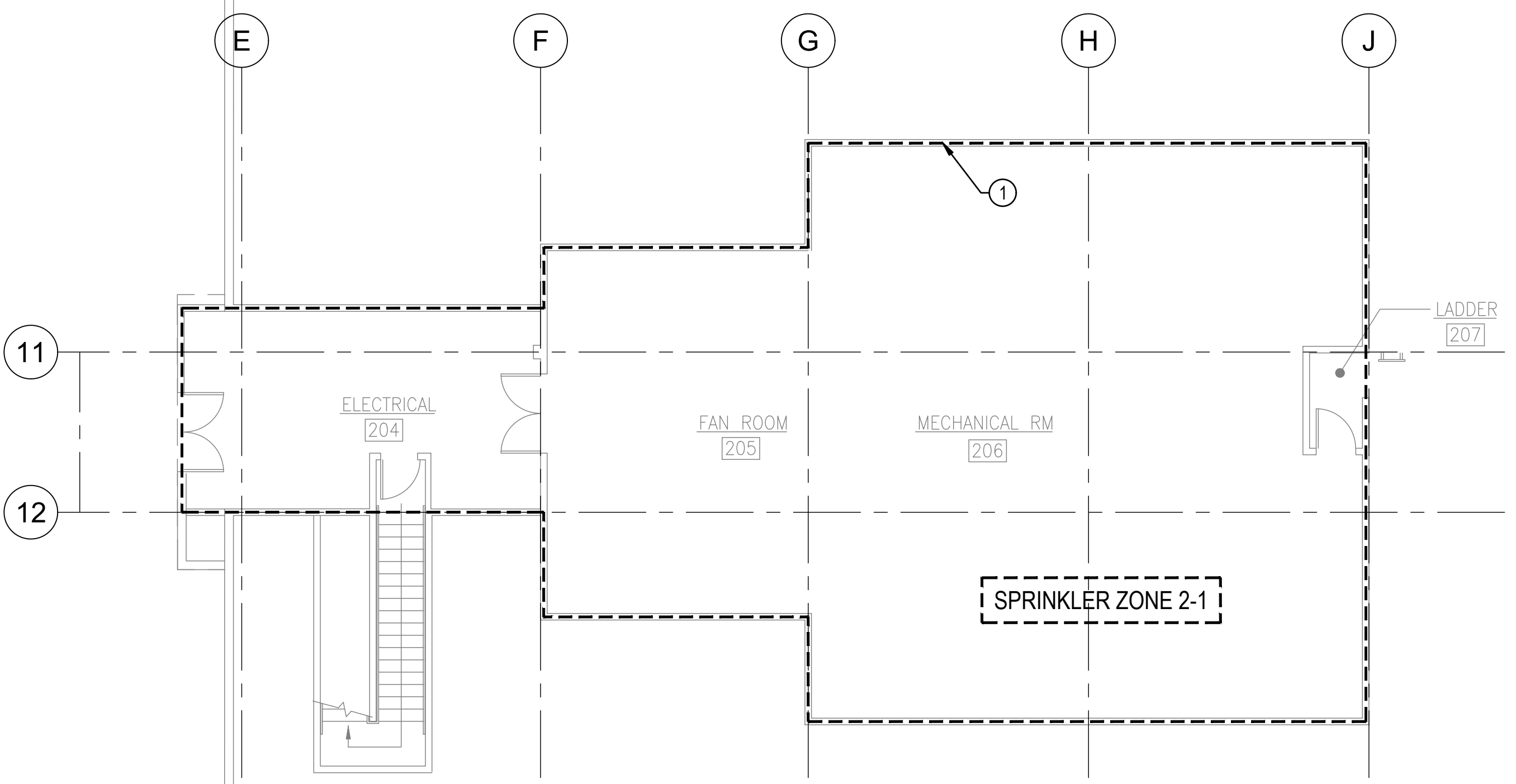
1 SECOND FLOOR TANK PENTHOUSE

1/8"=1'-0"



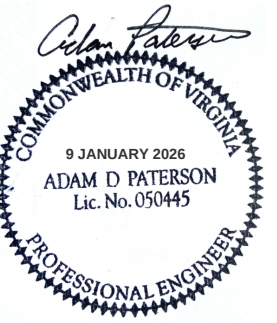
2 SECOND FLOOR NORTH PENTHOUSE

1/8"=1'-0"



3 SECOND FLOOR SOUTH PENTHOUSE

1/8"=1'-0"



|                                                                                                                                                                |  |                        |  |                                                        |  |                                     |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|--|--------------------------------------------------------|--|-------------------------------------|--|
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|                                                                                                                                                                |  |                        |  |                                                        |  |                                     |  |
|                                                                                                                                                                |  |                        |  |                                                        |  |                                     |  |
|                                                                                                                                                                |  | NO.                    |  | DATE                                                   |  | BY                                  |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |  | USDA                                                   |  | REVISIONS                           |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |  | FIRE SYSTEM UPGRADES                                   |  | Agricultural<br>Research<br>Service |  |
| DESIGNER<br>ROBERTSON, A                                                                                                                                       |  | PPM<br>T. VIGIL        |  | KEARNEYSVILLE, WEST VIRGINIA                           |  |                                     |  |
| CHECKED BY<br>PATERSON, A                                                                                                                                      |  | SAFETY & HEALTH<br>TBD |  | NATIONAL CENTER FOR COOL AND COLD WATER<br>AQUACULTURE |  |                                     |  |
| DRAWN BY<br>ROBERTSON, A                                                                                                                                       |  | REAL PROPERTY<br>TBD   |  | SOLICITATION NO.                                       |  | DWG. NO.                            |  |
|                                                                                                                                                                |  | DATE<br>2025-12-12     |  | SECOND FLOOR                                           |  | FX103                               |  |
|                                                                                                                                                                |  |                        |  |                                                        |  | SHEET 21 of 29                      |  |

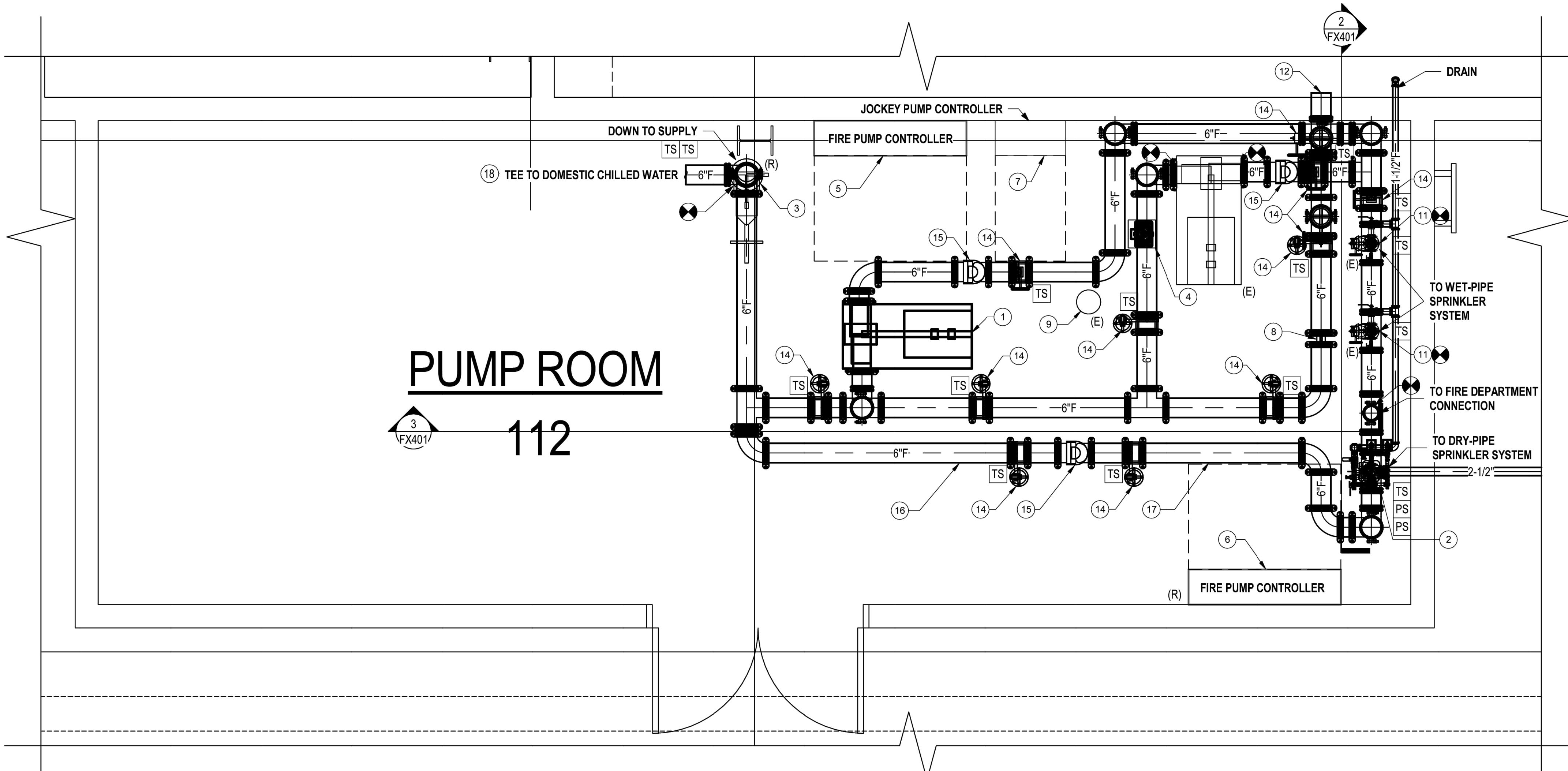


GENERAL NOTES

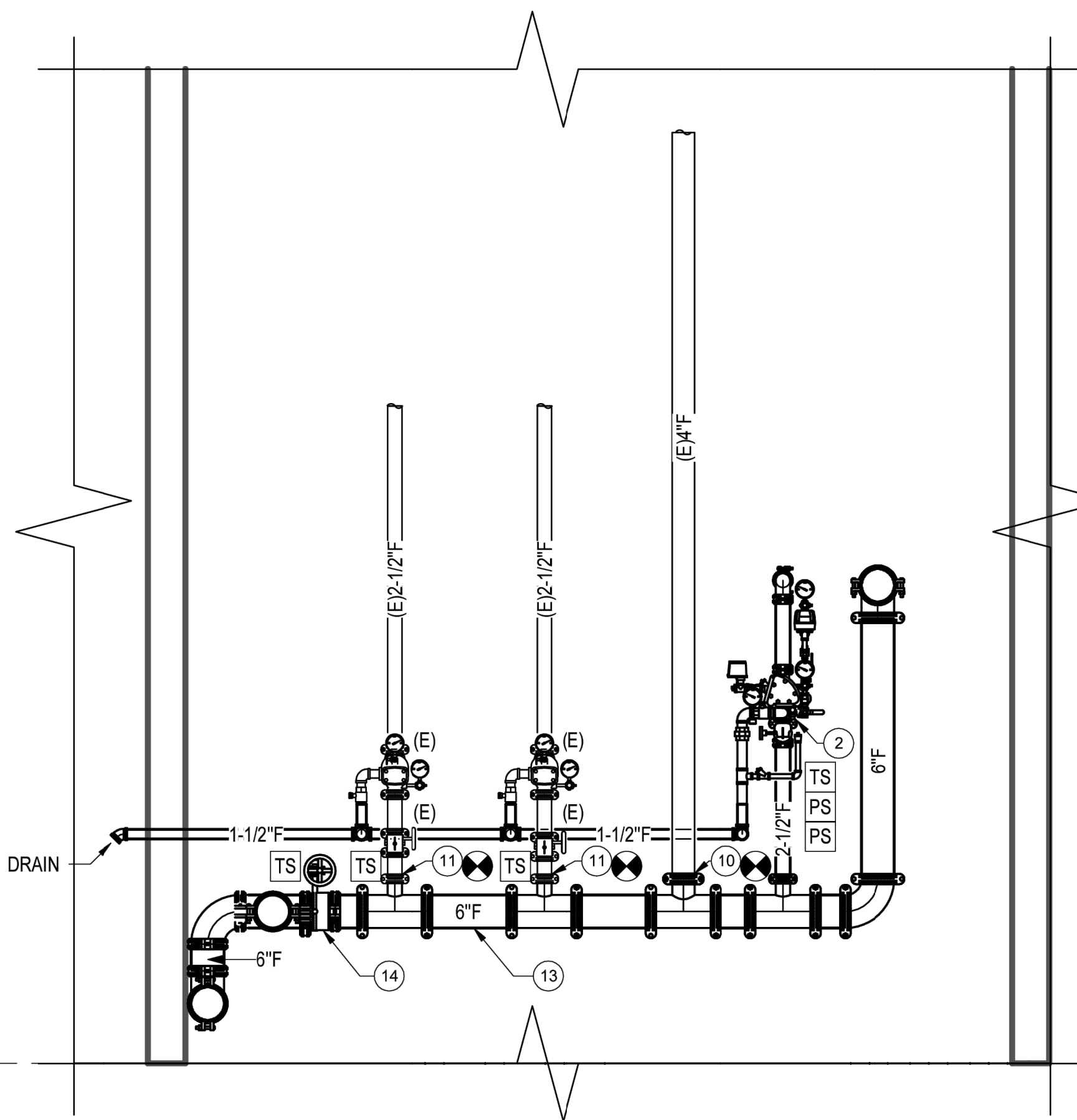
- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK SHALL BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST.", OR "(E)".

NOTES

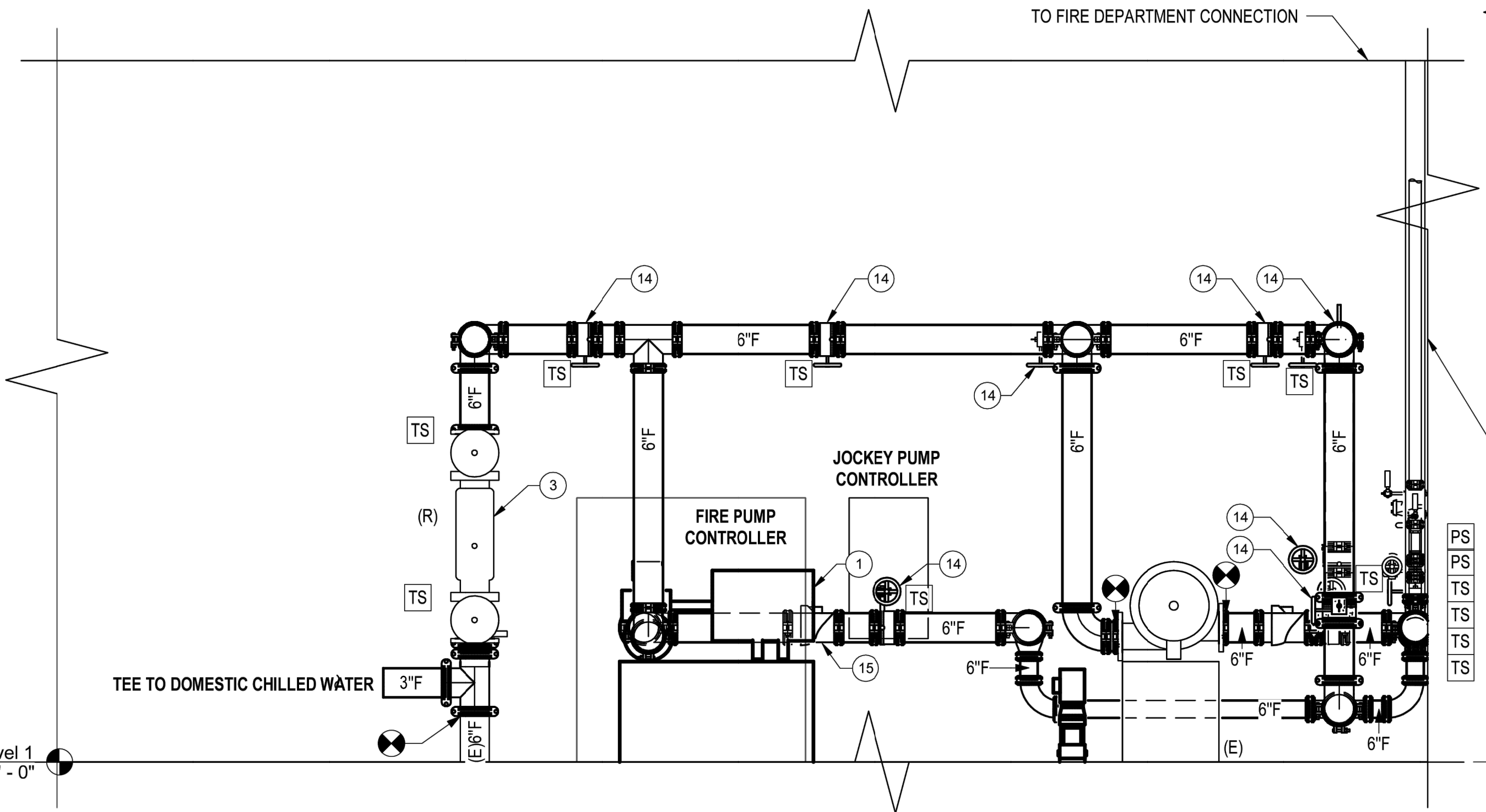
1. PROVIDE 500 GPM @ 65 PSI HORIZONTAL SPLIT CASE FIRE PUMP AND ASSOCIATED MODIFICATIONS AS SHOWN AND IN ACCORDANCE WITH NFPA 20 AND ARS 242.1. PROVIDE PEDESTAL TO MATCH EXISTING.
2. PROVIDE 2-1/2-INCH DRY PIPE SPRINKLER RISER WITH RISER MOUNTED AIR COMPRESSOR TO OUTDOOR TANK AREA. PROVIDE WATERFLOW PRESSURE SWITCH, VALVE TAMPER SWITCH, AND SUPERVISORY HIGH/LOW PRESSURE SWITCH.
3. REINSTALL EXISTING BACKFLOW PREVENTER IN THE VERTICAL ORIENTATION.
4. PROVIDE 10 GPM JOCKEY PUMP @ 145 PSI.
5. PROVIDE FLOOR MOUNTED FIRE PUMP CONTROLLER. COORDINATE CONNECTIONS. PROVIDE SENSING LINES AND GAUGES.
6. RELOCATE EXISTING FIRE PUMP CONTROLLER TO RETAIN WORKING CLEARANCE AND FACILITATE INSTALLATION OF JUNCTION BOXES. PROVIDE BATTERY TERMINAL CABINET FOR SECOND FIRE PUMP IN RELOCATED FIRE PUMP CONTROLLER.
7. PROVIDE WALL MOUNTED JOCKEY PUMP CONTROLLER.
8. PROVIDE FLOW METER FOR NON-EXTERNAL DISCHARGE FIRE PUMP TESTING.
9. DRAIN PUMP RELIEFS TO EXISTING FLOOR DRAIN.
10. CONNECT TO EXISTING PIPE AT LOCATION INDICATED.
11. CONNECT TO BASE OF EXISTING WET-PIPE SPRINKLER RISER.
12. PROVIDE 6" FIRE PUMP TEST HEADER.
13. PROVIDE NEW SPRINKLER RISER HEADER.
14. PROVIDE 6" BUTTERFLY VALVE.
15. PROVIDE 6" CHECK VALVE.
16. PROVIDE JOCKEY PUMP SUPPLY UPSTREAM OF BYPASS VALVES WITH WELDOLET CONNECTION. PROVIDE 1" BALL VALVE. LOCK OPEN ALL JOCKEY PUMP ISOLATION VALVES.
17. PROVIDE 1" CHECK VALVE AND 1" BALL VALVE FROM JOCKEY PUMP DISCHARGE. CONNECT DOWNSTREAM OF BYPASS VALVES WITH WELDOLET CONNECTION. LOCK OPEN ALL JOCKEY PUMP ISOLATION VALVES.
18. INSTALL EXISTING DOMESTIC WATER ISOLATION VALVE AND DOMESTIC BACKFLOW. ROUTE BACKFLOW VALVE DRAIN TO FLOOR DRAIN.



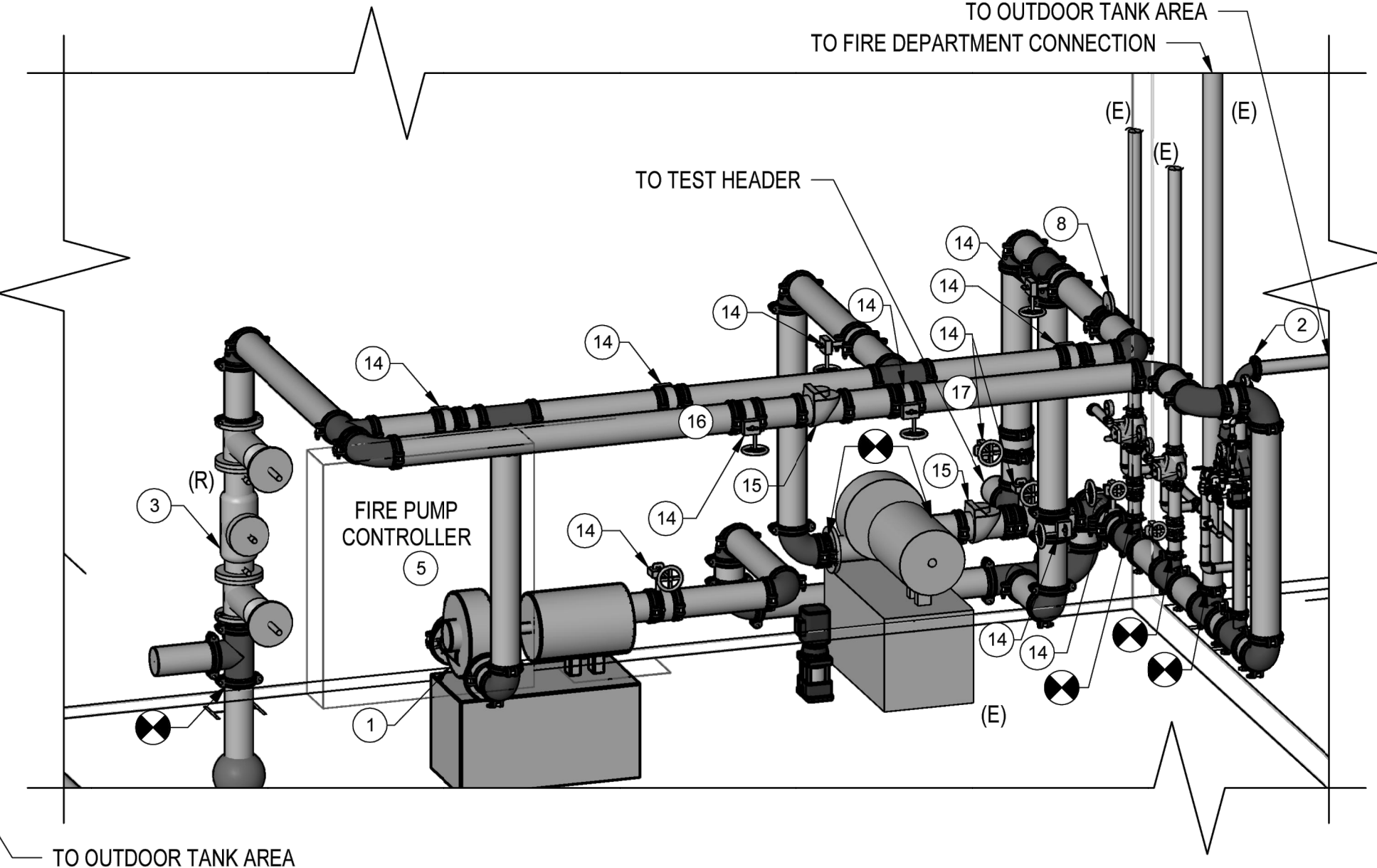
1 FIRE PUMP ROOM DETAIL  
FX401 1/2"=1'-0"



2 RISER DETAIL  
FX401 1/2"=1'-0"

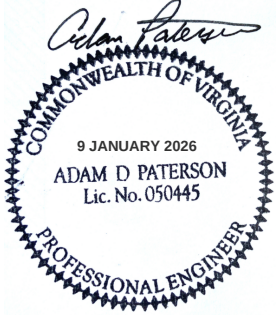


3 FIRE PUMP ROOM FRONT VIEW  
FX401 1/2"=1'-0"



4 FIRE PUMP ROOM 3D VIEW  
3/8"=1'-0"

|                                                                                                                                                        |  |                        |  |                    |    |                  |  |                   |                                                                                                                                |                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|--|--------------------|----|------------------|--|-------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <div>SUMMER CONSULTANTS INCORPORATED</div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div> |  | NO.                    |  | DATE               | BY | REVISIONS        |  | U<br>S<br>D<br>A  |  <div>Agricultural Research Service</div> | FIRE SYSTEM UPGRADES<br><br>KEARNEYVILLE, WEST VIRGINIA<br><br>NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE |
|                                                                                                                                                        |  |                        |  |                    |    |                  |  |                   |                                                                                                                                |                                                                                                                    |
| A-E FIRM                                                                                                                                               |  | USDA                   |  |                    |    |                  |  |                   |                                                                                                                                |                                                                                                                    |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                         |  | EPM<br>C. MORRIS       |  |                    |    |                  |  |                   |                                                                                                                                |                                                                                                                    |
| DESIGNER<br>ROBERTSON, A                                                                                                                               |  | PPM<br>T. VIGIL        |  |                    |    |                  |  |                   |                                                                                                                                |                                                                                                                    |
| CHECKED BY<br>PATERSON, A                                                                                                                              |  | SAFETY & HEALTH<br>TBD |  | SOLICITATION NO.   |    |                  |  |                   |                                                                                                                                |                                                                                                                    |
| DRAWN BY<br>ROBERTSON, A                                                                                                                               |  | REAL PROPERTY<br>TBD   |  | DATE<br>2025-12-12 |    | FIRE PUMP DETAIL |  | DWS. NO.<br>FX401 |                                                                                                                                |                                                                                                                    |
|                                                                                                                                                        |  |                        |  |                    |    |                  |  | SHEET 22 OF 29    |                                                                                                                                |                                                                                                                    |





| (E)PANEL LMPN9 ③                 |                           |           |                              |         |         |                        |                        |      |                         |               |                                             |          |                                      |      |
|----------------------------------|---------------------------|-----------|------------------------------|---------|---------|------------------------|------------------------|------|-------------------------|---------------|---------------------------------------------|----------|--------------------------------------|------|
| MAIN BUS SIZE (AMPS): 225        |                           |           | INTERRUPT. RATING (KAIC): 10 |         |         | NEUTRAL BUS SIZE: 100% |                        |      | MOUNTING TYPE: SURFACE  |               |                                             |          |                                      |      |
| MAIN BUS MATERIAL: COPPER        |                           |           | VOLTAGE: 120 / 208           |         |         | SECTION NO.: 1 OF 1    |                        |      | LOCATION: PUMP ROOM 112 |               |                                             |          |                                      |      |
| MAIN CIRCUIT BREAKER (AMPS):     |                           |           | PHASE: 3                     |         | WIRE: 4 |                        | ENCLOSURE TYPE: NEMA 1 |      |                         | SQUARE D NQDD |                                             |          |                                      |      |
| CKT NO.                          | DESCRIPTION               | LOAD TYPE | LOAD (KVA)                   | BREAKER |         |                        | BREAKER                |      | LOAD (KVA)              | LOAD TYPE     | DESCRIPTION                                 | CKT NO.  |                                      |      |
|                                  |                           |           |                              | AMPS    | POLES   |                        | POLES                  | AMPS |                         |               |                                             |          |                                      |      |
| 1                                | (E)TASK LIGHTING-RM119    | N         | 0.10                         | 20      | 1       | A                      |                        |      | 1.20                    | N             | (E)REVERSE SOMOSIS UNIT                     | 2        |                                      |      |
| 3                                | (E)UNIT HTR UH-12 (RM103) | M         | 0.01                         | 20      | 1       |                        | B                      |      | 1.20                    | N             |                                             | 4        |                                      |      |
| 5                                | (E)RO REPRESSURE PUMP     | M         | 1.18                         | 20      | 1       |                        |                        | C    | 0.50                    | N             | (E)CARBON FILTER<br>TIMER/RESISTIVITY METER | 6        |                                      |      |
| 7                                | (E)UNIT HTR UH-11(RM208)  | M         | 0.01                         | 20      | 1       | A                      |                        |      | 0.50                    | N             |                                             | 8        |                                      |      |
| 9                                | (E)UNIT HTR UH-8 (RM112)  | M         | 0.01                         | 20      | 1       |                        | B                      |      | 1                       | 20            | 1.08                                        | R        | (E)REC - RMS 121, 122, 123           | 10   |
| 11                               | (E)UNIT HTR UH-5 (RM 123) | M         | 0.01                         | 20      | 1       |                        |                        | C    | 1                       | 20            | 0.90                                        | R        | (E)REC - RMS 121, 122, 123           | 12   |
| 13                               | (E)UNIT HTR UH-7 (RM 208) | M         | 0.03                         | 20      | 1       | A                      |                        |      | 1                       | 20            | 0.90                                        | R        | (E)REC - RMS 121, 122, 123           | 14   |
| 15                               | (E)REC - RM 208           | R         | 0.54                         | 20      | 1       |                        | B                      |      | 1                       | 20            | 1.08                                        | R        | (E)REC - RMS 113, 116, 117, 118, 120 | 16   |
| 17                               | (E)REC - RM 208           | R         | 0.54                         | 20      | 1       |                        |                        | C    | 1                       | 20            | 0.90                                        | R        | (E)REC - RMS 113, 116, 117, 118, 120 | 18   |
| 19                               | (E)REC - RM 208           | R         | 0.54                         | 20      | 1       | A                      |                        |      | 1                       | 20            | 0.90                                        | R        | (E)REC - RMS 113, 116, 117, 118, 120 | 20   |
| 21                               | AIR COMPRESSOR ②          | M         | 0.83                         | 20      | 1       |                        | B                      |      | 1                       | 20            | 0.54                                        | R        | (E)REC - RMS 113, 116, 117, 118, 120 | 22   |
| 23                               | JOCKEY PUMP ①             | M         | 0.9                          | 20      | 3       |                        |                        | C    | 1                       | 20            | 0.54                                        | R        | (E)REC - RMS 112                     | 24   |
| 25                               |                           | M         | 0.9                          |         |         | A                      |                        | 2    | 20                      | 1.20          | N                                           | EXISTING | 26                                   |      |
| 27                               |                           | M         | 0.9                          |         |         | B                      |                        |      |                         | 1.20          | N                                           |          | 28                                   |      |
| 29                               | SPACE                     | N         | 0.00                         | --      | 1       |                        |                        | C    | 1                       | --            | 0.00                                        | N        | SPACE                                | 30   |
| 31                               | SPACE                     | N         | 0.00                         | --      | 1       | A                      |                        |      | 1                       | --            | 0.00                                        | N        | SPACE                                | 32   |
| 33                               | SPACE                     | N         | 0.00                         | --      | 1       |                        | B                      |      | 1                       | --            | 0.00                                        | N        | SPACE                                | 34   |
| 35                               | SPACE                     | N         | 0.00                         | --      | 1       |                        |                        | C    | 1                       | --            | 0.00                                        | N        | SPACE                                | 36   |
| 37                               | SPACE                     | N         | 0.00                         | --      | 1       | A                      |                        |      | 1                       | --            | 0.00                                        | N        | SPACE                                | 38   |
| 39                               | SPACE                     | N         | 0.00                         | --      | 1       |                        | B                      |      | 1                       | --            | 0.00                                        | N        | SPACE                                | 40   |
| 41                               | SPACE                     | N         | 0.00                         | --      | 1       |                        |                        | C    | 1                       | --            | 0.00                                        | N        | SPACE                                | 42   |
| CONNECTED LOAD PER PHASE (kVA) = |                           |           |                              |         |         | 6.3                    | 7.4                    | 5.5  | TOTAL CONNECTED LOAD =  |               |                                             | 19.1 KVA | 53.1                                 | AMPS |
| (AMPS) =                         |                           |           |                              |         |         | 52.3                   | 61.6                   | 45.5 | TOTAL DEMAND LOAD =     |               |                                             | 19.9 KVA | 55.2                                 | AMPS |
| PERCENT LOAD IMBALANCE=          |                           |           |                              |         |         | 15%                    | 0%                     | 26%  |                         |               |                                             |          |                                      |      |

LOAD TYPES: C = CONTINUOUS, R = RECEPTACLE, M = MOTOR, N = NON-CONTINUOUS

| (E)PANEL LMPE5③                  |                  |           |                              |         |         |                        |                        |      |                               |               |             |         |                              |    |          |            |
|----------------------------------|------------------|-----------|------------------------------|---------|---------|------------------------|------------------------|------|-------------------------------|---------------|-------------|---------|------------------------------|----|----------|------------|
| MAIN BUS SIZE (AMPS): 225        |                  |           | INTERRUPT. RATING (KAIC): 10 |         |         | NEUTRAL BUS SIZE: 100% |                        |      | MOUNTING TYPE: SURFACE        |               |             |         |                              |    |          |            |
| MAIN BUS MATERIAL: COPPER        |                  |           | VOLTAGE: 120 / 208           |         |         | SECTION NO.: 1 OF 1    |                        |      | LOCATION: SERVICE CORRIDOR 60 |               |             |         |                              |    |          |            |
| MAIN CIRCUIT BREAKER (AMPS):     |                  |           | PHASE: 3                     |         | WIRE: 4 |                        | ENCLOSURE TYPE: NEMA 1 |      |                               | SQUARE D NQDD |             |         |                              |    |          |            |
| CKT NO.                          | DESCRIPTION      | LOAD TYPE | LOAD (KVA)                   | BREAKER |         |                        | BREAKER                |      | LOAD (KVA)                    | LOAD TYPE     | DESCRIPTION | CKT NO. |                              |    |          |            |
|                                  |                  |           |                              | AMPS    | POLES   |                        | POLES                  | AMPS |                               |               |             |         |                              |    |          |            |
| 1                                | (E)EXH FAN EF-5  | N         | 0.86                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.86        | M       | (E)EXH FAN EF-16             | 2  |          |            |
| 3                                | (E)EXH FAN EF-6  | M         | 0.86                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 0.86        | M       | (E)EXH FAN EF-17             | 4  |          |            |
| 5                                | (E)EXH FAN EF-7  | M         | 0.86                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 0.86        | M       | (E)EXH FAN EF-19             | 6  |          |            |
| 7                                | (E)EXH FAN EF-8  | M         | 0.86                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.86        | M       | (E)EXH FAN EF-20             | 8  |          |            |
| 9                                | (E)EXH FAN EF-18 | M         | 0.86                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 1.18        | M       | (E)FTU-21                    | 10 |          |            |
| 11                               | (E)EXH FAN EF-21 | M         | 0.86                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 1.18        | M       | (E)FTU-22                    | 12 |          |            |
| 13                               | (E)FTU-1/FTU-2   | M         | 1.39                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.86        | M       | (E)FTU-23                    | 14 |          |            |
| 15                               | (E)FTU-3         | M         | 0.86                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 0.86        | M       | (E)FTU-24                    | 16 |          |            |
| 17                               | (E)FTU-4         | M         | 1.66                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 0.50        | N       | (E)RM 200 DCC PANEL          | 18 |          |            |
| 19                               | (E)FTU-5         | M         | 1.66                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.50        | N       | (E)RM 200 DCC PANEL          | 20 |          |            |
| 21                               | (E)FTU-6         | M         | 1.18                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 0.50        | N       | (E)RM 200 DCC PANEL          | 22 |          |            |
| 23                               | (E)FTU-7/FTU-8   | M         | 1.73                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 0.50        | N       | (E)RM 200 DCC PANEL          | 24 |          |            |
| 25                               | (E)FTU-9/FTU-10  | M         | 1.39                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.50        | N       | (E)AHU-1 DCC PANEL           | 26 |          |            |
| 27                               | (E)FTU-11        | M         | 0.86                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 0.50        | N       | (E)AHU-1 DCC PANEL           | 28 |          |            |
| 29                               | (E)FTU-12        | M         | 0.86                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 0.50        | C       | (E)FIRE ALARM PANEL (FACP) ① | 30 |          |            |
| 31                               | (E)FTU-13/FTU-14 | M         | 1.73                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.50        | C       | SECURITY PANEL ②             | 32 |          |            |
| 33                               | (E)FTU-15        | M         | 1.18                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 0.00        | N       | SPARE                        | 34 |          |            |
| 35                               | (E)FTU-16        | M         | 1.18                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 0.00        | N       | SPARE                        | 36 |          |            |
| 37                               | (E)FTU-17/FTU-18 | M         | 1.39                         | 20      | 1       | A                      |                        |      | 1                             | 20            | 0.00        | N       | SPARE                        | 38 |          |            |
| 39                               | (E)FTU-19        | M         | 0.86                         | 20      | 1       |                        | B                      |      | 1                             | 20            | 0.00        | N       | SPARE                        | 40 |          |            |
| 41                               | (E)FTU-20        | M         | 1.18                         | 20      | 1       |                        |                        | C    | 1                             | 20            | 0.00        | N       | SPARE                        | 42 |          |            |
| CONNECTED LOAD PER PHASE (KVA) = |                  |           |                              |         |         | 13.4                   | 10.6                   | 11.9 | TOTAL CONNECTED LOAD =        |               |             |         |                              |    | 35.8 KVA | 99.5 AMPS  |
| (AMPS) =                         |                  |           |                              |         |         | 111.5                  | 88.1                   | 98.9 | TOTAL DEMAND LOAD =           |               |             |         |                              |    | 37.2 KVA | 103.4 AMPS |
| PERCENT LOAD IMBALANCE=          |                  |           |                              |         |         | 0%                     | 21%                    | 11%  |                               |               |             |         |                              |    |          |            |

LOAD TYPES: C = CONTINUOUS, R = RECEPTACLE, M = MOTOR, N = NON-CONTINUOUS

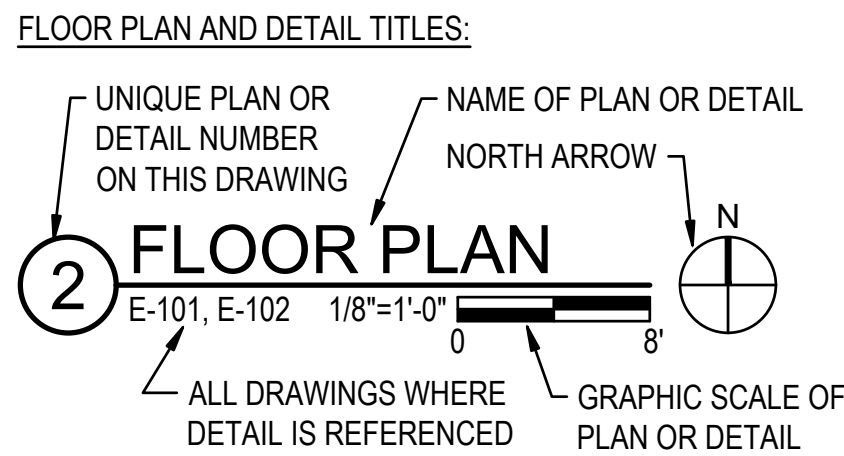
## SYMBOLS

|           |                                                                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ⌀         | DUPLEX RECEPTACLE, 20 AMPERE, 125 VOLT GROUNDING TYPE, NEMA 5-20R CONFIGURATION, RECESS MOUNTED IN 4" SQUARE x 2-1/8" DEEP PRESSED STEEL JUNCTION BOX AT 18" ABOVE FINISHED FLOOR.                                 |
| ⏏         | DUPLEX RECEPTACLE, 20 AMPERE, 125 VOLT GROUNDING TYPE, NEMA 5-20R CONFIGURATION, SURFACE MOUNTED IN 4" SQUARE x 2-1/8" DEEP PRESSED STEEL JUNCTION BOX AT 18" ABOVE FINISHED FLOOR.                                |
| ⌋         | JUNCTION BOX, 4" SQUARE x 2-1/8" DEEP GALVANIZED PRESSED STEEL, RECESS MOUNTED AT 18" ABOVE FINISHED FLOOR, WITH BLANK COVER PLATE.                                                                                |
| ⓪         | JUNCTION BOX, 4" SQUARE x 2-1/8" DEEP GALVANIZED PRESSED STEEL, MOUNTED ABOVE SUSPENDED TILE CEILING.                                                                                                              |
| ⌌         | CIRCUIT BREAKER TYPE PANELBOARD.                                                                                                                                                                                   |
| ■         | ELECTRICAL POINT OF CONNECTION.                                                                                                                                                                                    |
| T         | TRANSFORMER, SEE POWER RISER DIAGRAM FOR CHARACTERISTICS.                                                                                                                                                          |
| ⏏         | HEAVY DUTY ENCLOSED SAFETY SWITCH, SEE MOTOR AND EQUIPMENT SCHEDULE FOR SIZES AND CHARACTERISTICS.                                                                                                                 |
| Ⓜ         | MOTOR STARTER, SEE MOTOR AND EQUIPMENT SCHEDULE FOR SIZES AND CHARACTERISTICS.                                                                                                                                     |
| SM        | MANUAL MOTOR STARTER WITH THERMAL OVERLOAD RELAY, SURFACE MOUNTED IN DEVICE BOX ON UNIT UNLESS OTHERWISE NOTED.                                                                                                    |
| #10       | BRANCH CIRCUIT CONDUIT AND CONDUCTORS RUN EXPOSED. SIZE IN ACCORDANCE WITH N.E.C. RACEWAY SHALL BE MINIMUM 3/4" CONDUIT, AND WIRE SHALL BE MINIMUM SIZE #12 AWG COPPER, UNLESS OTHER SIZES ARE SPECIFICALLY NOTED. |
| EM2 1,3,5 | HOME RUN TO PANEL. NUMBER OF ARROWS INDICATE THE NUMBER OF CIRCUITS.                                                                                                                                               |
| —/—/—     | NUMBER OF SLASH MARKS INDICATE THE NUMBER OF CONDUCTORS IN RACEWAY. NO SLASH MARKS INDICATE 2 CIRCUIT CONDUCTORS PLUS A GROUND CONDUCTOR.                                                                          |
| —G—       | GREEN GROUND CONDUCTOR RUN IN RACEWAY.                                                                                                                                                                             |
| EM2       | PANEL NAME TO WHICH CIRCUITS RUN.                                                                                                                                                                                  |
| 1,3,5     | CIRCUIT BREAKERS IN PANEL SERVING THESE CIRCUITS.                                                                                                                                                                  |
| #10       | WIRE SIZE IN AWG. SIZE NOT SHOWN INDICATES #12 AWG.                                                                                                                                                                |
| #10       | BRANCH CIRCUIT CONDUIT AND CONDUCTORS RUN CONCEALED IN WALLS AND CEILING, ALL OTHER CHARACTERISTICS SAME AS SHOWN ABOVE.                                                                                           |
| #10       | BRANCH CIRCUIT CONDUIT AND CONDUCTORS RUN CONCEALED IN CONCRETE, ALL OTHER CHARACTERISTICS SAME AS SHOWN ABOVE.                                                                                                    |
| —EX—      | EXISTING CONDUIT AND CONDUCTORS TO REMAIN.                                                                                                                                                                         |
| —EC—      | EMPTY 1" CONDUIT WITH NYLON PULL STRING.                                                                                                                                                                           |
| —W—       | SURFACE METAL RACEWAY, WIREMOLD SERIES 700 UNLESS SPECIFICALLY NOTED OTHERWISE.                                                                                                                                    |

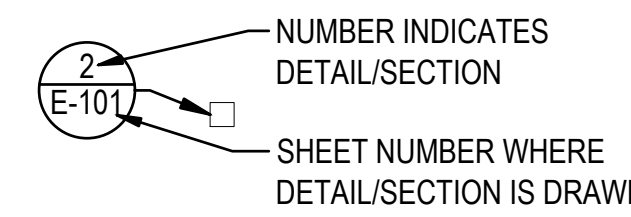
## ABBREVIATIONS

|        |                                                 |
|--------|-------------------------------------------------|
| A      | AMPERES.                                        |
| AF     | FRAME SIZE IN AMPERES.                          |
| AFF    | ABOVE FINISHED FLOOR.                           |
| AT     | TRIP SIZE IN AMPERES.                           |
| ATS    | AUTOMATIC TRANSFER SWITCH.                      |
| AWG.   | AMERICAN WIRE GAUGE.                            |
| BATT   | BATTERY BACK-UP.                                |
| BLDG.  | BUILDING.                                       |
| C/B    | CIRCUIT BREAKER.                                |
| CBM    | CERTIFIED BALLAST MANUFACTURERS.                |
| DDC    | DIRECT DIGITAL CONTROL.                         |
| EC     | EMPTY CONDUIT.                                  |
| EM     | EMERGENCY PANEL.                                |
| EMT    | ELECTRICAL METALLIC TUBING.                     |
| ETR    | EXISTING TO REMAIN.                             |
| FA     | FIRE ALARM.                                     |
| FAAP   | FIRE ALARM ANNUNCIATOR PANEL.                   |
| FACP   | FIRE ALARM CONTROL PANEL.                       |
| FLA    | FULL LOAD AMPERES.                              |
| FVNR   | FULL VOLTAGE NON-REVERSING.                     |
| GFI    | GROUND FAULT INTERRUPTER.                       |
| GRD.   | GROUND CONDUCTOR.                               |
| H-O-A  | HAND-OFF-AUTOMATIC SELECTOR SWITCH.             |
| IDS    | INTRUSION DETECTION PANEL.                      |
| KAIC   | THOUSAND AMPERE INTERRUPTING CAPACITY.          |
| KW     | KILO-WATTS.                                     |
| M.L.O. | MAIN LUGS ONLY.                                 |
| N.C.   | NORMALLY CLOSED.                                |
| N.E.C. | NATIONAL ELECTRICAL CODE.                       |
| NEMA   | NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION. |
| NFSS   | NON-FUSED SAFETY SWITCH.                        |
| N.O.   | NORMALLY OPEN.                                  |
| P      | NUMBER OF POLES.                                |
| PH.    | NUMBER OF PHASES.                               |
| PVC    | POLY-VINYL CHLORIDE.                            |
| RGS    | RIGID GALVANIZED STEEL.                         |
| SPD    | SURGE PROTECTIVE DEVICE.                        |
| U.L.   | UNDERWRITER'S LABORATORIES.                     |
| V      | VOLTAGE.                                        |

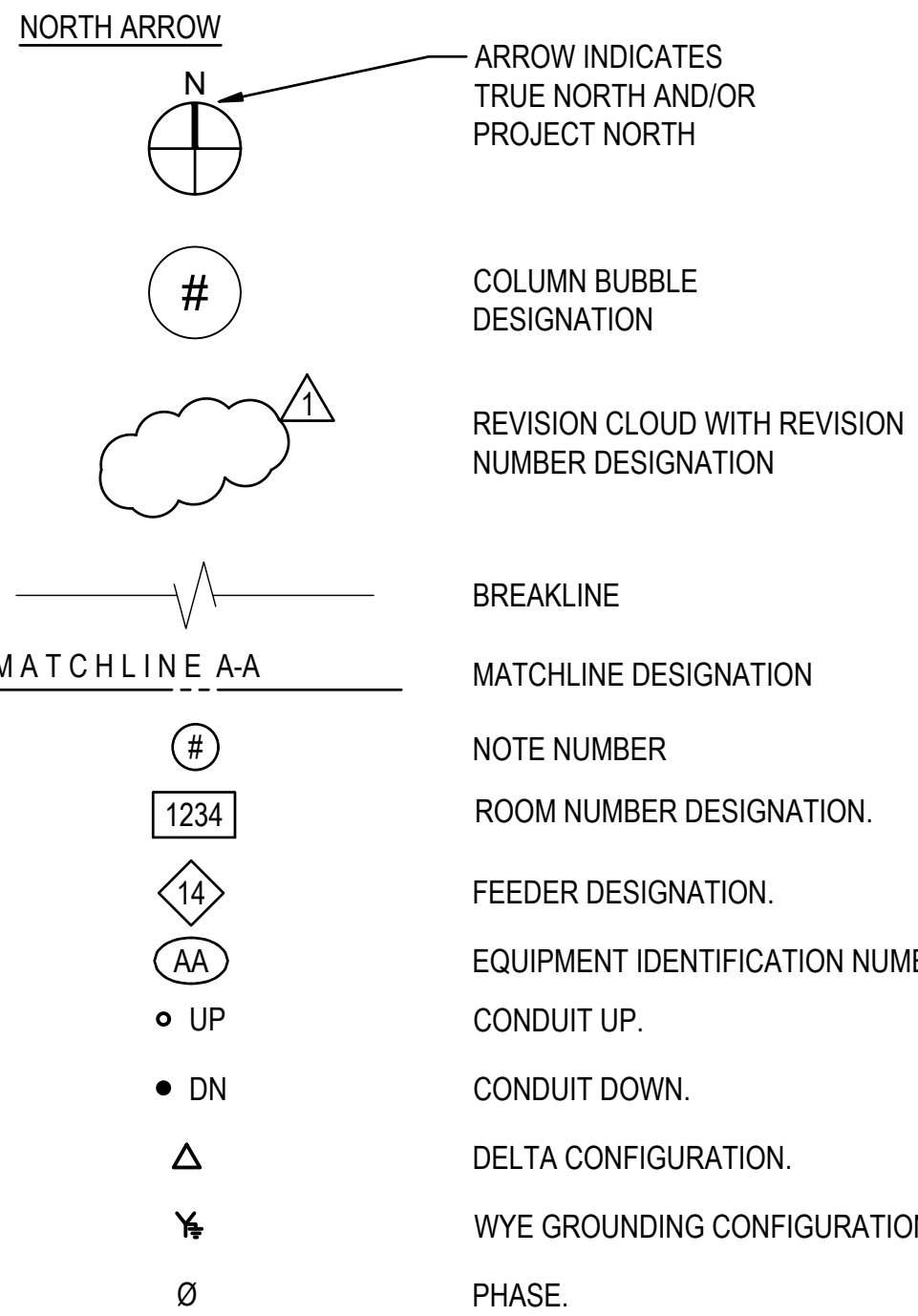
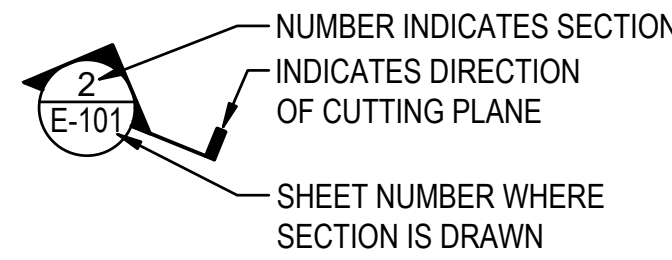
## GENERAL SYMBOLS



### 2-PART REFERENCE DETAIL IDENTIFICATION:



### SECTION IDENTIFICATION



## GENERAL NOTES

- VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".
- FOR INFORMATION ONLY.
- PROVIDE CIRCUIT BREAKER INDICATED AND LABEL AS NOTED.
- PROVIDE TYPEWRITTEN PANELBOARD SCHEDULE.

## NOTES

| (E)PANEL RP③                     |                             |           |                              |         |         |                        |                        |      |                        |               |             |         |                             |    |          |           |
|----------------------------------|-----------------------------|-----------|------------------------------|---------|---------|------------------------|------------------------|------|------------------------|---------------|-------------|---------|-----------------------------|----|----------|-----------|
| MAIN BUS SIZE (AMPS): 225        |                             |           | INTERRUPT. RATING (KAIC): 10 |         |         | NEUTRAL BUS SIZE: 100% |                        |      | MOUNTING TYPE: SURFACE |               |             |         |                             |    |          |           |
| MAIN BUS MATERIAL: COPPER        |                             |           | VOLTAGE: 120 / 208           |         |         | SECTION NO.: 1 OF 1    |                        |      | LOCATION: TANK ROOM    |               |             |         |                             |    |          |           |
| MAIN CIRCUIT BREAKER (AMPS): 150 |                             |           | PHASE: 3                     |         | WIRE: 4 |                        | ENCLOSURE TYPE: NEMA 1 |      |                        | SQUARE D NQDD |             |         |                             |    |          |           |
| CKT NO.                          | DESCRIPTION                 | LOAD TYPE | LOAD (KVA)                   | BREAKER |         |                        | BREAKER                |      | LOAD (KVA)             | LOAD TYPE     | DESCRIPTION | CKT NO. |                             |    |          |           |
|                                  |                             |           |                              | AMPS    | POLES   |                        | POLES                  | AMPS |                        |               |             |         |                             |    |          |           |
| 1                                | (E)TANK RM LTG VIA RELAY R1 | N         | 0.40                         | 20      | 1       | A                      |                        |      | 1                      | 20            | 0.30        | R       | (E)TANK RM LTG VIA RELAY R5 | 2  |          |           |
| 3                                | (E)TANK RM LTG VIA RELAY R2 | N         | 0.40                         | 20      | 1       |                        | B                      |      | 1                      | 20            | 0.30        | R       | (E)TANK RM LTG VIA RELAY R6 | 4  |          |           |
| 5                                | (E)TANK RM LTG VIA RELAY R3 | N         | 0.40                         | 20      | 1       |                        |                        | C    | 1                      | 20            | 1.20        | R       | (E)TANK RM LTG VIA RELAY R7 | 6  |          |           |
| 7                                | (E)TANK RM LTG VIA RELAY R4 | N         | 0.40                         | 20      | 1       | A                      |                        |      | 1                      | 20            | 1.20        | R       | (E)TANK RM LTG              | 8  |          |           |
| 9                                | (E)LTG CONTROL PANEL XFMR   | C         | 0.20                         | 20      | 1       |                        | B                      |      | 1                      | 20            | 1.20        | R       | (E)TANK RM LTG              | 10 |          |           |
| 11                               | (E)RECPT-100,103,104        | R         | 1.20                         | 20      | 1       |                        |                        | C    | 1                      | 20            | 1.30        | R       | (E)LTG ROOMS 101-105        | 12 |          |           |
| 13                               | (E)RECPT-TANK ROOM          | R         | 1.00                         | 20      | 1       | A                      |                        |      | 1                      | 20            | 0.90        | R       | (E)EF-2                     | 14 |          |           |
| 15                               | (E)RECPT-EXTERIOR           | R         | 1.20                         | 20      | 1       |                        | B                      |      | 1                      | 20            | 1.65        | R       | (E)UH-2                     | 16 |          |           |
| 17                               | (E)RECPT-TANK ROOM          | R         | 0.60                         | 20      | 1       |                        |                        | C    | 1                      | --            | 1.65        | N       | SPACE                       | 18 |          |           |
| 19                               | (E)RECPT-TANK ROOM          | R         | 0.40                         | 20      | 1       | A                      |                        |      | 1                      | 20            | 0.00        | N       | SPARE                       | 20 |          |           |
| 21                               | (E)RECPT-TANK ROOM          | R         | 1.20                         | 20      | 1       |                        | B                      |      | 1                      | 30            | 3.00        | R       | (E)WATER HEATER             | 22 |          |           |
| 23                               | (E)RECPT-TANK ROOM          | R         | 0.60                         | 20      | 1       |                        |                        | C    | 1                      | 20            | 0.90        | R       | (E)AC-1                     | 24 |          |           |
| 25                               | (E)RECPT-TANK ROOM          | R         | 0.60                         | 20      | 1       | A                      |                        |      | 1                      | --            | 0.00        | N       | SPACE                       | 26 |          |           |
| 27                               | (E)FIRE ALARM PANEL ①       | C         | 0.20                         | 20      | 1       |                        | B                      |      | 1                      | 20            | 0.00        | N       | SPARE                       | 28 |          |           |
| 29                               | (E)TELEPHONE CABINET        | C         | 0.20                         | 20      | 1       |                        |                        | C    | 1                      | 20            | 0.00        | N       | SPARE                       | 30 |          |           |
| 31                               | SECURITY PANEL ②            | C         | 0.20                         | 20      | 1       | A                      |                        |      | 1                      | 20            | 0.00        | N       | SPARE                       | 32 |          |           |
| 33                               | SPARE                       | N         | 0.00                         | 20      | 1       |                        | B                      |      | 1                      | 20            | 0.00        | N       | SPARE                       | 34 |          |           |
| 35                               | (E)FEED SYSTEM              | N         | 0.20                         | 20      | 1       |                        |                        | C    | 1                      | 20            | 0.00        | N       | SPARE                       | 36 |          |           |
| 37                               | SPARE                       | N         | 0.00                         | 20      | 1       | A                      |                        |      | 1                      | 20            | 0.00        | N       | SPARE                       | 38 |          |           |
| 39                               | (E)SUMP PUMP                | M         | 1.60                         | 20      | 1       |                        | B                      |      | 1                      | 20            | 1.40        | R       | (E)RECPT-EYE WASH HTR       | 40 |          |           |
| 41                               | SPARE                       | N         | 0.00                         | 20      | 1       |                        |                        | C    | 1                      | 20            | 0.00        | N       | SPARE                       | 42 |          |           |
|                                  |                             |           |                              |         |         |                        |                        |      |                        |               |             |         |                             |    |          |           |
| CONNECTED LOAD PER PHASE (kVA) = |                             |           |                              |         |         |                        |                        |      |                        | 5.4           | 12.4        | 8.3     | TOTAL CONNECTED LOAD =      |    | 26.0 KVA | 72.2 AMPS |
| (AMPS) =                         |                             |           |                              |         |         |                        |                        |      |                        | 45.0          | 102.9       | 68.8    | TOTAL DEMAND LOAD =         |    | 21.5 KVA | 59.8 AMPS |
| PERCENT LOAD IMBALANCE =         |                             |           |                              |         |         |                        |                        |      |                        | 56%           | 0%          | 33%     |                             |    |          |           |



G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.

G2. ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "E".



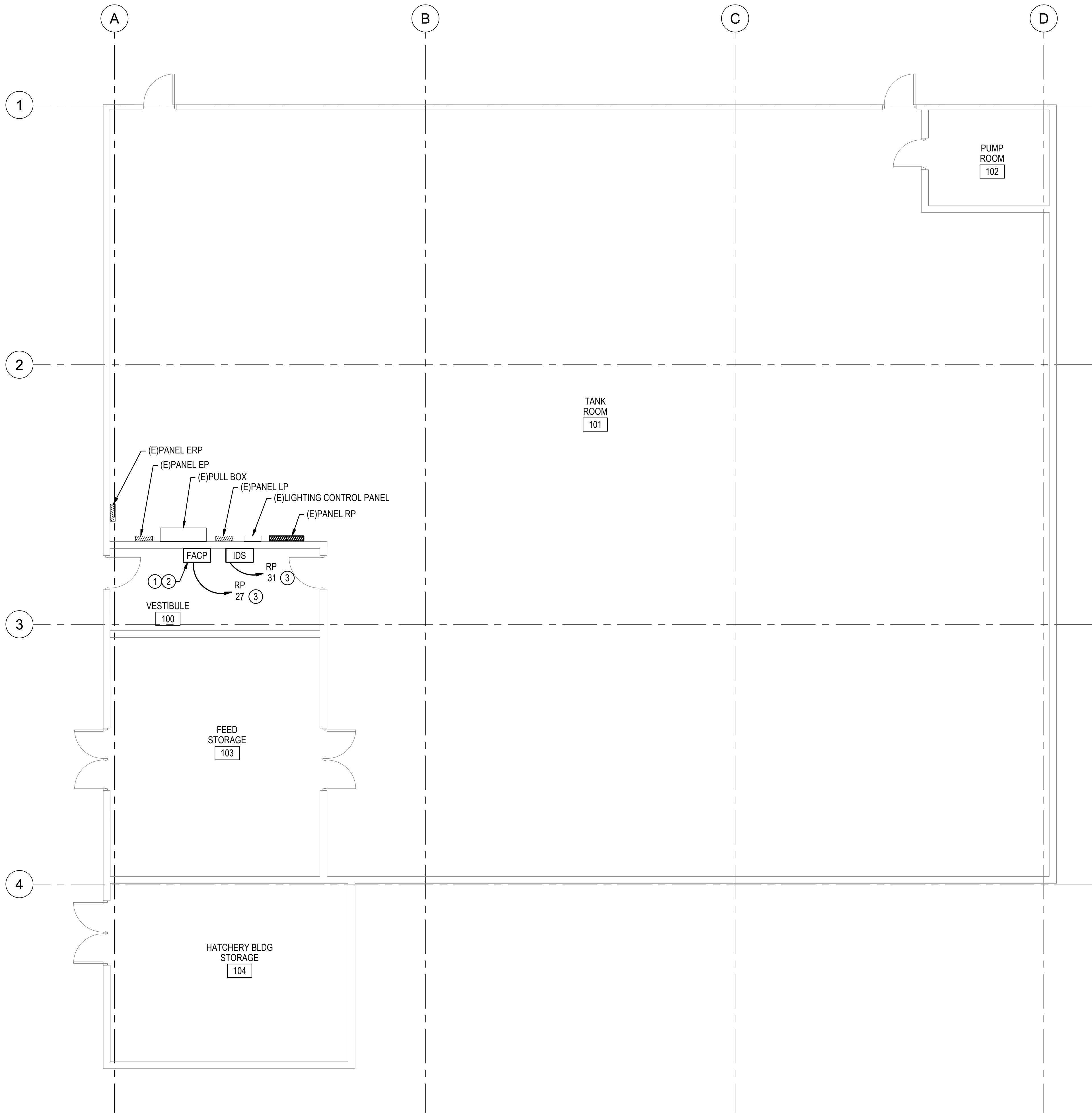
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| <div>SUMMER<br/>CONSULTANTS<br/>INCORPORATED</div> <div>7900 Westpark Dr. / Suite A405 / McLean, VA 22102<br/>(703) 556-8820 / www.summerconsultants.com</div> |  |  |  |  |  |  |  |  |
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ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

FILE NAME: 212.012 E-101.DWG  
OUR REF: 212.012  
PLOT DATE: 12/3/2025 1:19:20 PM



FILE NAME: 212.012 E-102.DWG  
OUR REF: 212.012  
PLOT DATE: 12/3/2025 1:19:23 PM



1 BROODSTOCK BUILDING  
3/16"=1'-0"  
0 4' 8' 12'

## GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".
- G3. DEMOLITION DEFINITION: DISCONNECTION AND REMOVAL OF ASSOCIATED CONDUCTORS SHALL CONSIST OF THE REMOVAL OF THE CONDUCTORS BACK TO THE NEXT EXISTING DEVICE TO REMAIN. NEW CONDUIT AND CONDUCTORS SHALL BE PROVIDED TO RECONNECT ANY EXISTING TO REMAIN DEVICES INTERRUPTED BY THE DEMOLITION UNDER THIS PROJECT.

## NOTES

1. DISCONNECT FIRE ALARM CONTROL PANEL. TAG, PROTECT AND RETAIN ALL ASSOCIATED CONDUIT AND CONDUCTORS BACK TO SOURCE.
2. CONNECT FIRE ALARM CONTROL PANEL TO CONDUIT AND CONDUCTORS RETAINED DURING DEMOLITION. CONNECT TO OPERATE AS BEFORE.
3. FOR INFORMATION ONLY.



|                                                                                                                                                                                                                                                                                        |                                |                      |                        |                                            |  |                                  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------|------------------------|--------------------------------------------|--|----------------------------------|--|
| <div>SUMMER<br/>CONSULTANTS<br/>INCORPORATED</div> <div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div>                                                                                                                         |                                |                      |                        |                                            |  |                                  |  |
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|                                                                                                                                                                                                                                                                                        |                                |                      |                        |                                            |  |                                  |  |
| NO.                                                                                                                                                                                                                                                                                    |                                | DATE                 |                        | BY                                         |  | DESCRIPTION                      |  |
| REVISIONS                                                                                                                                                                                                                                                                              |                                |                      |                        |                                            |  |                                  |  |
| U<br>S<br>D<br>A                                                                                                                                                                                                                                                                       | A-E FIRM                       |                      | USDA                   |                                            |  |                                  |  |
|                                                                                                                                                                                                                                                                                        | PROJECT MANAGER<br>PATERSON, A |                      | EPM<br>C. MORRIS       |                                            |  |                                  |  |
|                                                                                                                                                                                                                                                                                        | DESIGNER<br>DI SAVINO, E       |                      | PPM<br>T. VIGIL        |                                            |  |                                  |  |
|                                                                                                                                                                                                                                                                                        | CHECKED BY<br>URDA, K          |                      | SAFETY & HEALTH<br>TBD |                                            |  |                                  |  |
| DRAWN BY<br>DI SAVINO, E                                                                                                                                                                                                                                                               |                                | REAL PROPERTY<br>TBD |                        | SOLICITATION NO.<br><br>DATE<br>2025-12-12 |  | BROODSTOCK BUILDING<br><br>E-102 |  |
|                                                                                                                                                                                                                                                                                        |                                |                      |                        |                                            |  | SHEET 25 OF 29                   |  |
| <div><div>Agricultural<br/>Research<br/>Service</div><div>FIRE SYSTEM UPGRADES<br/><br/>KEARNEYSVILLE, WEST VIRGINIA<br/><br/>NATIONAL CENTER FOR COOL AND COLD WATER<br/>AQUACULTURE</div></div> |                                |                      |                        |                                            |  |                                  |  |

ISSUED FOR CONSTRUCTION - 9 JANUARY 2026

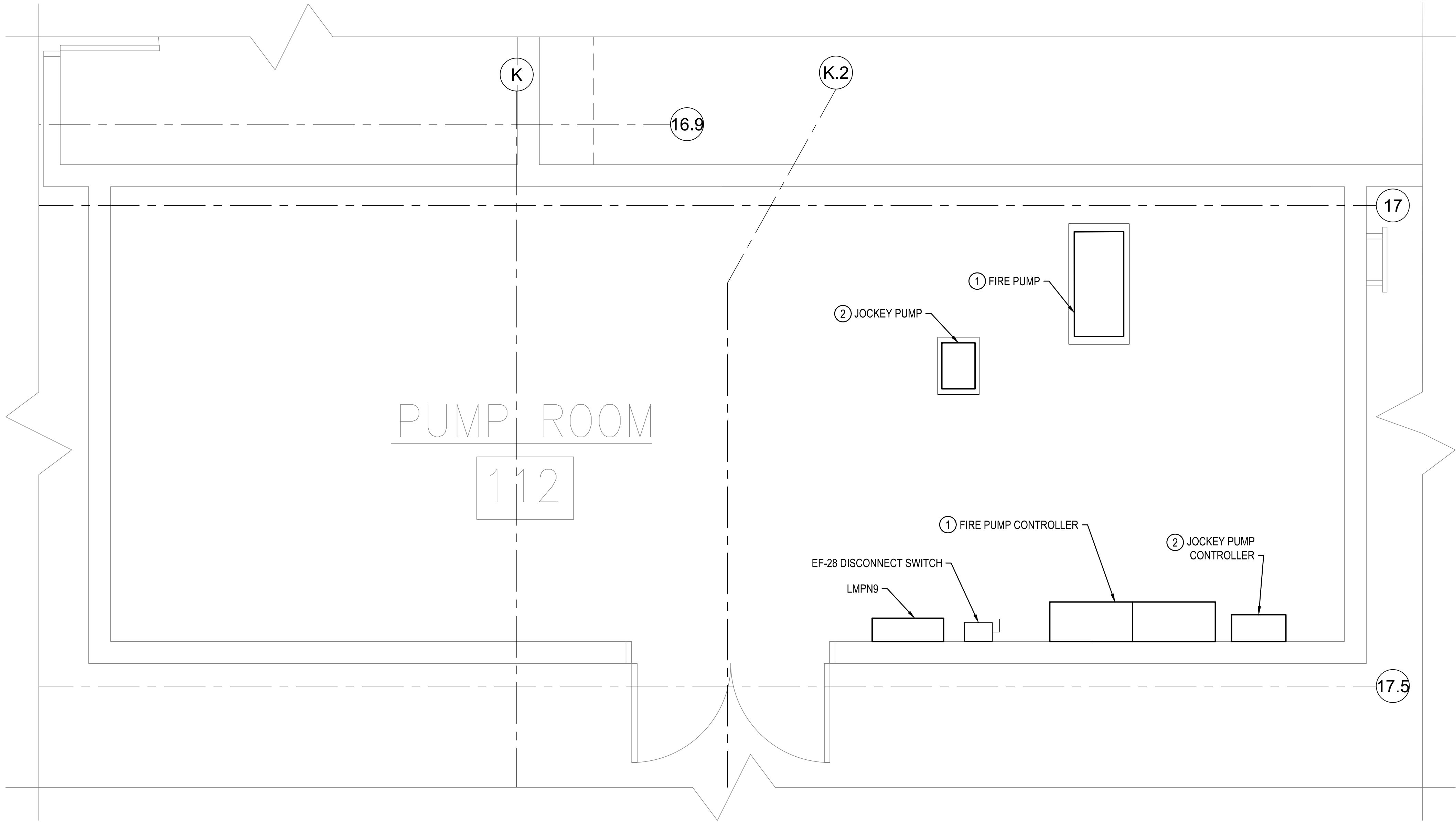


NOTES

1. DISCONNECT FIRE PUMP. REMOVE ASSOCIATED CONDUIT AND CONDUCTORS BACK TO CONTROLLER. TAG, PROTECT AND RETAIN ASSOCIATED FIRE PUMP CONTROLLER. TAG, PROTECT AND RETAIN ALL ASSOCIATED CONDUIT AND CONDUCTORS FROM FIRE PUMP CONTROLLER BACK TO SOURCE.
2. DISCONNECT AND REMOVE JOCKEY PUMP. REMOVE ASSOCIATED CONTROLLER AND ASSOCIATED CONDUIT AND CONDUCTORS BACK TO SOURCE.

GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".
- G3. DEMOLITION DEFINITION: DISCONNECTION AND REMOVAL OF ASSOCIATED CONDUCTORS SHALL CONSIST OF THE REMOVAL OF THE CONDUCTORS BACK TO THE NEXT EXISTING DEVICE TO REMAIN. NEW CONDUIT AND CONDUCTORS SHALL BE PROVIDED TO RECONNECT ANY EXISTING TO REMAIN DEVICES INTERRUPTED BY THE DEMOLITION UNDER THIS PROJECT.



1 DETAIL - PUMP ROOM - DEMOLITION  
E-101 1/2"=1'-0" 0 2' 4'



|                                                                                                                                                                |  |                        |    |                                                                                                                                                                                                         |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>SUMMERCONSULTANTSINCORPORATED</div><div>7900 Westpark Dr.   Suite A405   McLean, VA 22102<br/>(703) 556-8820   www.summerconsultants.com</div></div> |  |                        |    |                                                                                                                                                                                                         |  |
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| NO.                                                                                                                                                            |  | DATE                   | BY | DESCRIPTION                                                                                                                                                                                             |  |
| REVISIONS                                                                                                                                                      |  |                        |    |                                                                                                                                                                                                         |  |
| A-E FIRM                                                                                                                                                       |  | USDA                   |    | <div><div>ds</div><div>Agricultural Research Service</div></div> <div>FIRE SYSTEM UPGRADES</div> <div>KEARNEYSVILLE, WEST VIRGINIA</div> <div>NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE</div> |  |
| PROJECT MANAGER<br>PATERSON, A                                                                                                                                 |  | EPM<br>C. MORRIS       |    |                                                                                                                                                                                                         |  |
| DESIGNER<br>DI SAVINO, E                                                                                                                                       |  | PPM<br>T. VIGIL        |    |                                                                                                                                                                                                         |  |
| CHECKED BY<br>URDA, K                                                                                                                                          |  | SAFETY & HEALTH<br>TBD |    | SOLICITATION NO.                                                                                                                                                                                        |  |
| DRAWN BY<br>DI SAVINO, E                                                                                                                                       |  | REAL PROPERTY<br>TBD   |    | DATE<br>2025-12-12                                                                                                                                                                                      |  |
|                                                                                                                                                                |  |                        |    | DETAILS                                                                                                                                                                                                 |  |
|                                                                                                                                                                |  |                        |    | DWG. NO.<br>E-401                                                                                                                                                                                       |  |
|                                                                                                                                                                |  |                        |    | SHEET 26 of 29                                                                                                                                                                                          |  |

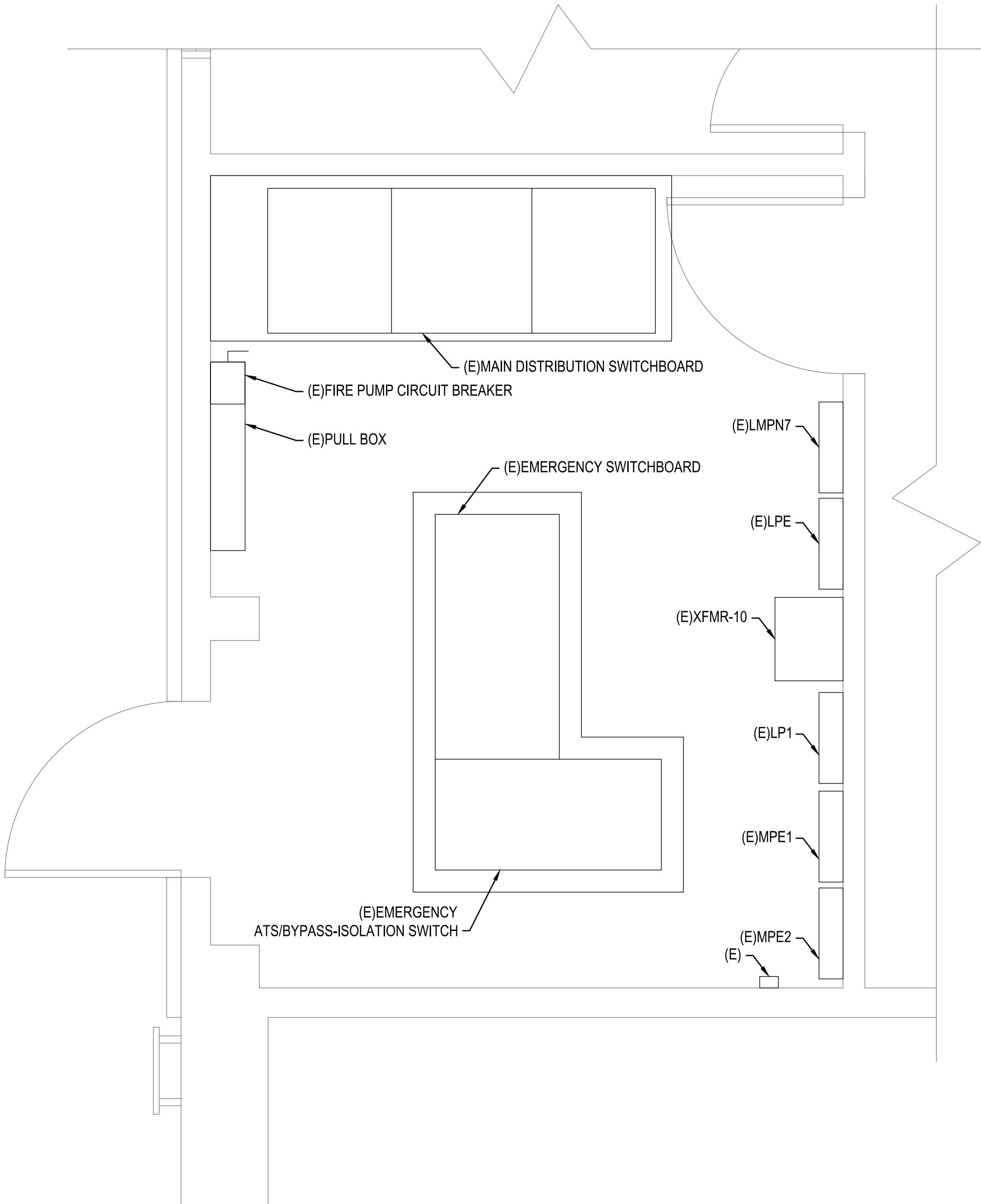


NOTES

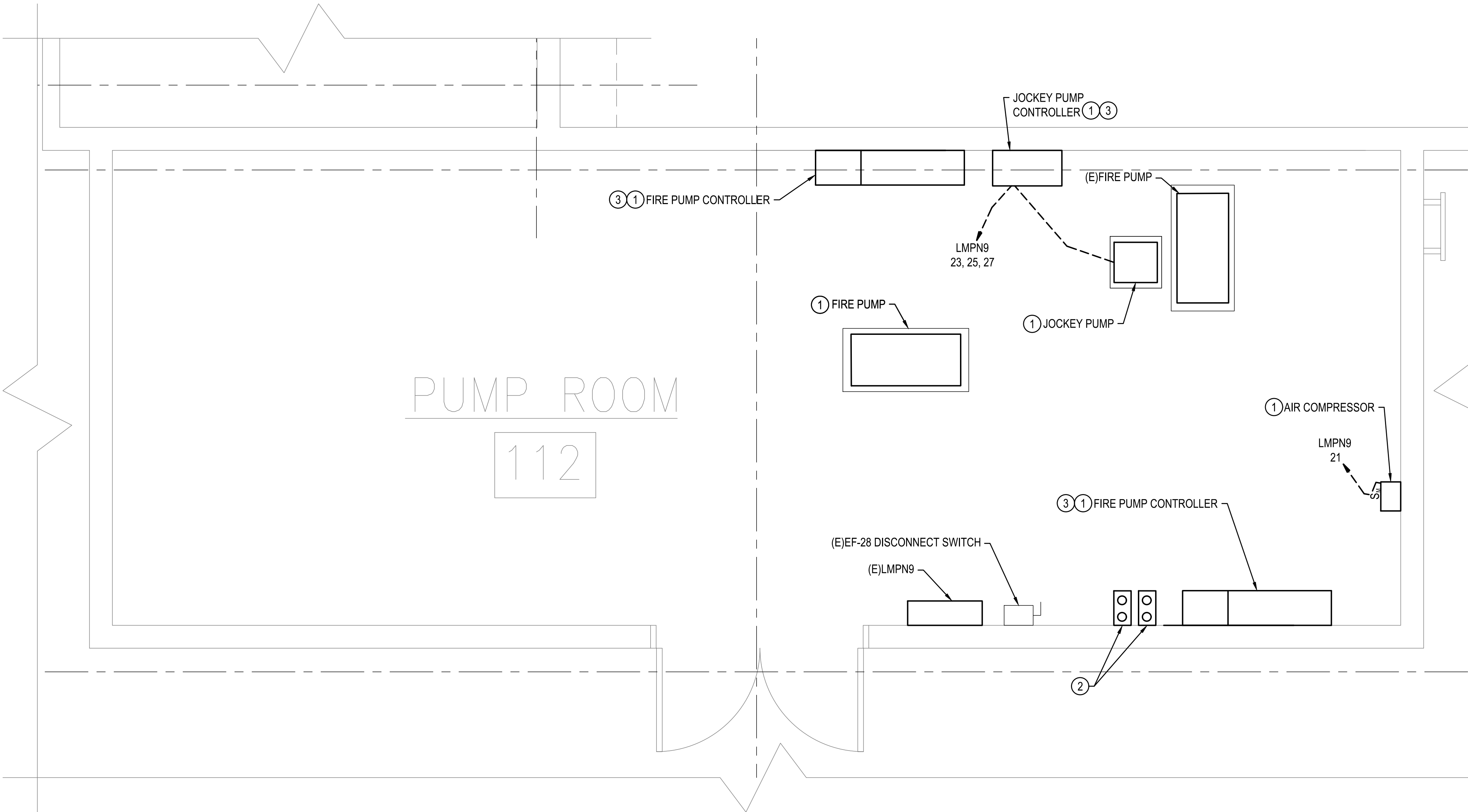
1. SEE MOTOR AND EQUIPMENT SCHEDULE FOR MOTOR AND FEEDER CHARACTERISTICS.
2. PROVIDE 6" W x 24" L x 12" D JUNCTION BOX A MINIMUM OF 12 INCHES ABOVE THE FLOOR. JUNCTION BOX SHALL BE A MINIMUM NEMA TYPE 2 ENCLOSURE. EXTEND EMERGENCY AND NORMAL FEEDS FOR THE FIRE PUMP RETAINED DURING DEMOLITION TO JUNCTION BOX. SPLICE FEEDERS IN JUNCTION BOX TO SERVE THE TWO FIRE PUMP CONTROLLERS.
3. PROVIDE 36" DEPTH WORKING CLEARANCE IN FRONT OF EQUIPMENT.

GENERAL NOTES

- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".



1  
DETAIL -  
ELECTRIC ROOM  
E-101



2  
DETAIL -  
PUMP ROOM  
E-101



|                                                                                                                                                                |                 |                 |           |                  |                                                     |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------|------------------|-----------------------------------------------------|----------------|
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|                                                                                                                                                                |                 |                 |           |                  |                                                     |                |
|                                                                                                                                                                |                 |                 |           |                  |                                                     |                |
|                                                                                                                                                                | NO.             | DATE            | BY        | DESCRIPTION      |                                                     |                |
| REVISIONS                                                                                                                                                      |                 |                 |           |                  |                                                     |                |
| U<br>S<br>D<br>A                                                                                                                                               | A-E FIRM        |                 | USDA      |                  |                                                     |                |
|                                                                                                                                                                | PROJECT MANAGER |                 | EPM       |                  | FIRE SYSTEM UPGRADES                                |                |
|                                                                                                                                                                | PATERSON, A     |                 | C. MORRIS |                  | KEARNEYSVILLE, WEST VIRGINIA                        |                |
|                                                                                                                                                                | DESIGNER        |                 | PPM       |                  | NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE |                |
| DI SAVINO, E                                                                                                                                                   |                 | T. VIGIL        |           |                  |                                                     |                |
| CHECKED BY                                                                                                                                                     |                 | SAFETY & HEALTH |           | SOLICITATION NO. | DWS. NO.                                            |                |
| URDA, K                                                                                                                                                        |                 | TBD             |           |                  |                                                     |                |
| DRAWN BY                                                                                                                                                       |                 | REAL PROPERTY   |           | DATE             | DETAILS                                             |                |
| DI SAVINO, E                                                                                                                                                   |                 | TBD             |           |                  |                                                     |                |
|                                                                                                                                                                |                 |                 |           | 2025-12-12       |                                                     | E-402          |
|                                                                                                                                                                |                 |                 |           |                  |                                                     | SHEET 27 of 29 |



NOTES

1.

DISCONNECT FIRE ALARM CONTROL PANEL. REMOVE CONDUIT AND CONDUCTORS BACK TO CEILING ABOVE. PROVIDE JUNCTION BOX ABOVE CEILING. TAG, PROTECT AND RETAIN ALL ASSOCIATED CONDUIT AND CONDUCTORS FROM JUNCTION BOX BACK TO SOURCE.
2.

EXTEND CONDUIT AND CONDUCTORS FROM JUNCTION BOX PROVIDED TO FIRE ALARM CONTROL PANEL. CONNECT FIRE ALARM CONTROL PANEL TO CIRCUIT RETAINED DURING DEMOLITION. CONNECT TO OPERATE AS BEFORE.
3.

FOR INFORMATION ONLY.

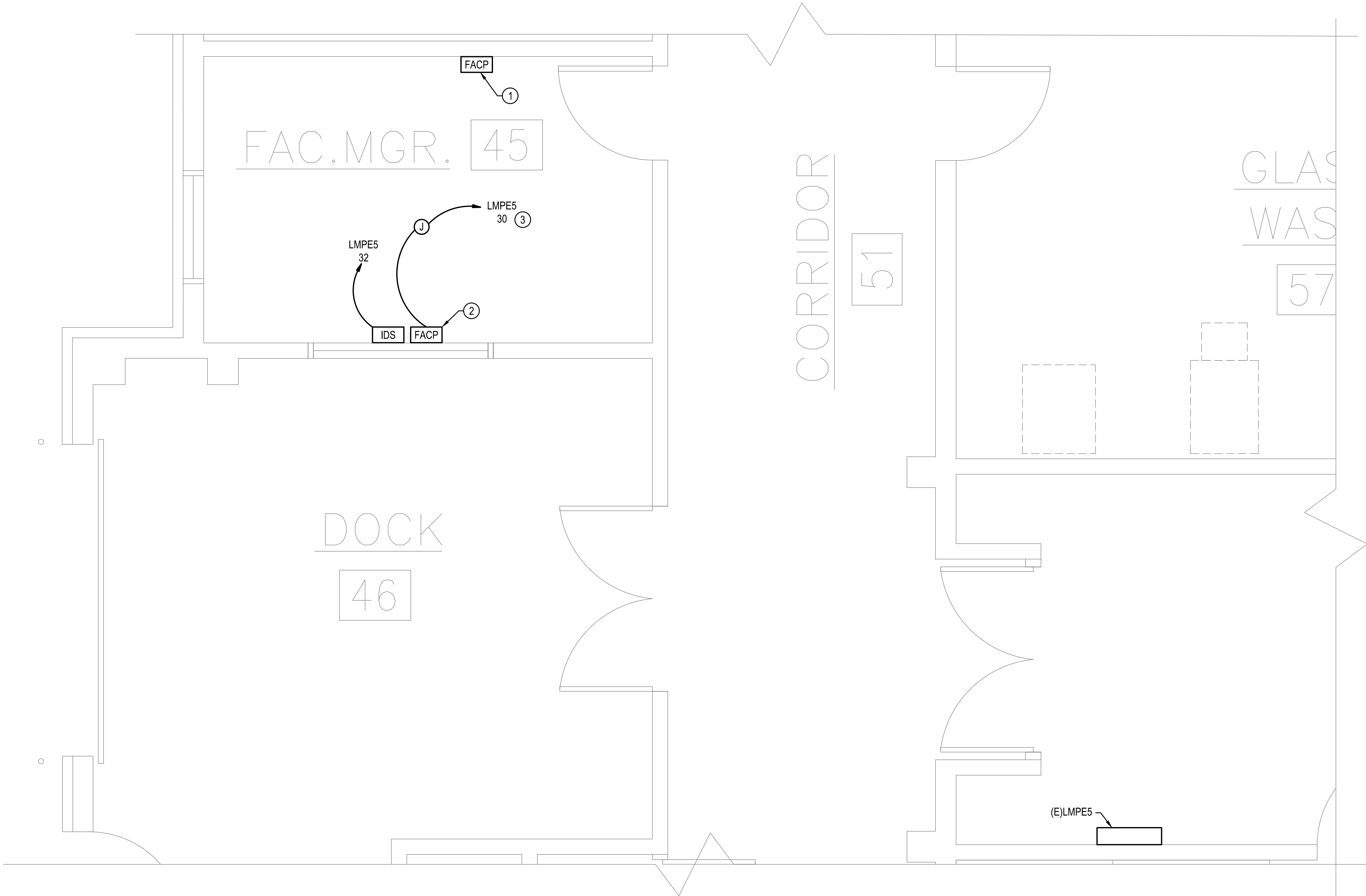
GENERAL NOTES

- G1.

VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2.

ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "(E)".
- G3.

DEMOLITION DEFINITION: DISCONNECTION AND REMOVAL OF ASSOCIATED CONDUCTORS SHALL CONSIST OF THE REMOVAL OF THE CONDUCTORS BACK TO THE NEXT EXISTING DEVICE TO REMAIN. NEW CONDUIT AND CONDUCTORS SHALL BE PROVIDED TO RECONNECT ANY EXISTING TO REMAIN DEVICES INTERRUPTED BY THE DEMOLITION UNDER THIS PROJECT.



DETAIL -  
FACILITY MANAGER ROOM  
1 E-101

1/2"=1'-0"  
0 2' 4'



|                                                                                                                                                                                            |  |                               |  |                                                     |    |                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------|--|-----------------------------------------------------|----|----------------|
| <div><div>SUMMER</div><div>CONSULTANTS</div><div>INCORPORATED</div><div>7900 Westpark Dr.   Suite A405   McLean, VA 22102</div><div>(703) 556-8820   www.summerconsultants.com</div></div> |  | NO.                           |  | DATE                                                | BY | DESCRIPTION    |
|                                                                                                                                                                                            |  | REVISIONS                     |  |                                                     |    |                |
|                                                                                                                                                                                            |  | U<br>S<br>D<br>A              |  |                                                     |    |                |
|                                                                                                                                                                                            |  | Agricultural Research Service |  |                                                     |    |                |
| A-E FIRM                                                                                                                                                                                   |  | USDA                          |  | FIRE SYSTEM UPGRADES                                |    |                |
| PROJECT MANAGER                                                                                                                                                                            |  | EPM                           |  | HEARNESVILLE, WEST VIRGINIA                         |    |                |
| PATERSON, A                                                                                                                                                                                |  | C. MORRIS                     |  | NATIONAL CENTER FOR COOL AND COLD WATER AQUACULTURE |    |                |
| DESIGNER                                                                                                                                                                                   |  | PPM                           |  |                                                     |    |                |
| DI SAVINO, E                                                                                                                                                                               |  | T. VIGIL                      |  |                                                     |    |                |
| CHECKED BY                                                                                                                                                                                 |  | SAFETY & HEALTH               |  | SOLICITATION NO.                                    |    | DWG. NO.       |
| URDA, K                                                                                                                                                                                    |  | TBD                           |  |                                                     |    | E-403          |
| DRAWN BY                                                                                                                                                                                   |  | REAL PROPERTY                 |  | DATE                                                |    | SHEET 28 OF 29 |
| DI SAVINO, E                                                                                                                                                                               |  | TBD                           |  | 2025-12-12                                          |    |                |



- G1. VERIFY LOCATION AND DIMENSIONS OF EXISTING EQUIPMENT AND COORDINATE ALL WORK PRIOR TO THE START OF CONSTRUCTION.
- G2. ALL WORK MUST BE NEW AND PROVIDED UNDER THIS CONTRACT UNLESS SPECIFICALLY MARKED "EXISTING", "EXIST", OR "EJ".
- G3. DEMOLITION DEFINITION: DISCONNECTION AND REMOVAL OF ASSOCIATED CONDUCTORS SHALL CONSIST OF THE REMOVAL OF THE CONDUCTORS BACK TO THE NEXT EXISTING DEVICE TO REMAIN. NEW CONDUIT AND CONDUCTORS SHALL BE PROVIDED TO RECONNECT ANY EXISTING TO REMAIN DEVICES INTERRUPTED BY THE DEMOLITION UNDER THIS PROJECT.

1. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER SERVING FIRE PUMP AT THE GENERATOR ENCLOSURE. TAG, PROTECT AND RETAIN CONDUIT AND CONDUCTORS FOR FIRE PUMP FOR REUSE. PROVIDE CIRCUIT BREAKER INDICATED, CONNECT TO CONDUCTORS RETAINED DURING DEMOLITION. CONNECT TO OPERATE AS BEFORE.



ISSUED FOR CONSTRUCTION - 9 JANUARY 2026